CSE 4615:: Wireless Networks

Overview of Wireless Access Networks

Lecture 2

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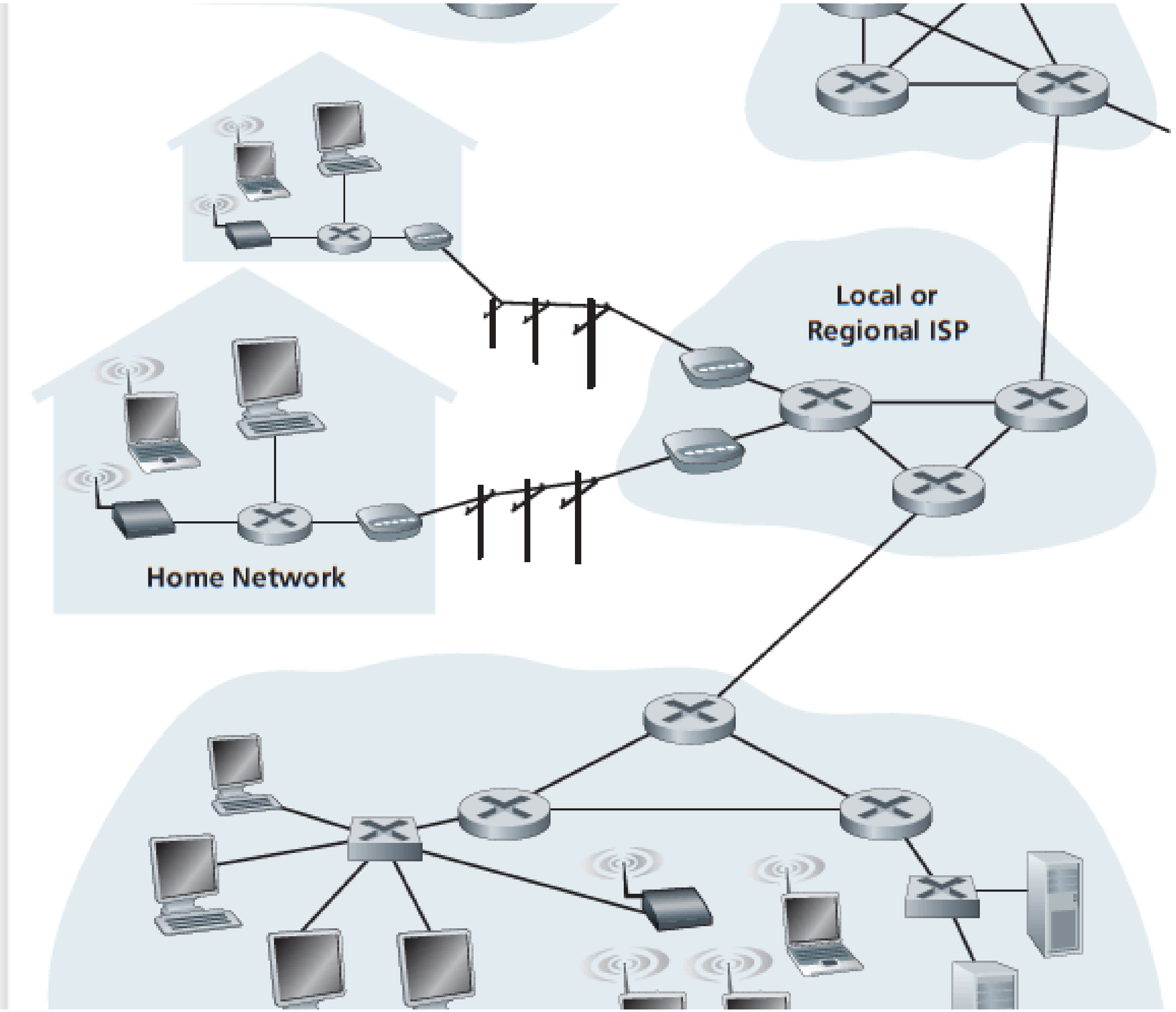
Internet – the bigger picture



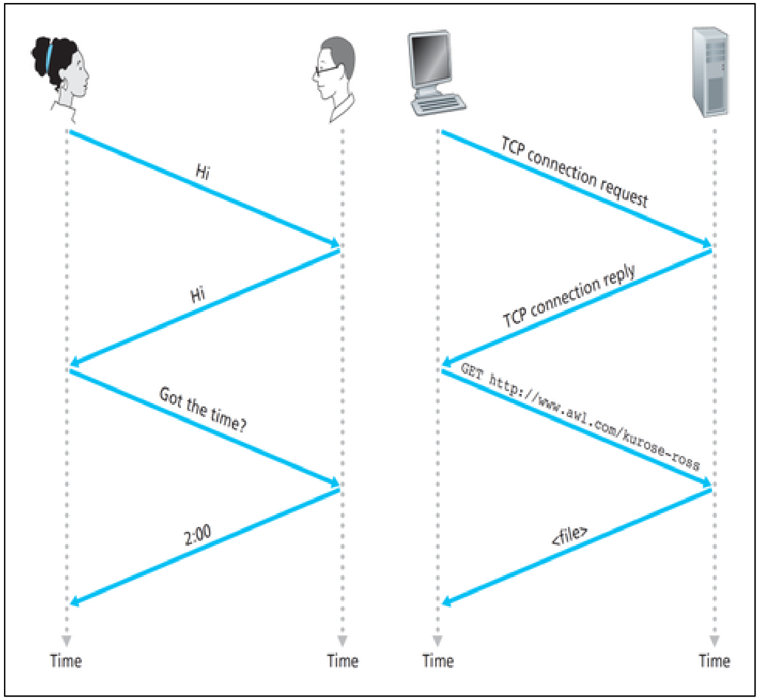
Networks
↓
Computer Networks
↓
internetworks
↓
The Internet

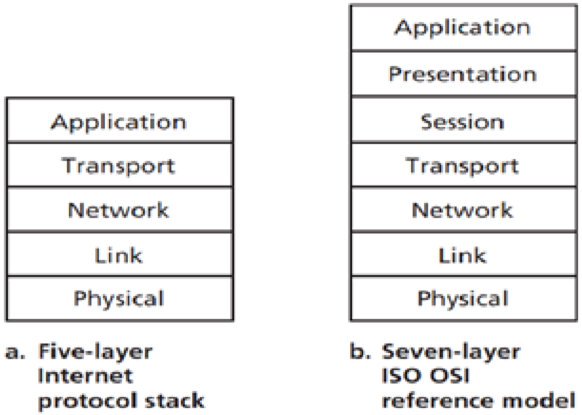
Communication Links

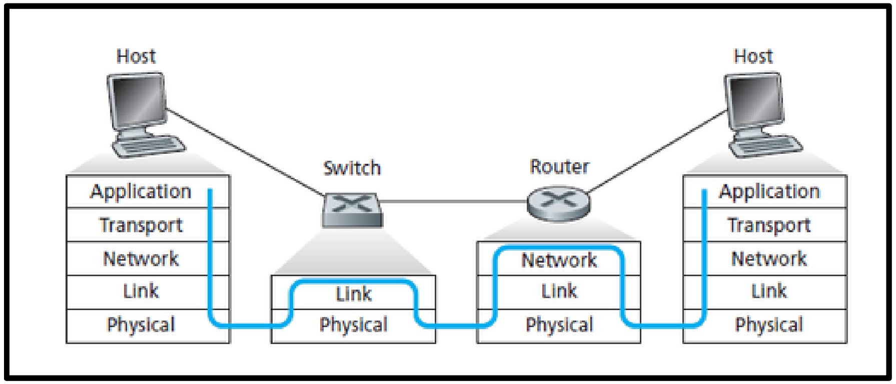
End systems / hosts

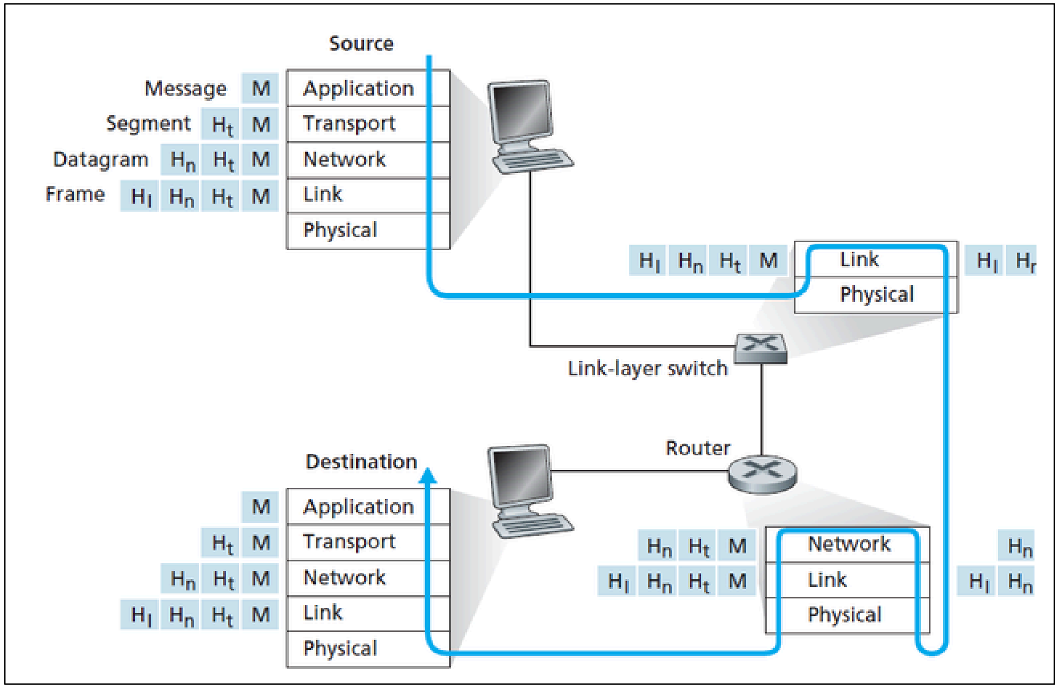
Switches / routers

Protocols and Standards

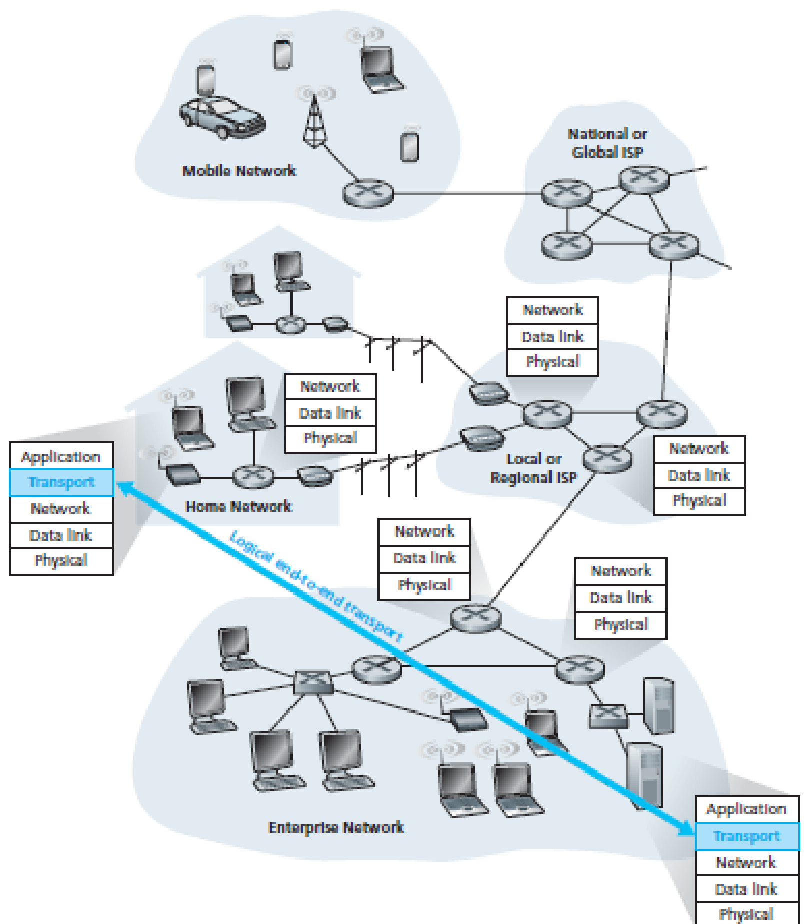




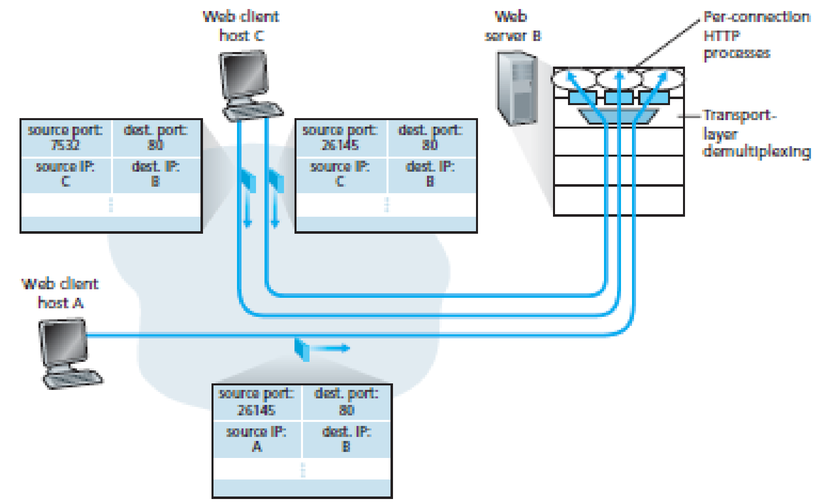


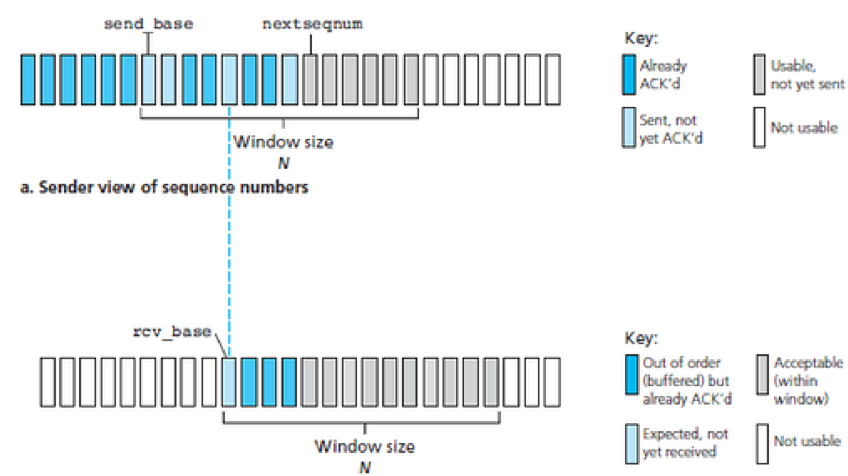


The Transport Layer

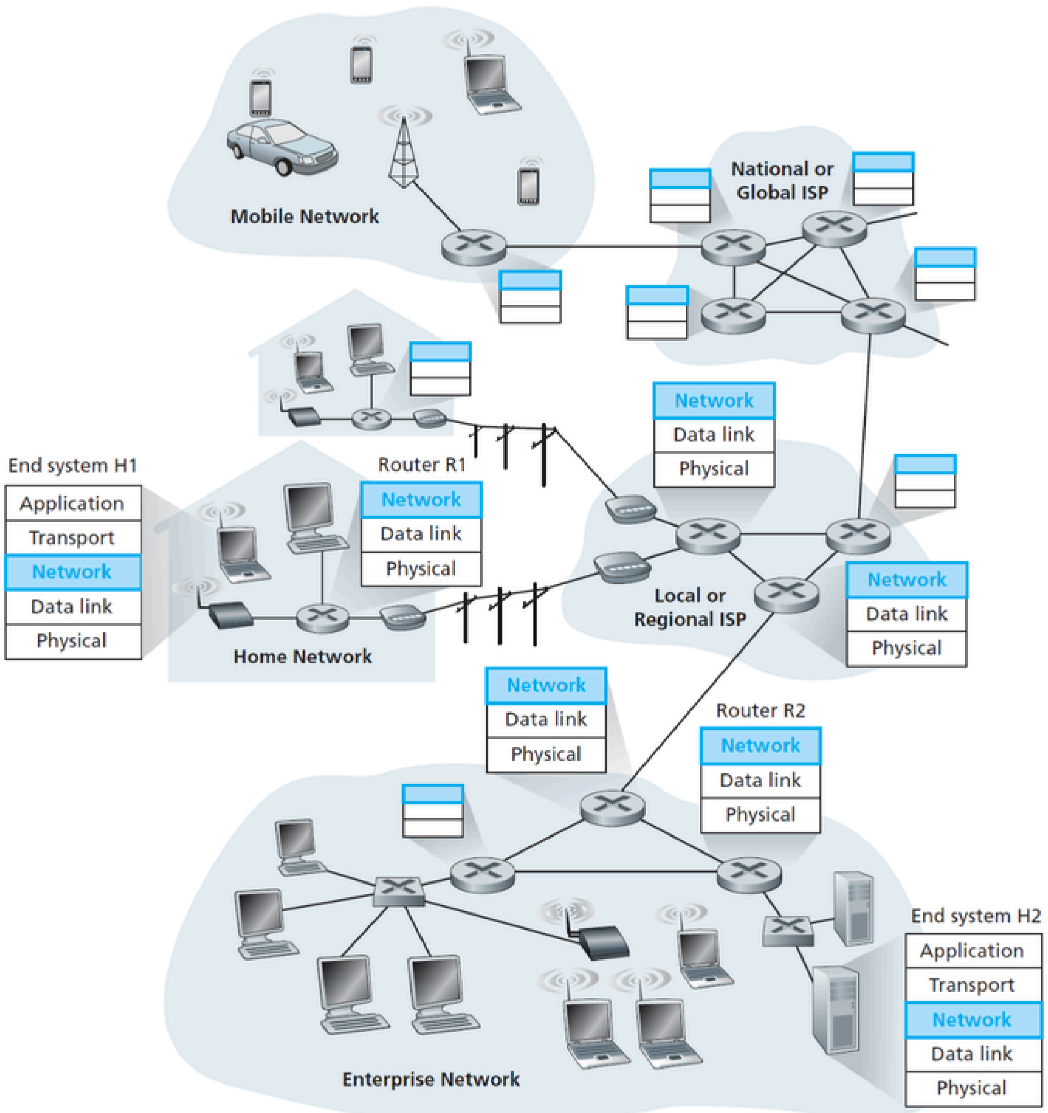


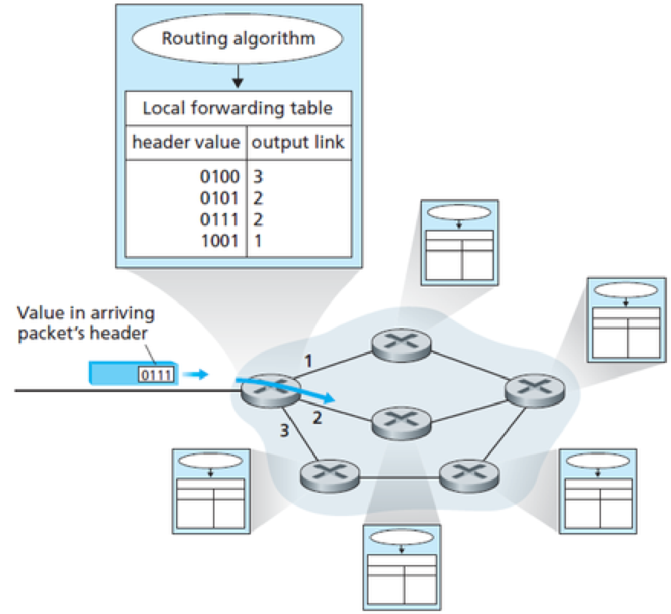
- Multiplexing / de-multiplexing
- Reliability & in-order delivery
- Congestion control
- Flow Control





The Network Layer



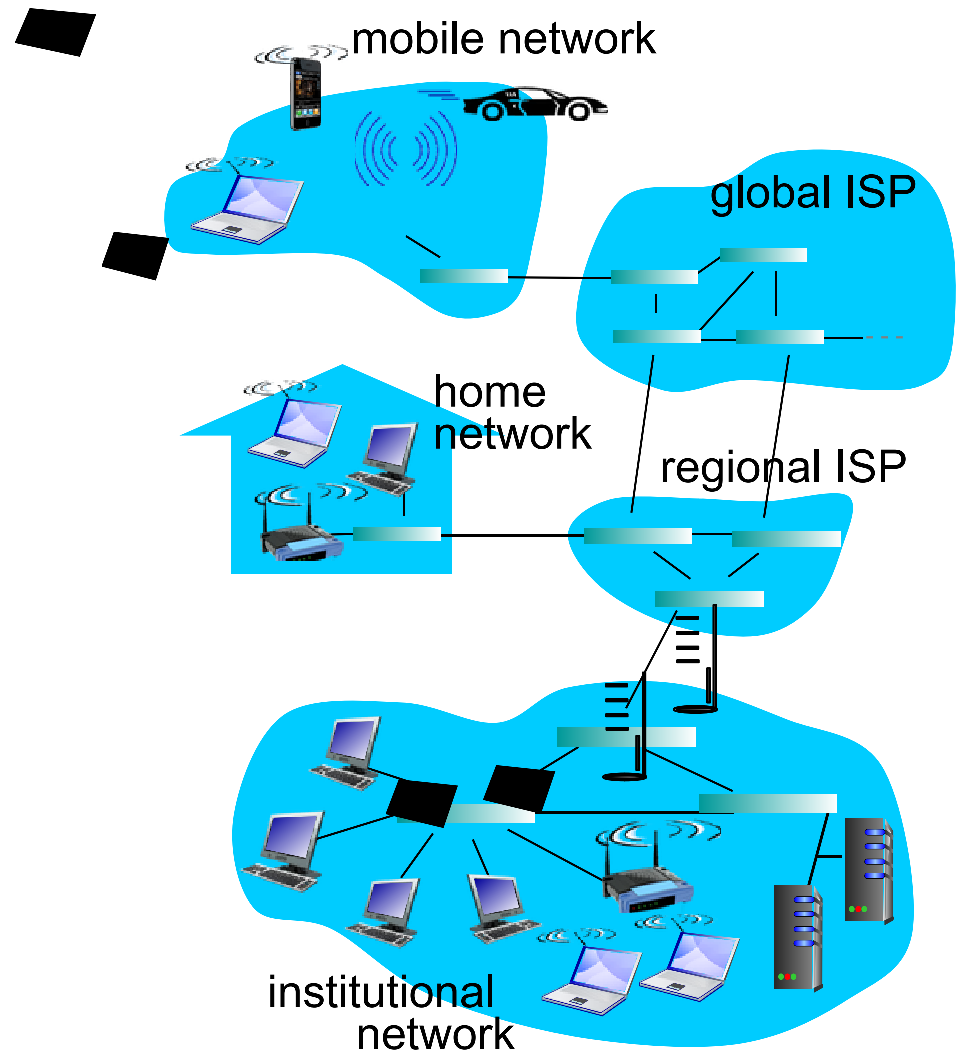


- Routing
  - Routing Metric
    - Hop-count, BW
    - ETX, ETT
- Routing Protocol
  - RIP, OSPF
  - BGP
- Packet Forwarding
- QoS

A closer look at network structure

A closer look at network structure:

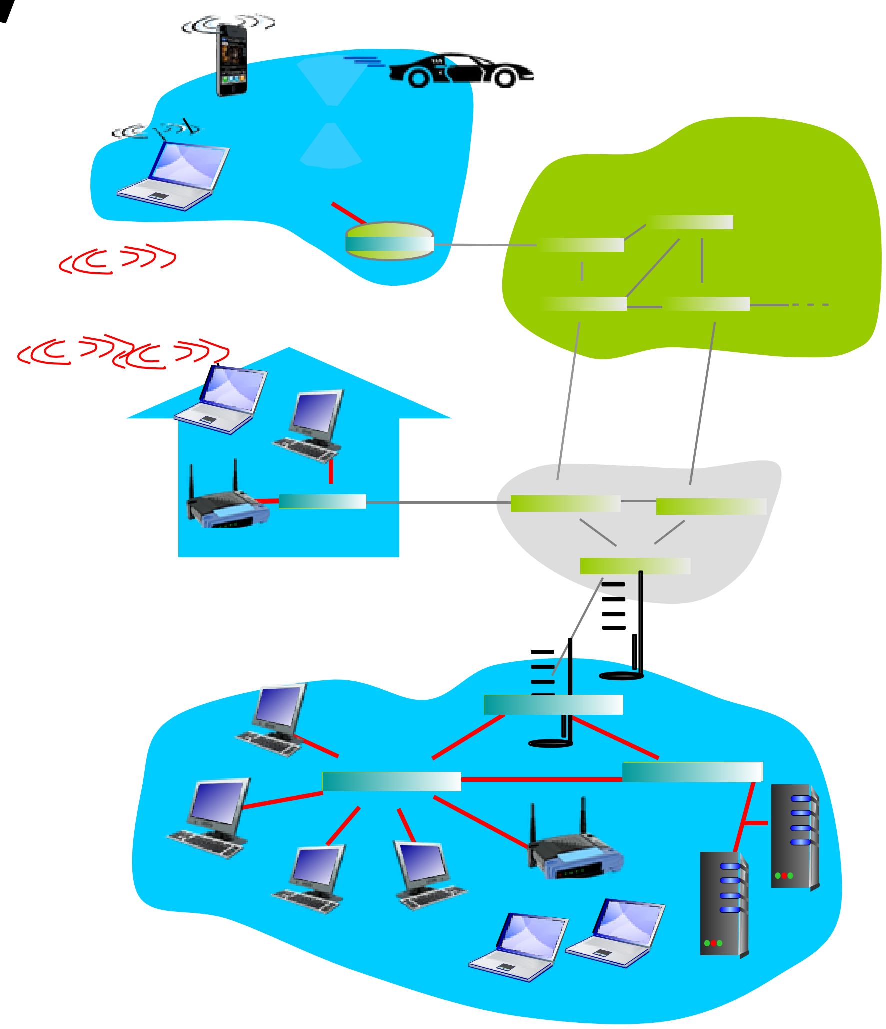
- *network edge:*
  - hosts: clients and servers
  - servers often in data centers
- *physical media:* wired, wireless communication links
- *network core:*
  - interconnected routers
  - network of networks



Access networks and physical media

Q: How to connect end systems to edge router?

- residential access nets
- institutional access networks (school, company)
- mobile access networks



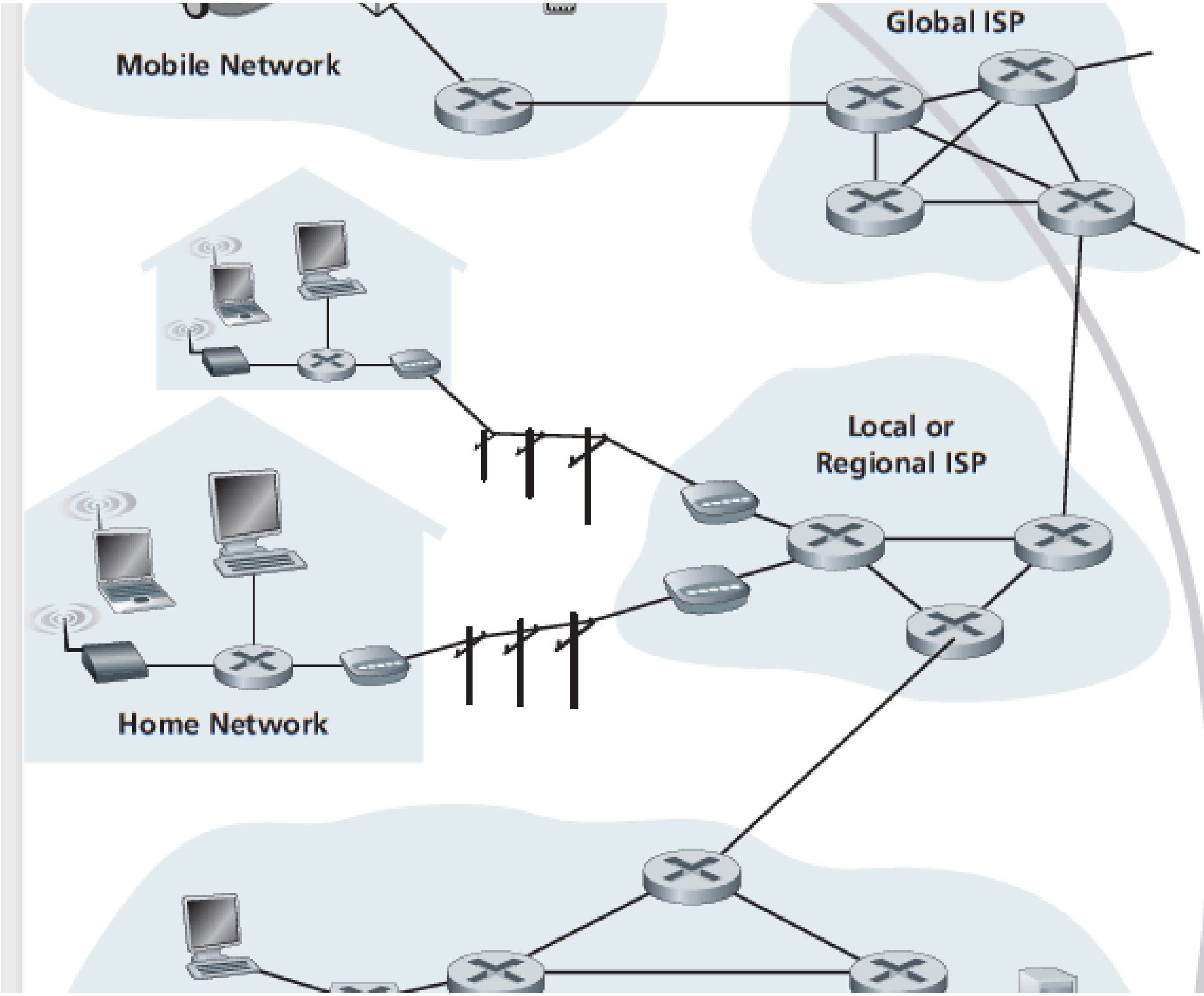
Access Networks

Access Networks

Nuts and Bolts of Internet

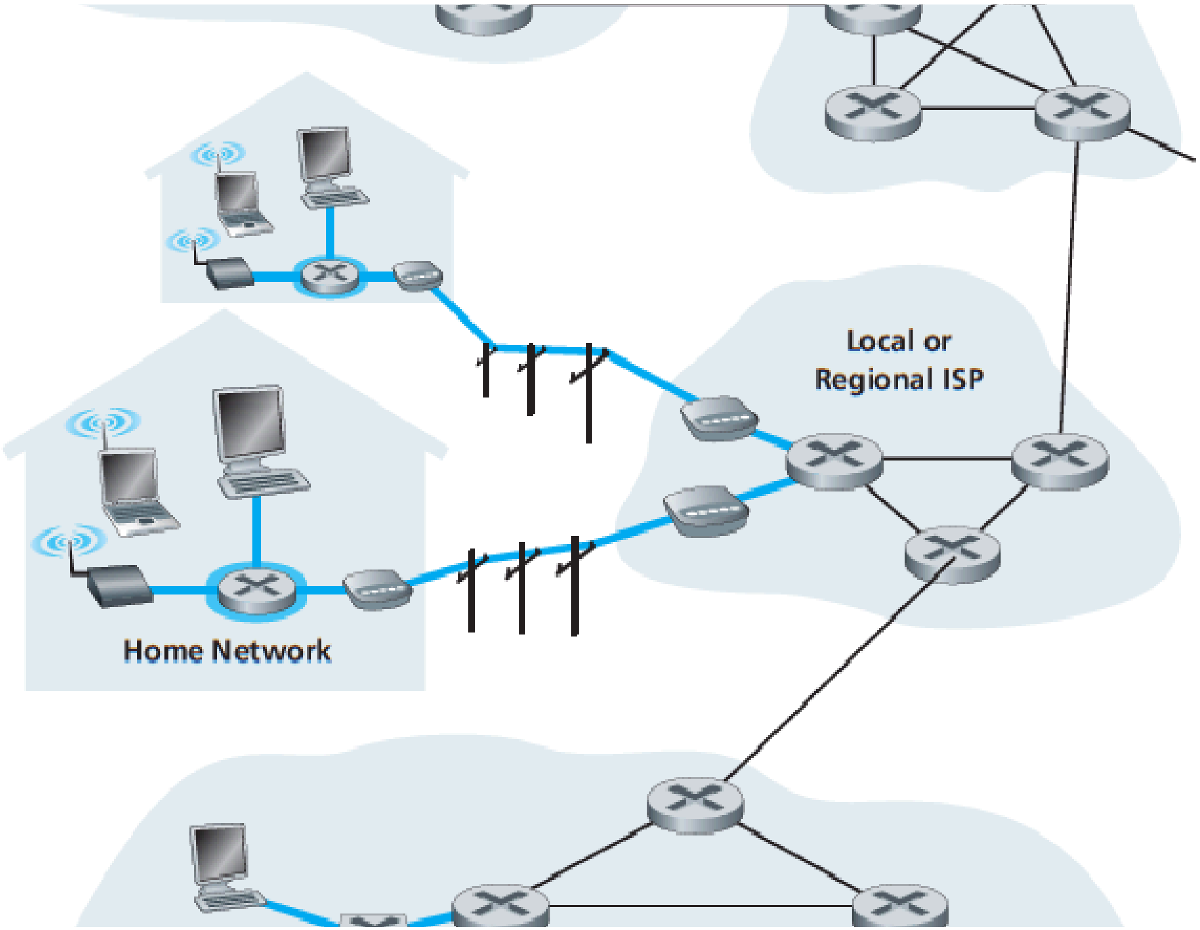
Access Networks

End System Interaction

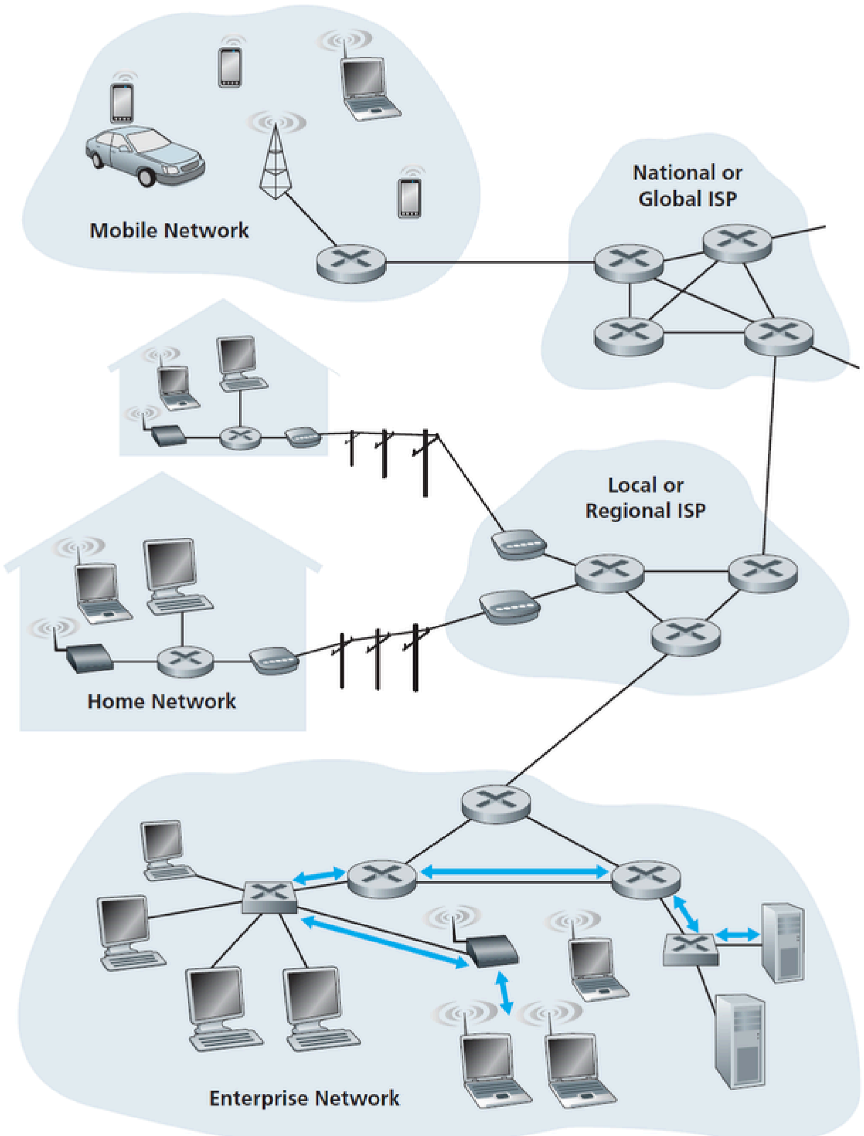


Access Networks

The network that **physically connects an end system to the first router** (also known as the “edge router”) **on a path from the end system to any other distant end system**



Access Networks
Taxonomy based on Technology Used



- Access Networks
  - Ethernet
  - WiFi-Based Access Networks
    - Wireless LAN
      - 802.11a/b/g
    - Ad-hoc Networks
      - Mesh Networks
      - DTN/ICN
      - VANET
      - 6LowPANs
        - WSNs
    - WiMAX-Based Access Net
    - Cellular Networks
      - 3G Cellular Networks and beyond
    - etc…

Access Networks

Taxonomy based on Technology Used

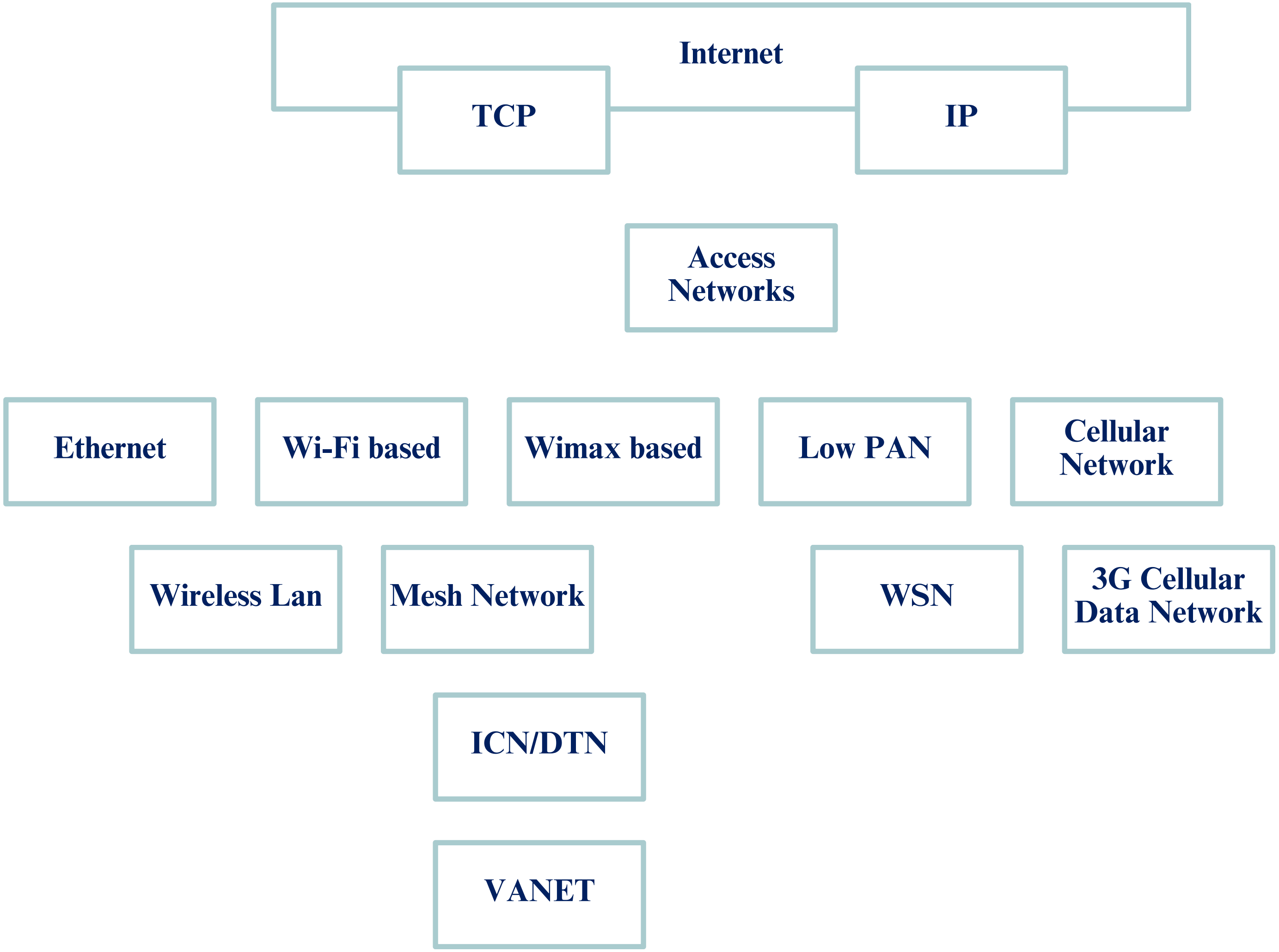
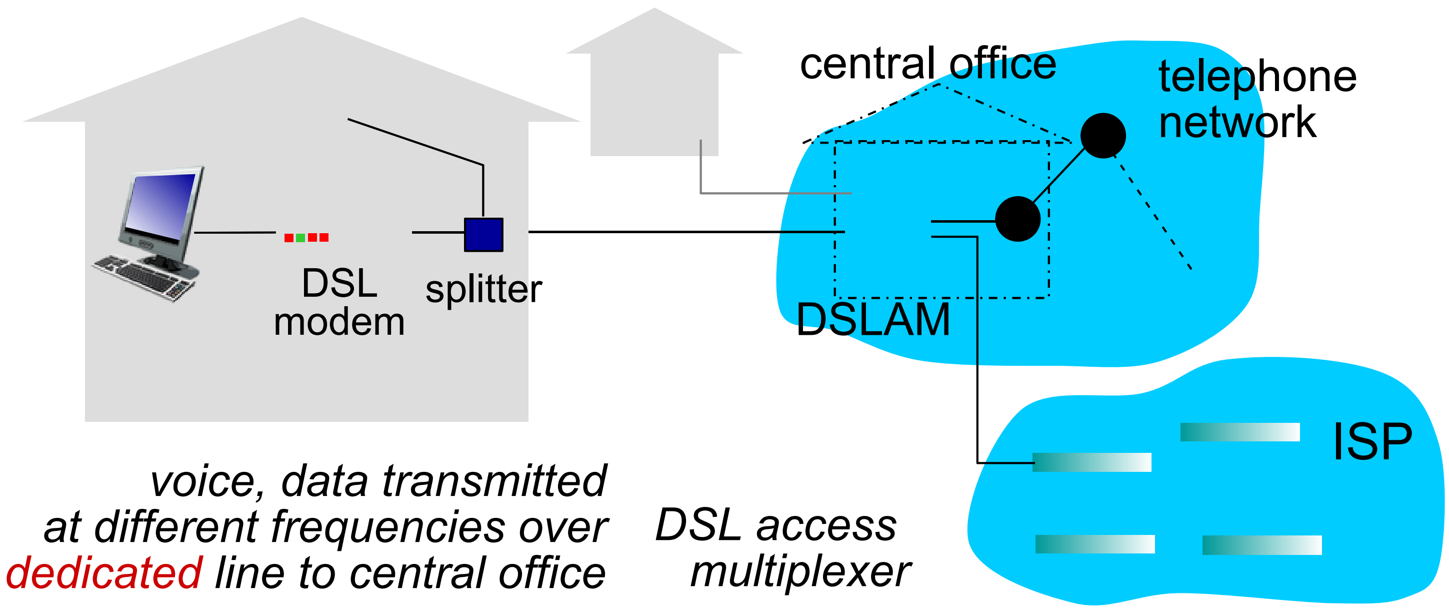


Figure: a taxonomy of access networks (approximate)

Access Networks

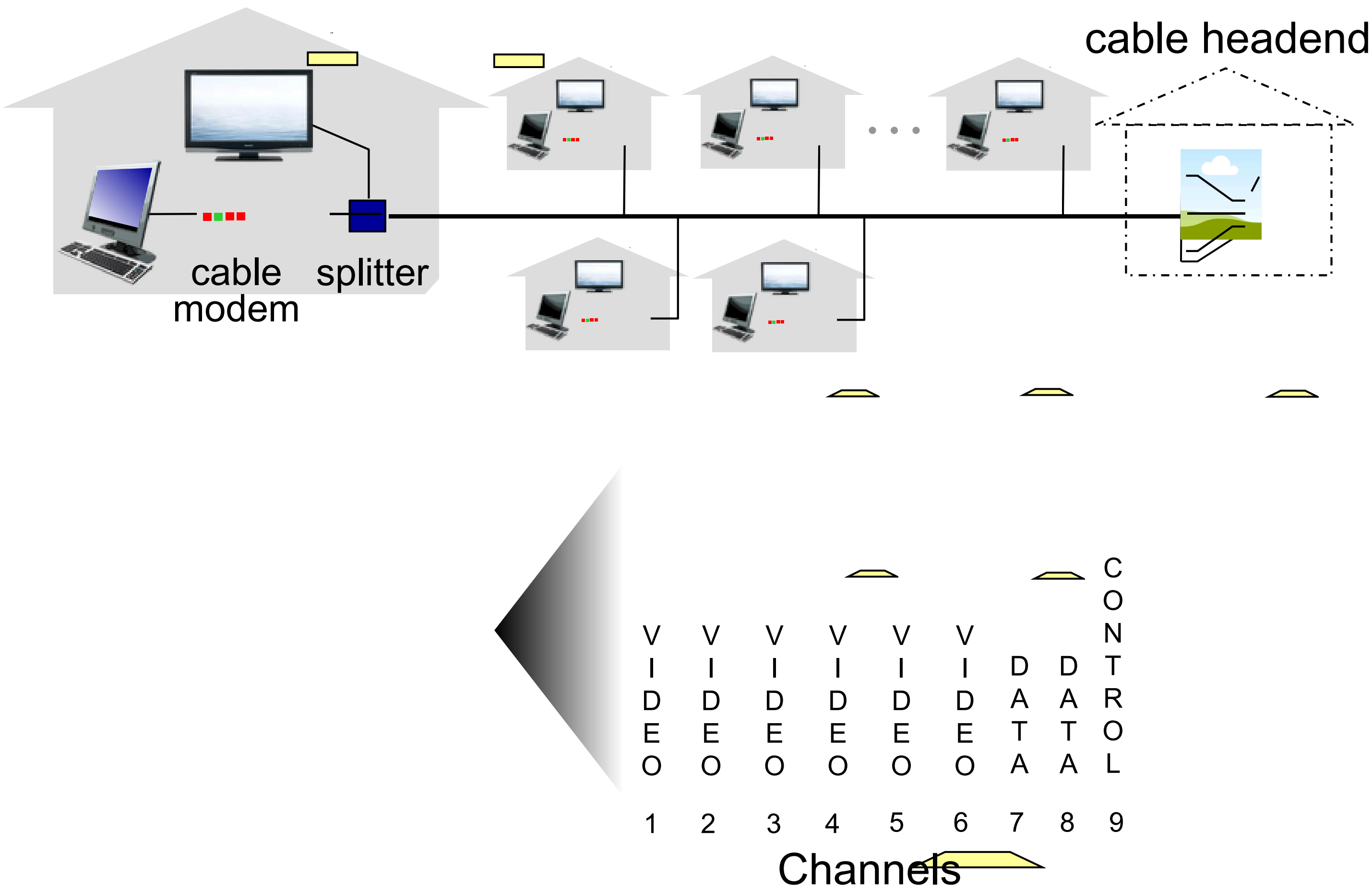
Wired Access Networks

Access net: digital subscriber line (DSL)



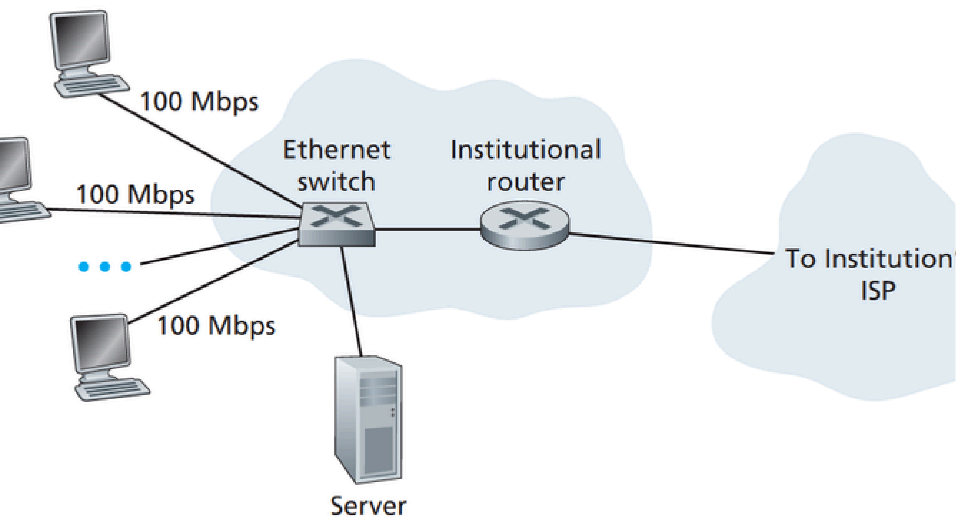
- use *existing* telephone line to central office DSLAM
  - data over DSL phone line goes to Internet
  - voice over DSL phone line goes to telephone net
- < 2.5 Mbps upstream transmission rate (typically < 1 Mbps)
- < 24 Mbps downstream transmission rate (typically < 10 Mbps)

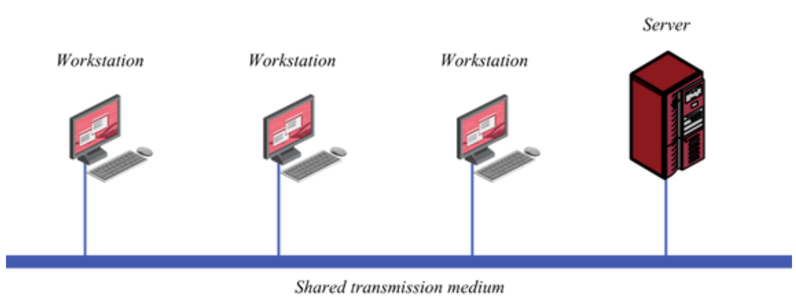
Cable Network

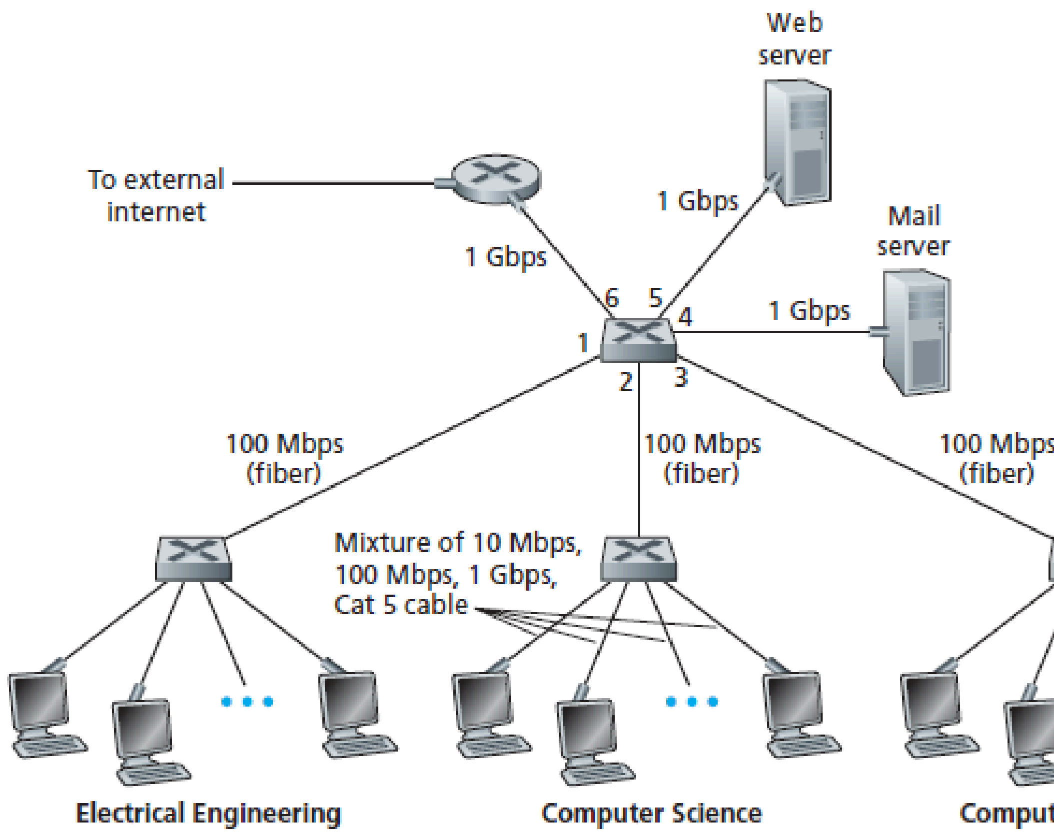


frequency division multiplexing: different channels transmitted in different frequency bands

Ethernet – Traditional Wired Access







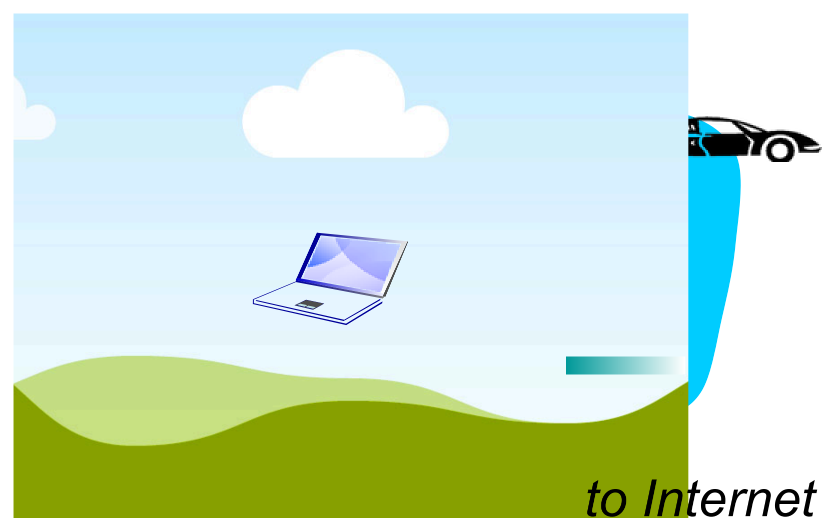
Wireless Access Networks

Wireless Access Networks

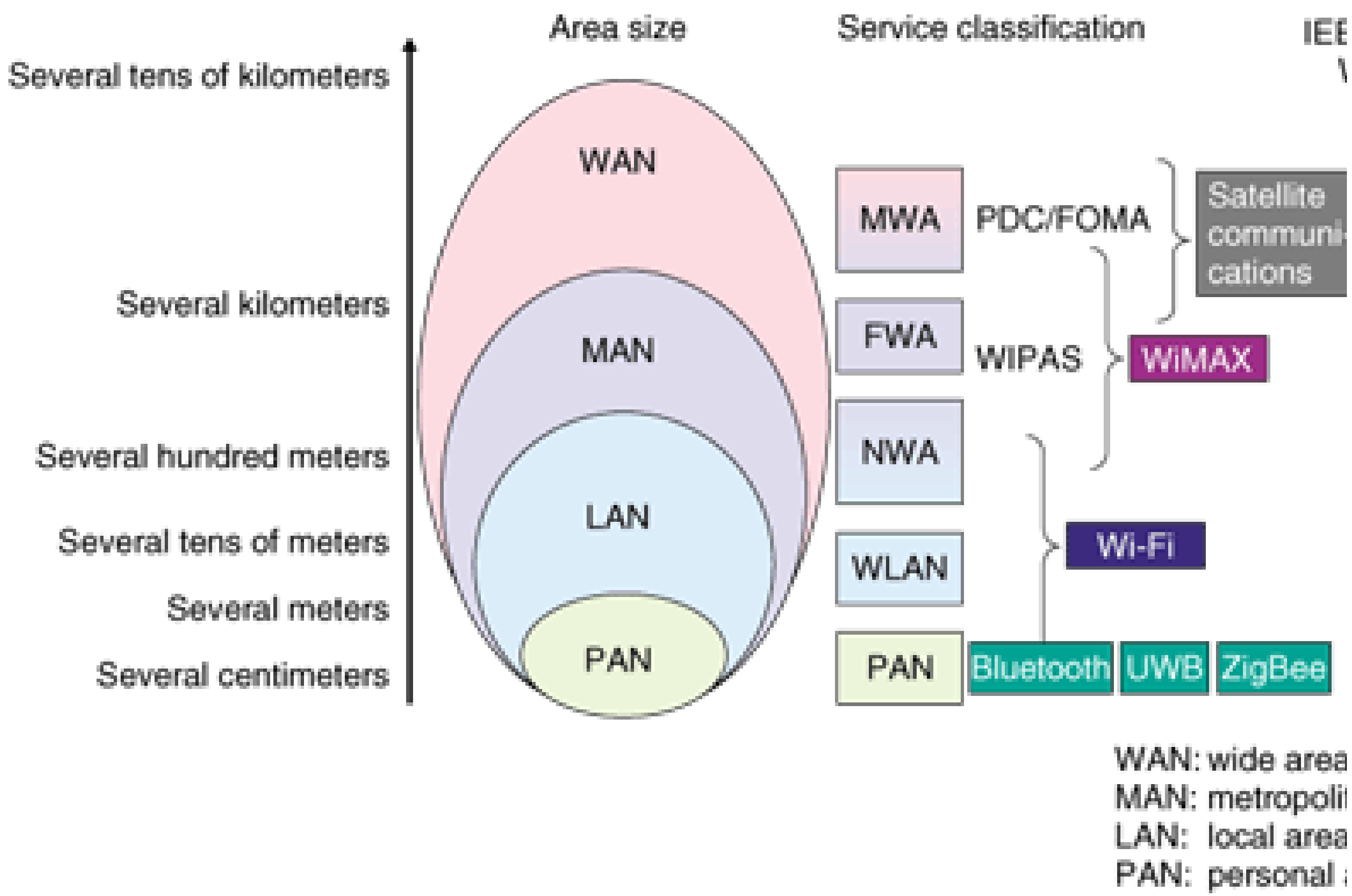
- shared *wireless* access network connects end system to router
  - via base station aka “access point”

wireless LANs:
- within building (100 ft)
- 802.11b/g (WiFi): 11, 54 Mbps transmission rate

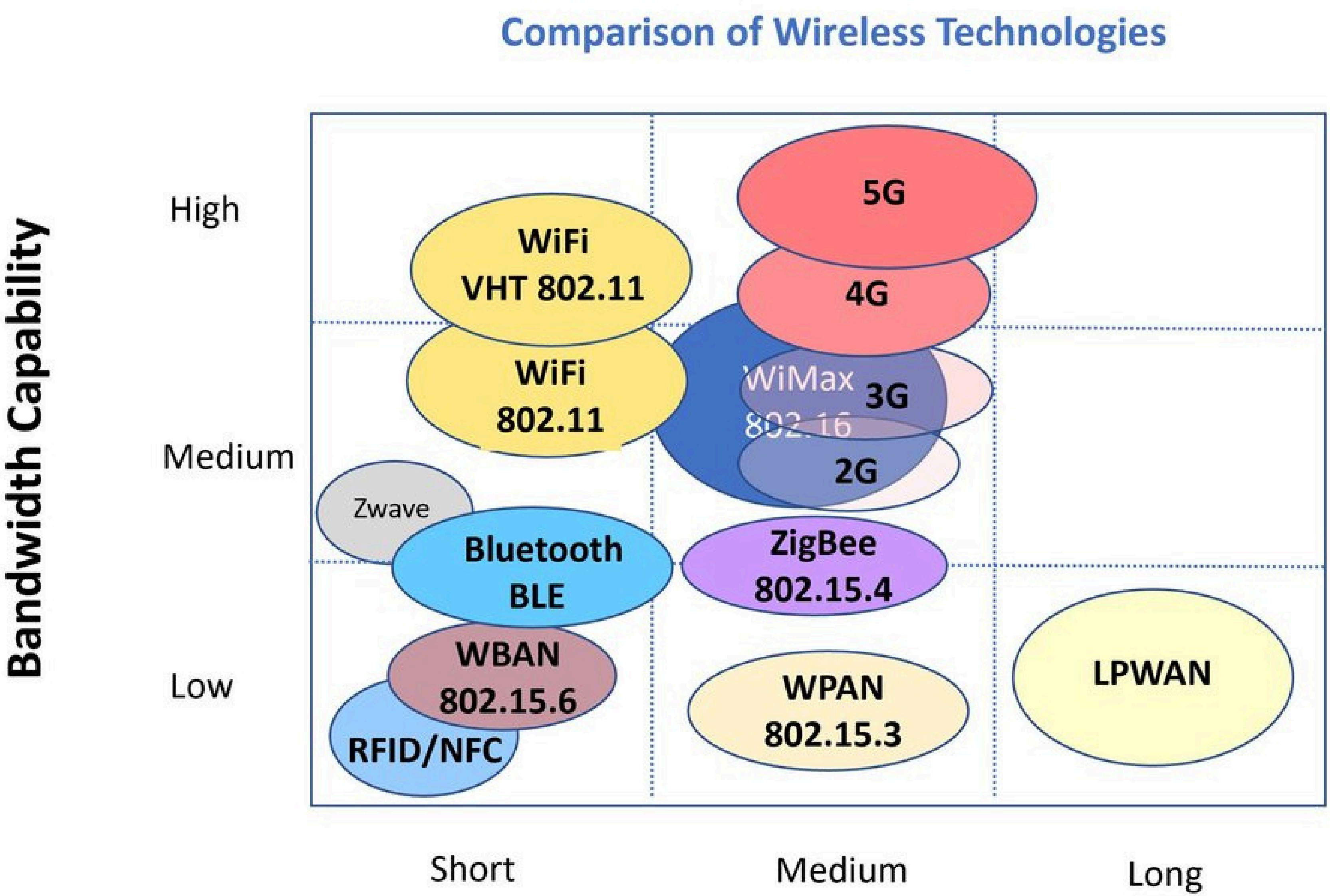
wide-area wireless access
- provided by telco (cellular) operator, 10’s km
- between 1 and 10 Mbps
- 3G, 4G: LTE



Wireless Access Networks



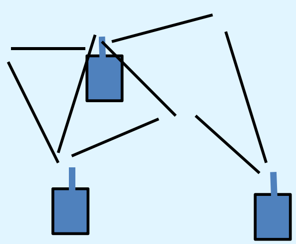
Wireless Access Networks: Range Comparison



Wireless Access Networks

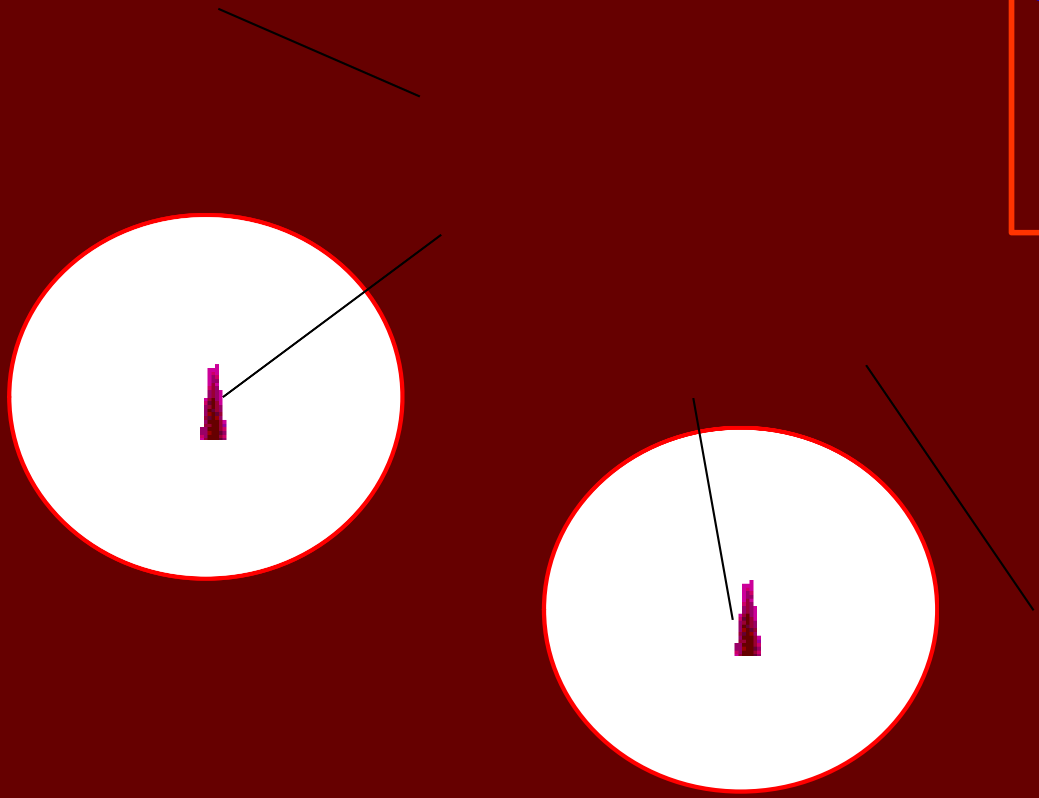
Personal Area Network (PAN)	*Local Area Network (LAN)*	*Metropolitan Area Network (MAN)*
Bluetooth	802.11b 802.11a 802.11g	802.16 802.16a 802.16e
Medium data rates	High data rates	Very high data rates
1 Mbps to 2 Mbps	11 Mbps to 54 Mbps	Quality of service, up tc 280 Mbps
Very short range	Short range	Medium range
3 m (~10 feet)	100 m (~300 feet)	50 km (~31 miles)
Notebook to PC to peripherals devices to systems	Computer to computer and the internet	LAN or computer to high speed wire line internet

Milestones in Wireless Communications

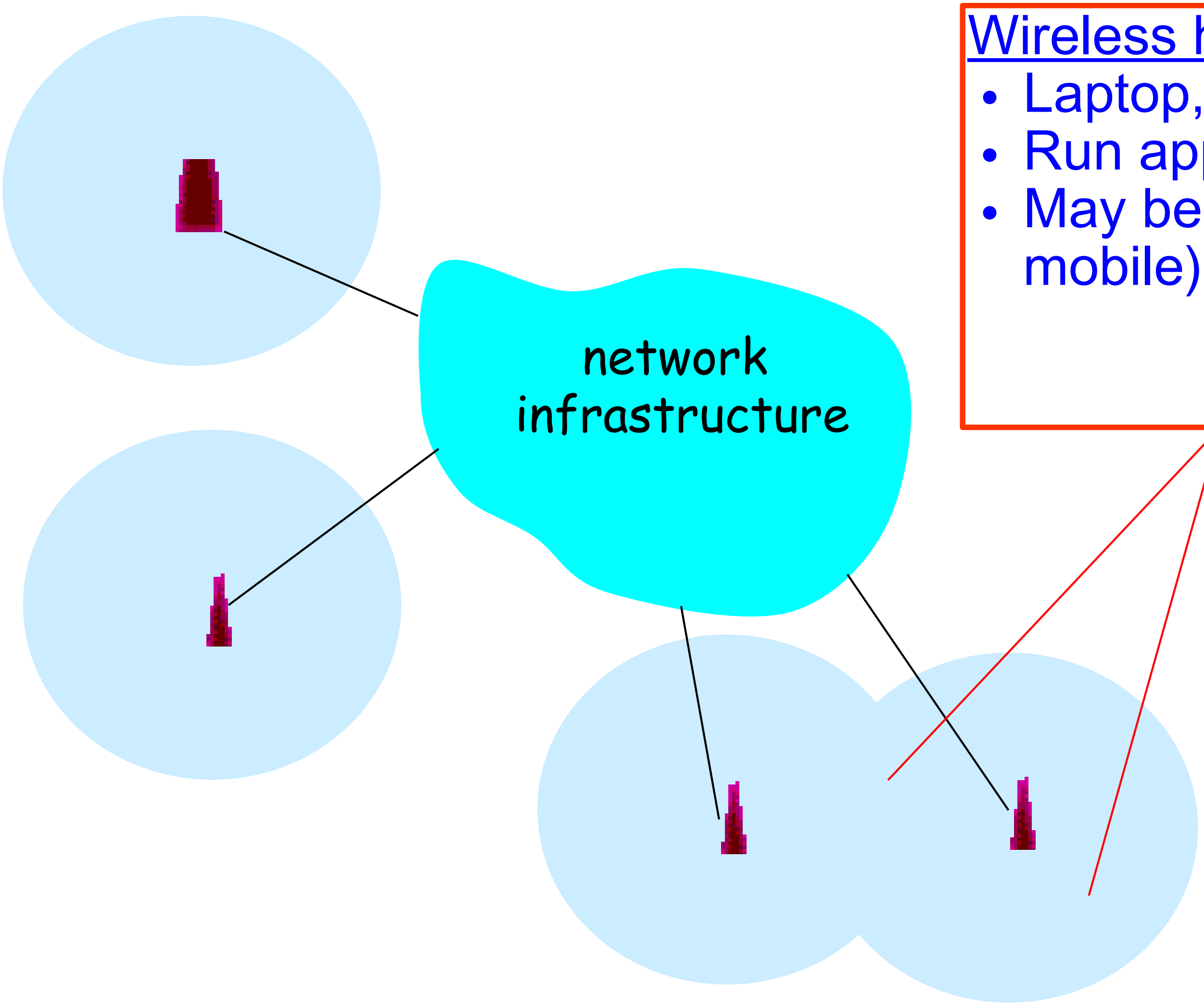


Wireless link

- Typically used to connect mobile(s) to base station
- Also used as backbone link
- Multiple access protocol coordinates link access

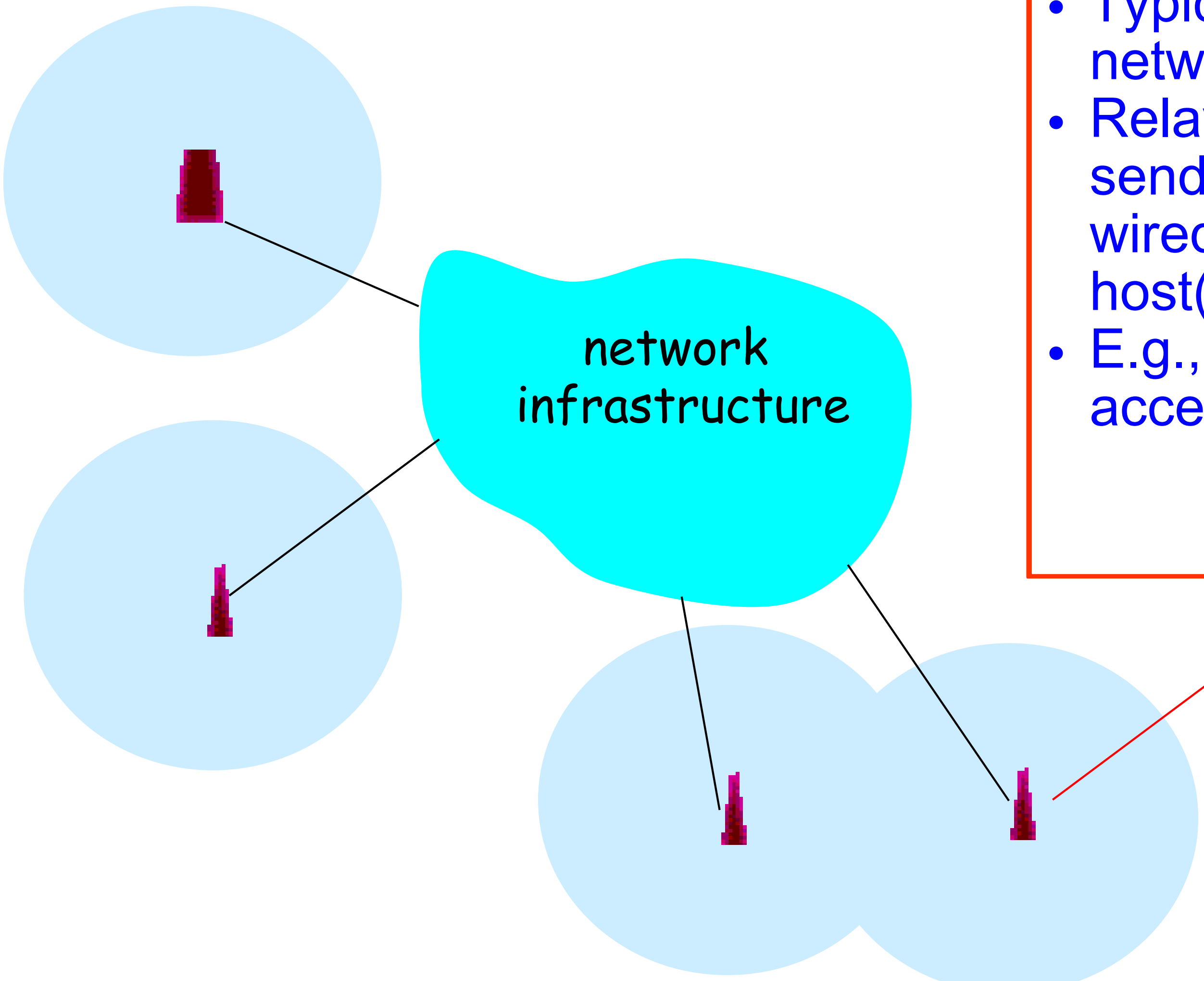


Wireless Network: Wireless Hosts



Wireless host
- Laptop, PDA, IP phone
- Run applications
- May be stationary (non-mobile) or mobile

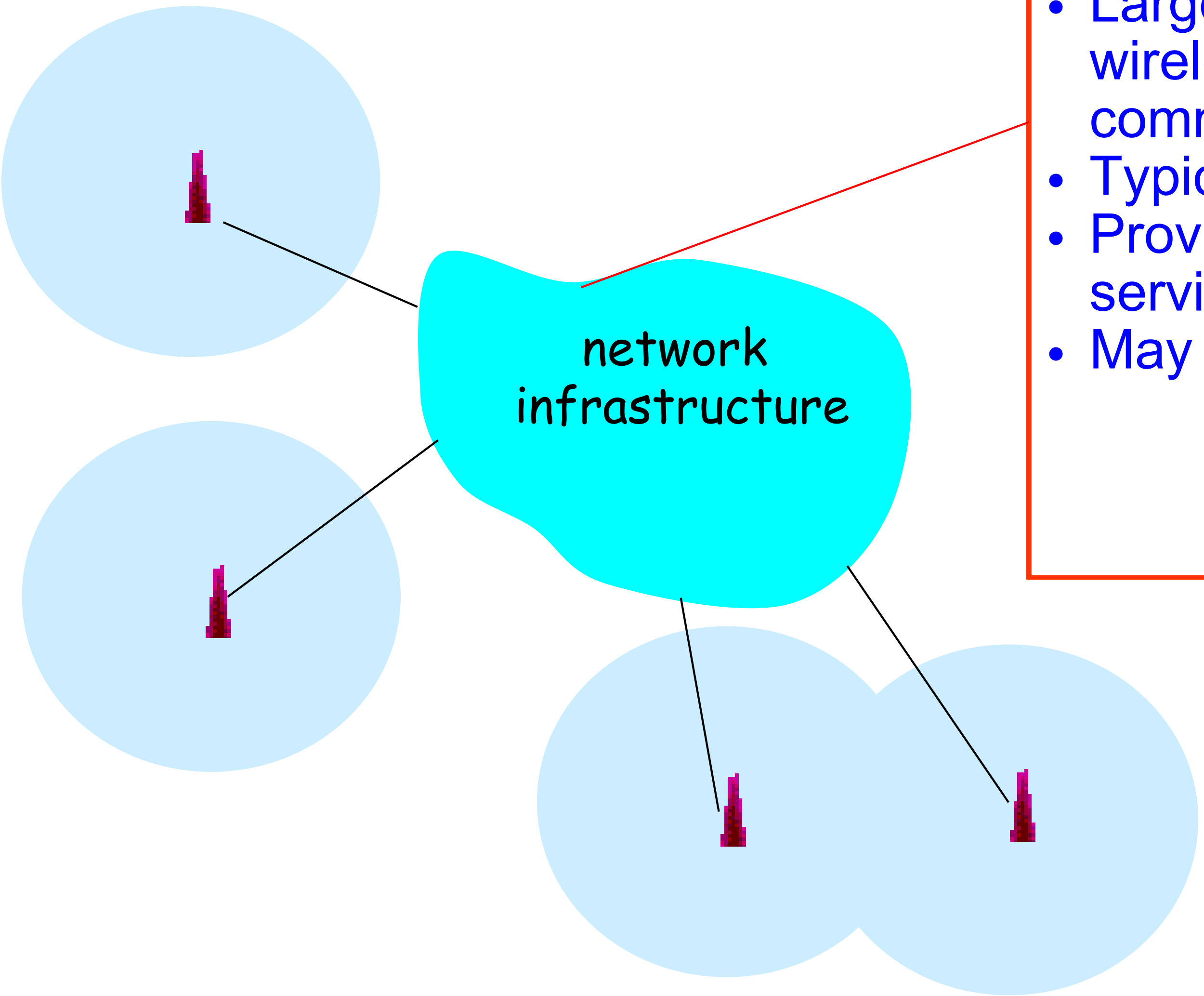
Wireless Network: Base Station



Base station

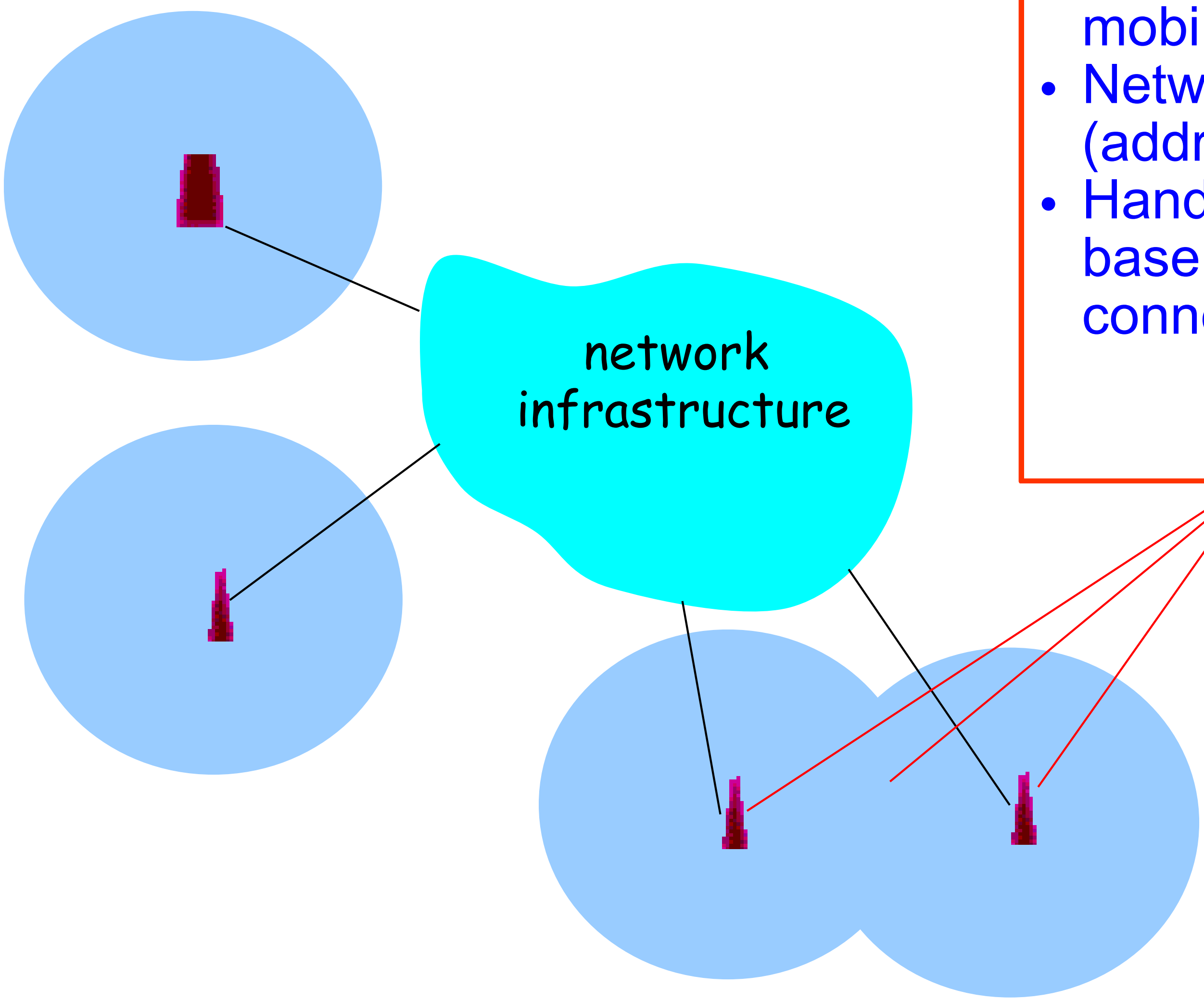
- Typically connected to wired network
- Relay responsible for sending packets between wired network and wireless host(s) in its "area"
- E.g., cell towers, 802.11 access points

Wireless Network: Infrastructure



Network infrastructure
- Larger network with which a wireless host wants to communicate
- Typically a wired network
- Provides traditional network services
- May not always exist

Wireless Access Networks:: Infrastructure Modes
Scenario #1: Infrastructure Mode

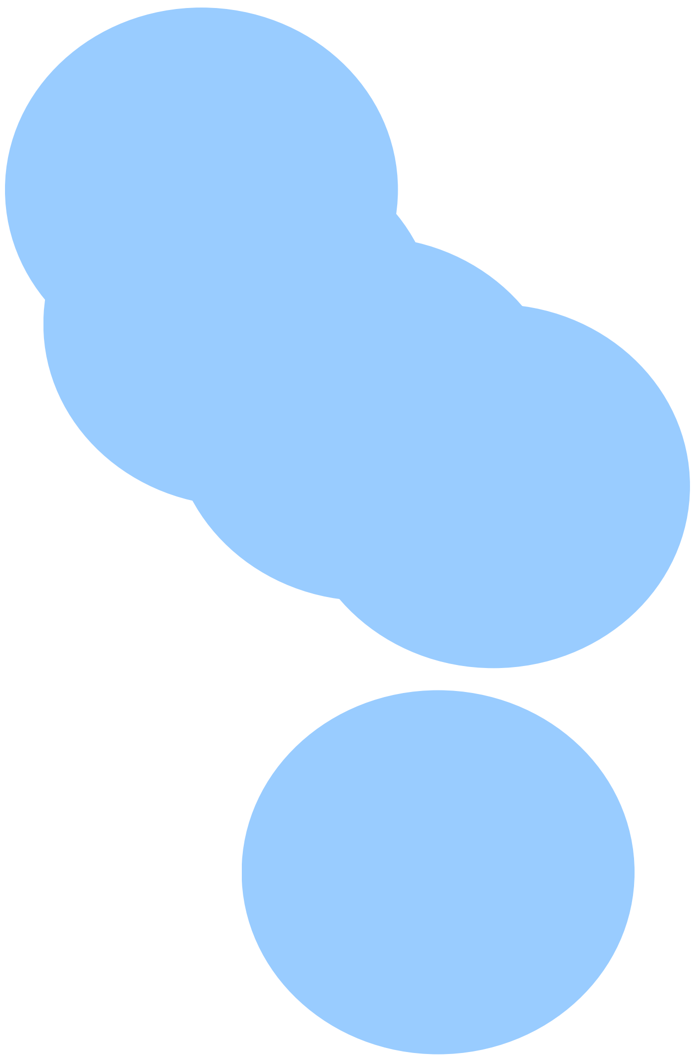


Infrastructure mode
- Base station connects mobiles into wired network
- Network provides services (addressing, routing, DNS)
- Handoff: mobile changes base station providing connection to wired network

Client-server communication

Wireless Access Networks:: Infrastructure Modes
Scenario #2: Infrastructure-less (Ad Hoc) Mode

Peer-to-peer communication



Ad hoc mode
- No base stations
- Nodes can only transmit to other nodes within link coverage
- Nodes self-organize and route among themselves

Wireless Access Networks:: Infrastructure Modes

	Infrastructure-based
Single-hop	• Base station connected to larger wired network ∘ WiFi based wireless LAN ∘ Cellular telephony networks

Wireless Access Networks:: Infrastructure Modes

	Infrastructure-based	Infrastructure-less
Single-hop	• Base station connected to larger wired network ◦ WiFi based wireless LAN ◦ Cellular telephony networks	• One node coordinates the transmissions of the others ◦ Bluetooth ◦ Ad-hoc 802.11 (e.g., P2P, WiFi Direct)

Wireless Access Networks:: Infrastructure Modes

	Infrastructure-based	Infrastructure-less
Single-hop	• Base station connected to larger wired network ◦ WiFi based wireless LAN ◦ Cellular telephony networks ◦ WiMAX	• One node coordinates the transmissions of the others ◦ Bluetooth ◦ Ad-hoc 802.11 (e.g., P2P, WiFi Direct)
Multi-hop	• Base station exists • Some nodes must relay through other nodes ◦ Wireless Mesh Networks (semi-structured)	

Wireless Access Networks:: Infrastructure Modes

	Infrastructure-based	Infrastructure-less
Single-hop	• Base station connected to larger wired network ∘ WiFi based wireless LAN ∘ Cellular telephony networks	• One node coordinates the transmissions of the others ∘ Bluetooth ∘ Ad-hoc 802.11 (e.g., P2P, WiFi Direct)
Multi-hop	• Base station exists • Some nodes must relay through other nodes ∘ WiMAX ∘ Wireless Mesh Networks(semi-structured)	• No base station exists • Some nodes must relay through others ∘ Ad hoc wireless Networks ∘ Wireless Sensor Networks ∘ Mobile Ad-hoc Networks /Vehicular Ad-hoc Networks ∘ Delay Tolerant Networks

Wireless Access Networks

Single-hop Infrastructure-based Wireless Systems

1. Cellular Networks
2. Wireless LANs
3. WiMAX

Single-hop Infrastructure-less Wireless Systems

4. Bluetooth

Multi-hop Infrastructure-based Wireless Systems

5. Wireless Mesh Networks (semi-structured)

Multi-hop Infrastructure-less Wireless Systems

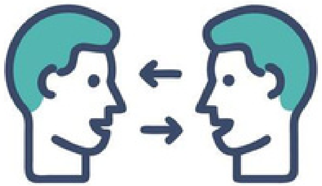
6. Ad hoc wireless Networks

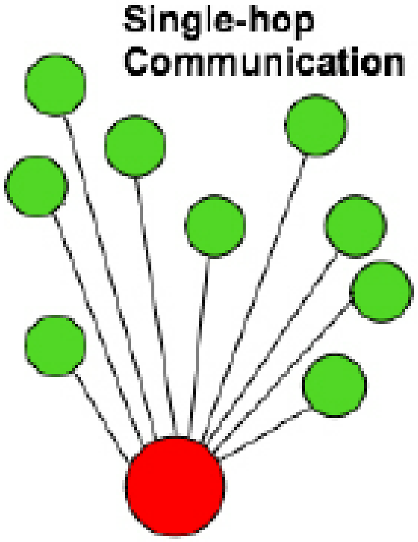
6.1. Wireless Sensor Networks

6.2. Vehicular Ad-hoc Networks

6.3. Delay Tolerant Networks

Single-hop Infrastructure-base Wireless Access Networks





1. Cellular Networks
2. Wireless LANs
3. WiMAX

Single-hop Infrastructure-based:: Cellular Networks

- **3G cellular data network connects radio access networks to the public Internet.**

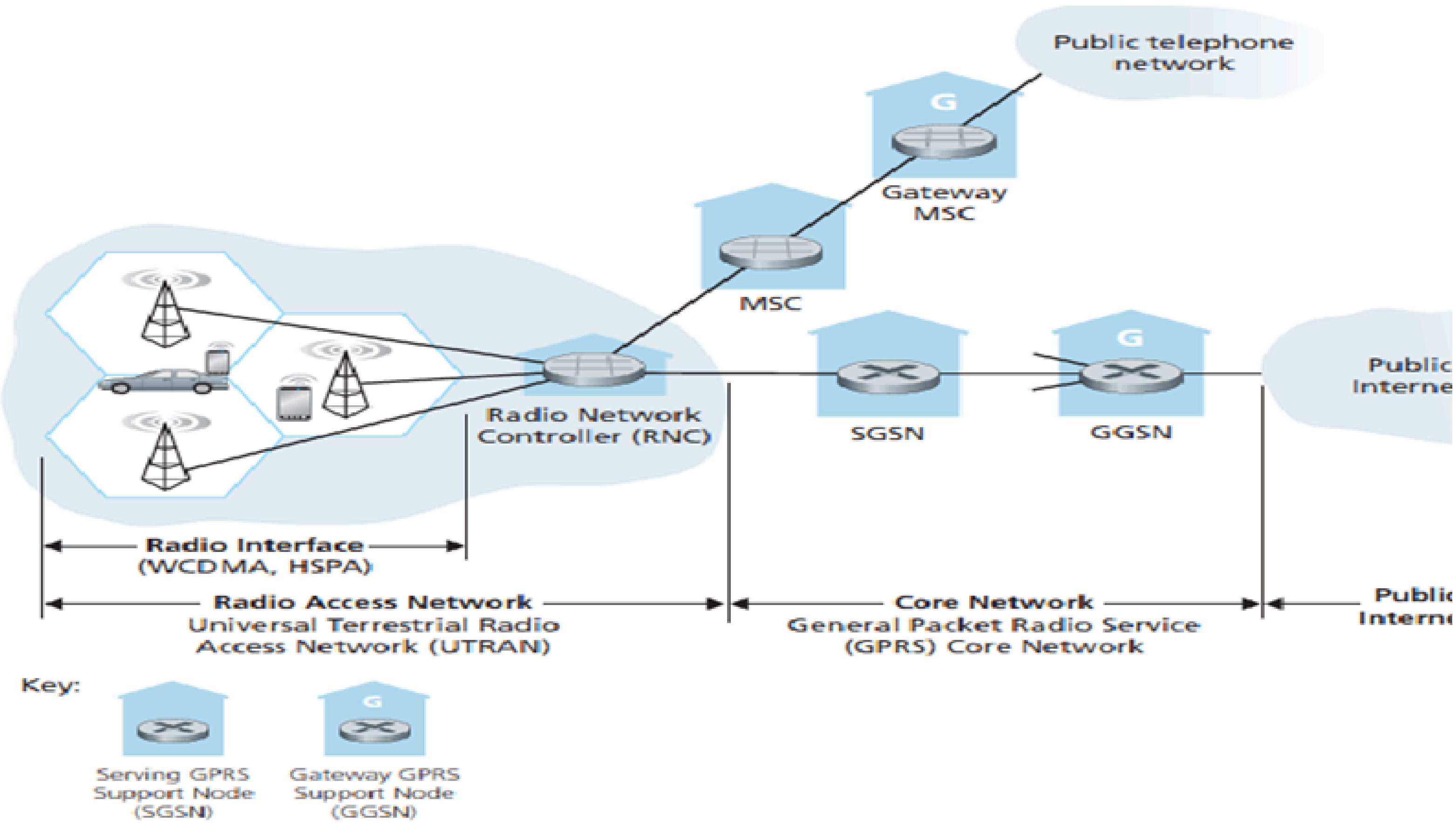
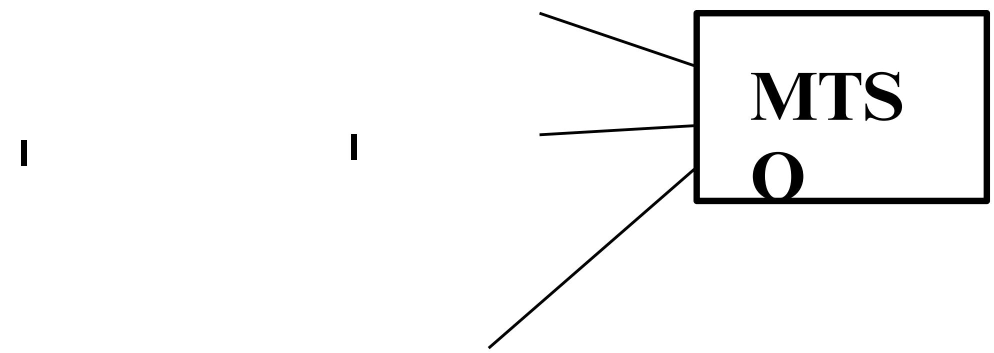


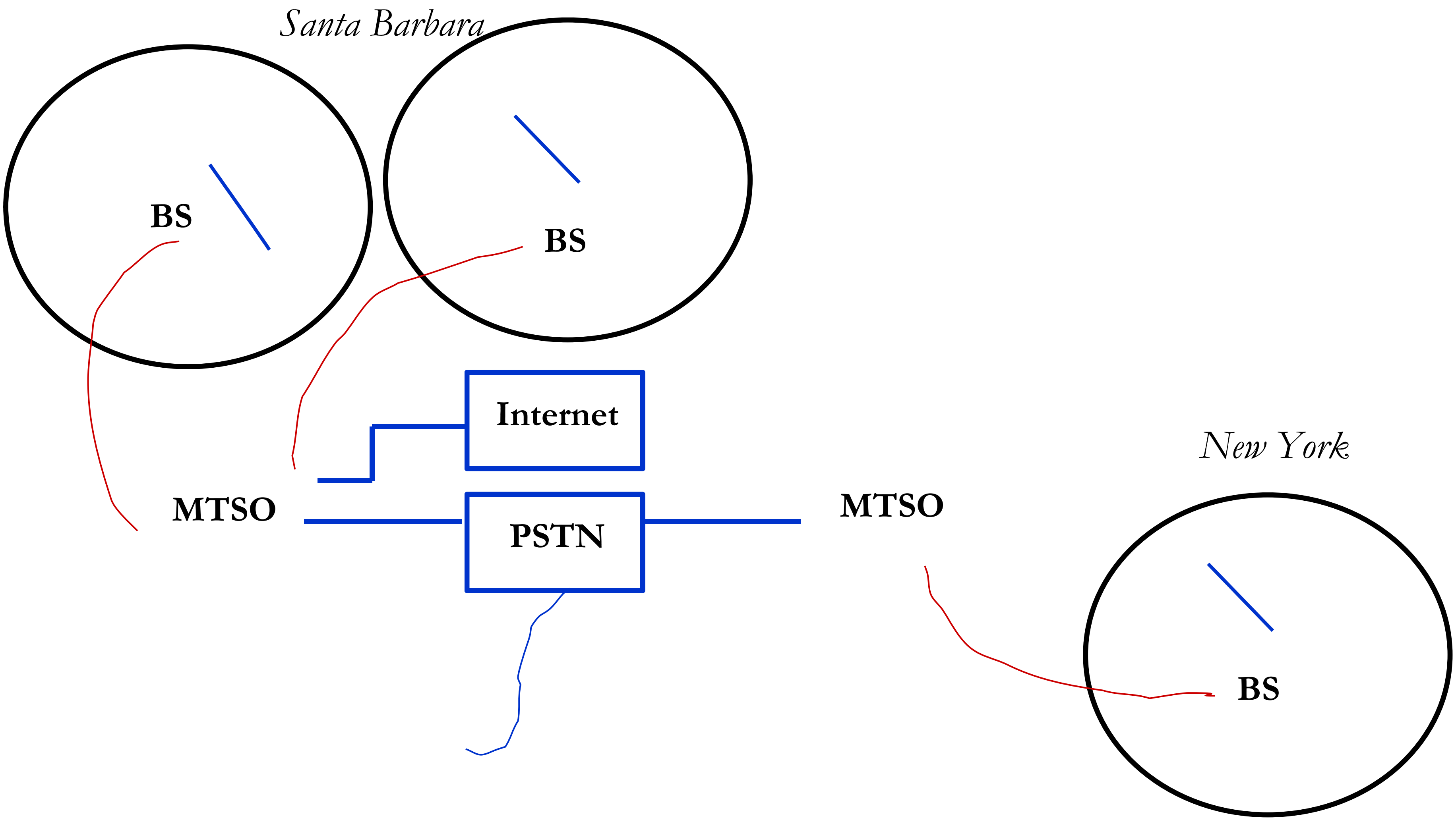
Figure: 3G cellular data network architecture

Cellular Networks:
Reuse channels to maximize capacity

- Geographic region divided into cells
- Frequencies/timeslots/codes reused at spatially-separated locations.
- Co-channel interference between same color cells.
- Base stations/MTSOs coordinate handoff and control functions
- Shrinking cell size increases capacity, as well as networking burden



Cellular Networks

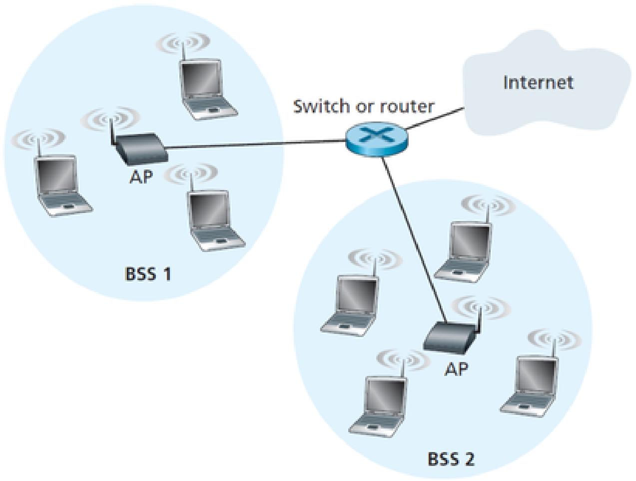


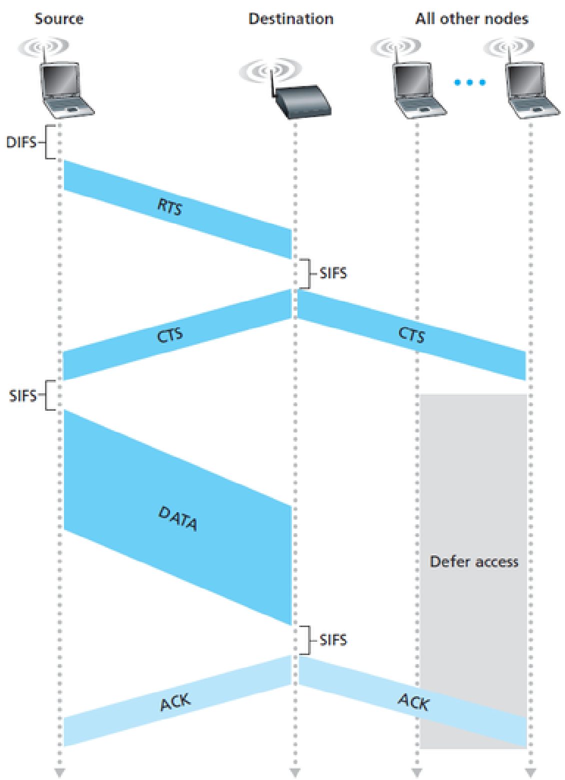
Cellular Design:
Voice and Data

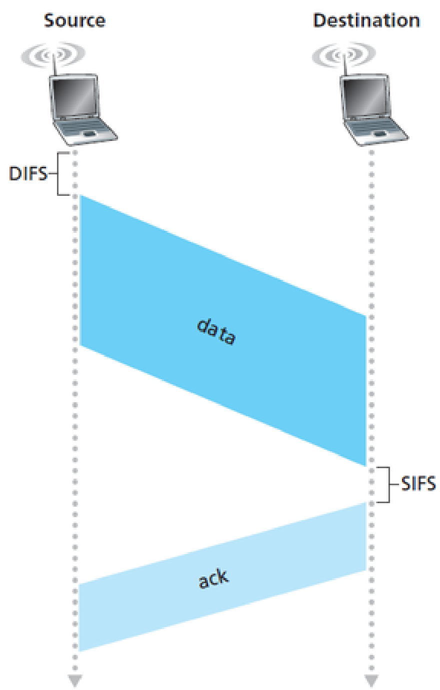
- Data is bursty, whereas voice is continuous
  - Typically require different access and routing strategies
- 3G "widens the data pipe":
  - 384 Kbps.
  - Standard based on wideband CDMA
  - Packet-based switching for both voice and data
- Evolution of wireless networks:
  - GSM
  - IS-95(CDMA)
  - CDMA 2000 / W-CDMA // IEEE 802.11a/g/b

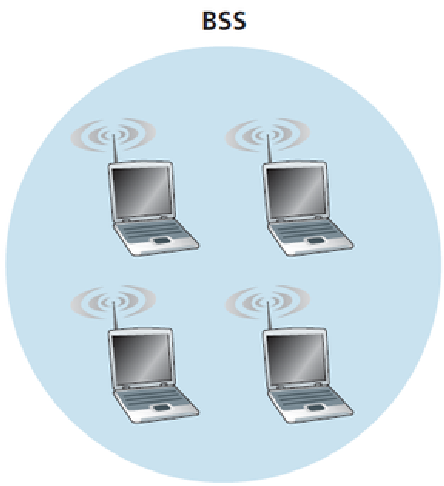
Single-hop Infrastructure-based:: Wireless LANs

Standard	Frequency Range	Data Rate
802.11b	2.4–2.485 GHz	up to 11 Mbps
802.11a	5.1–5.8 GHz	up to 54 Mbps
802.11g	2.4–2.485 GHz	up to 54 Mbps









Single-hop Infrastructure-based:: Wireless LANs

- WLANs connect “local” computers (100m range)
- Breaks data into packets
- Channel access is shared (random access)
- Backbone Internet provides best-effort service
    - Poor performance in some apps (e.g. video)

Single-hop Infrastructure-based:: Wireless LAN Standards

- 802.11b
  - Standard for 2.4GHz ISM band (80 MHz)
  - Frequency hopped spread spectrum
  - 1.6-10 Mbps, 500 ft range
- 802.11a
  - Standard for 5GHz NII band (300 MHz)
  - OFDM with time division
  - 20-70 Mbps, variable range
  - Similar to HiperLAN in Europe
- 802.11g
  - Standard in 2.4 GHz and 5 GHz bands
  - OFDM
  - Speeds up to 54 Mbps

Single-hop Infrastructure-based:: Wireless LAN Standards

Data rate :

- IEEE 802.11b: 11 Mbps
- IEEE 802.11g: 54 Mbps
- IEEE 802.11n (Wi-Fi 4): Up to 600 Mbps
- IEEE 802.11ac (Wi-Fi 5): Up to 6.9 Gbps
- IEEE 802.11ax (Wi-Fi 6/6E): Up to 9.6 Gbps
- IEEE 802.11be (Wi-Fi 7) (emerging): Targeting >30 Gbps

Multi-hop Infrastructure-based:: WiMAX

(Worldwide Interoperability for Microwave Access)

- A wireless communications standard - designed to provide 30 to 40 megabit-per-second data rates, with the 2011 update providing up to 1 Gbit/s for fixed stations.
- It refers to interoperable implementations of the IEEE 802.16 family of wireless-networks standards.

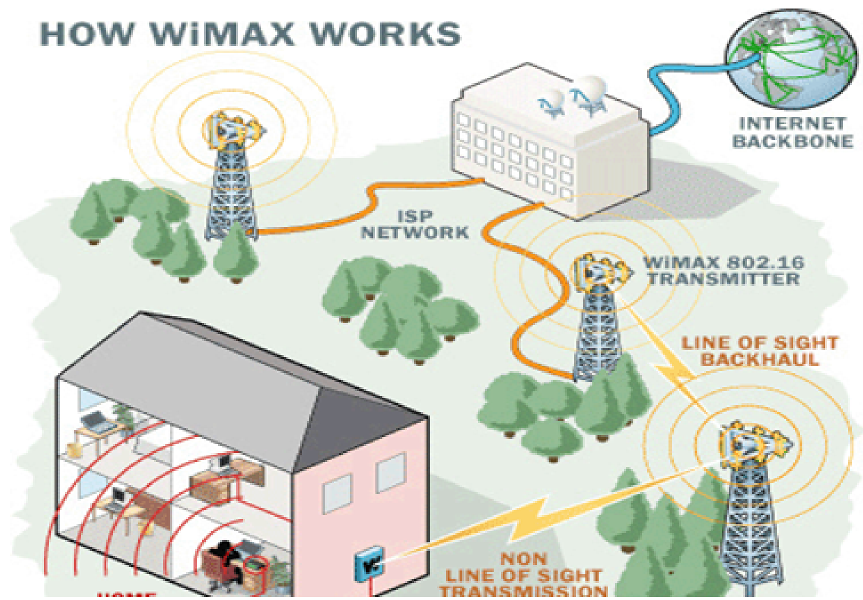


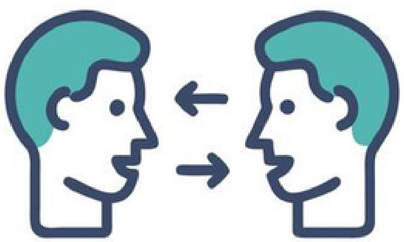
Figure: Working Procedure of WiMAX

Multi-hop Infrastructure-based:: WiMAX

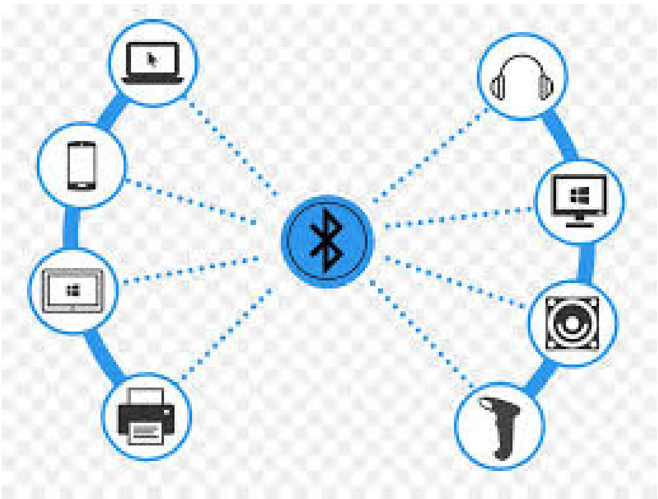
- Provide portable mobile broadband connectivity across cities and countries
- Wireless alternative to cable and digital subscriber line (DSL) for "last mile" broadband access
- Provide data, telecommunications (VoIP) and IPTV services (triple play)
- A source of Internet connectivity as part of a business continuity plan

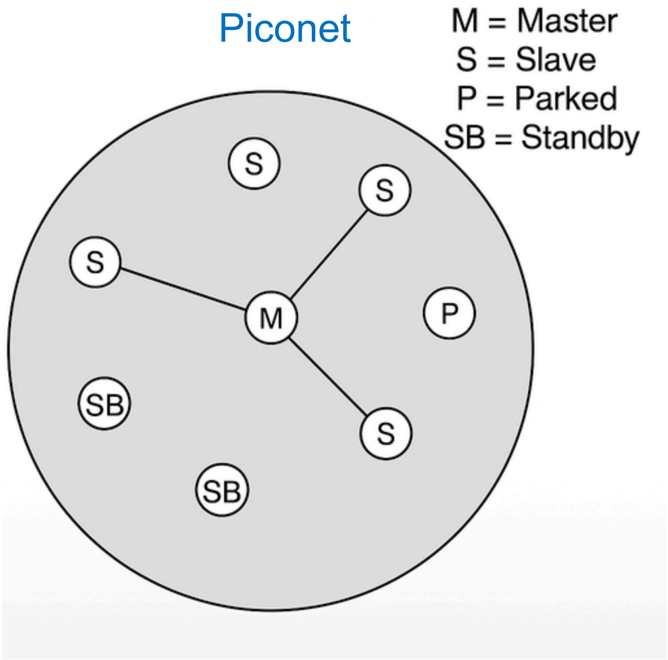
Single-hop Infrastructure-less Wireless Access Networks

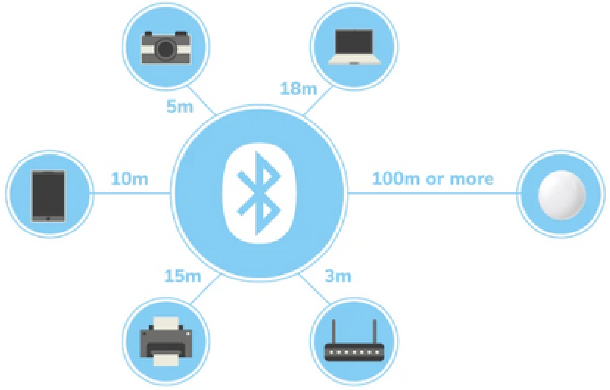
4. Bluetooth



Single-hop Infrastructure-less:: Bluetooth Networks







Master – Slave Communication

only the Master can initiate and control communication within a piconet.

Parked Node: still synchronized **not actively participating**

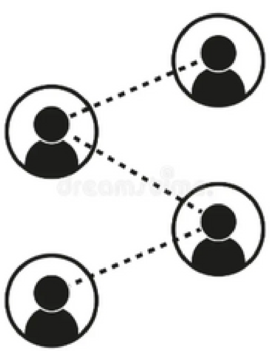
Standby Node: **not connected**

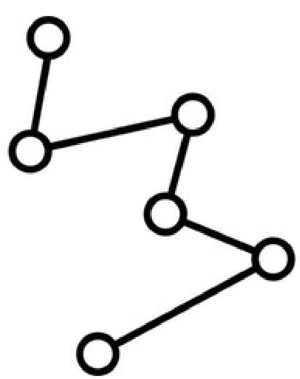
Single-hop Infrastructure-less:: Bluetooth Networks

Property	Description
Frequency Band	2.4 GHz ISM band (2.402 – 2.480 GHz)
Technology	Frequency Hopping Spread Spectrum (FHSS), 1600 hops/sec
Network Topology	Piconet (1 master + up to 7 active slaves)
Data Rate	Up to 721 Kbps (Classic)
Range	~10 meters (Class 2), up to 100 meters (Class 1)
Roles	Master (initiates & controls communication), Slave (responds to master)
Security	Authentication, Authorization, Encryption
Voice Support	Up to 3 simultaneous SCO voice channels
Device States	Active, Parked, Standby

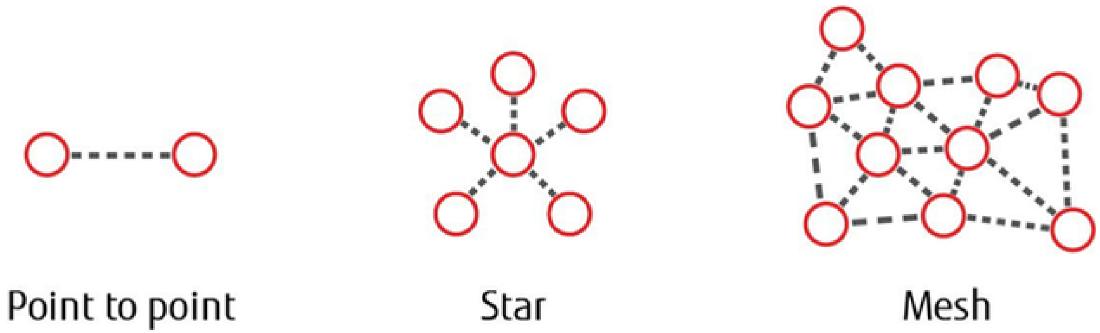
Multi-hop Infrastructure-base Wireless Access Networks

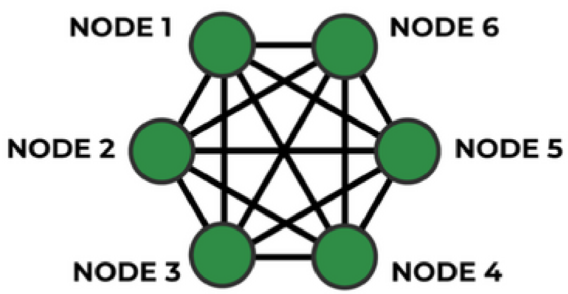
5. Wireless Mesh Networks (semi-structured)

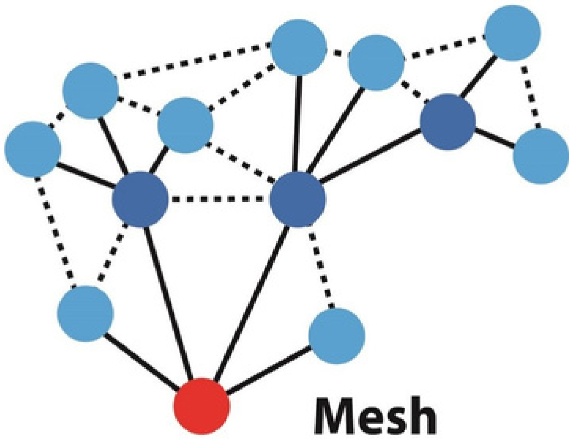




Multi-hop Infrastructure-based:: Wireless Mesh Networks

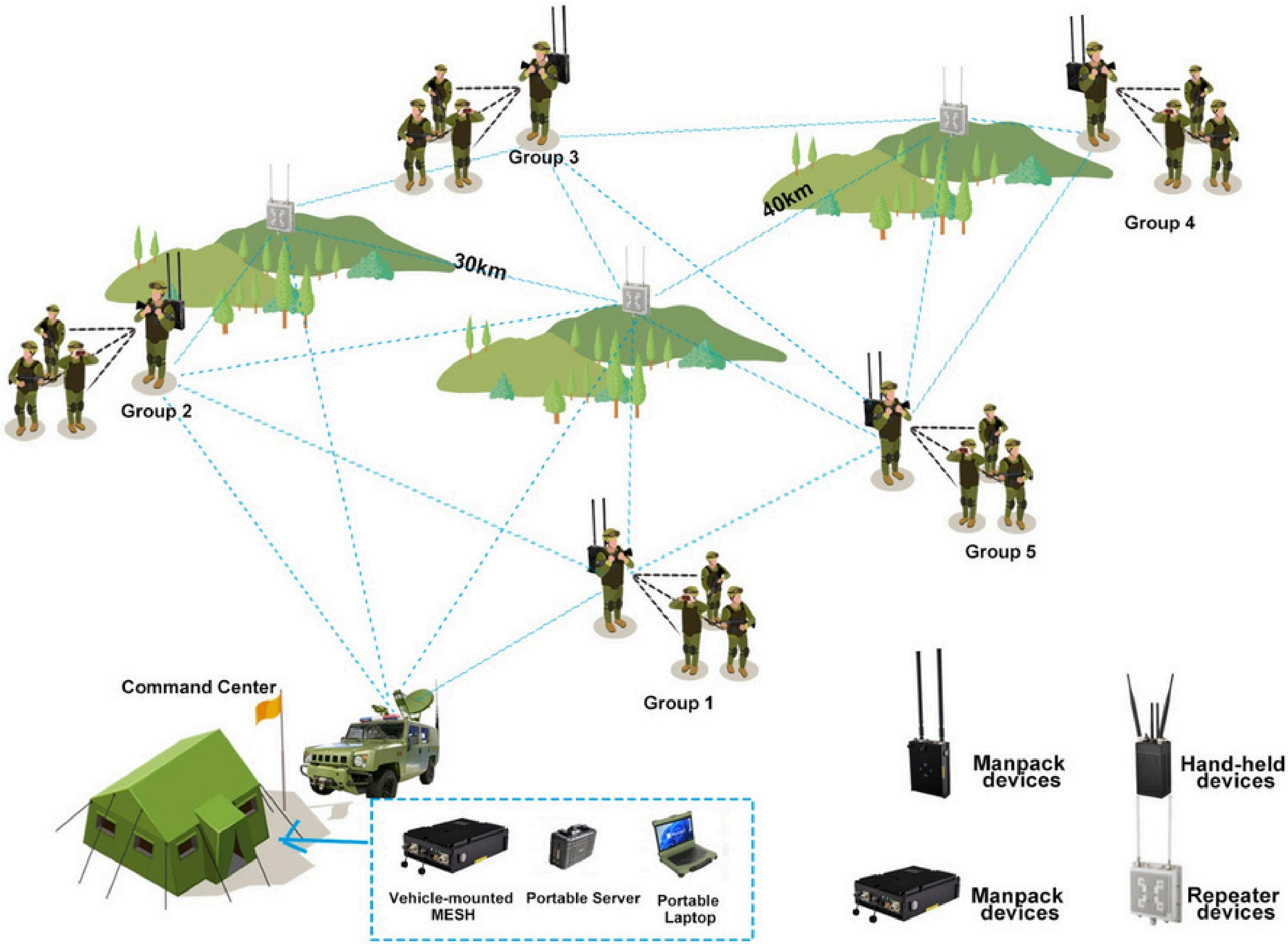






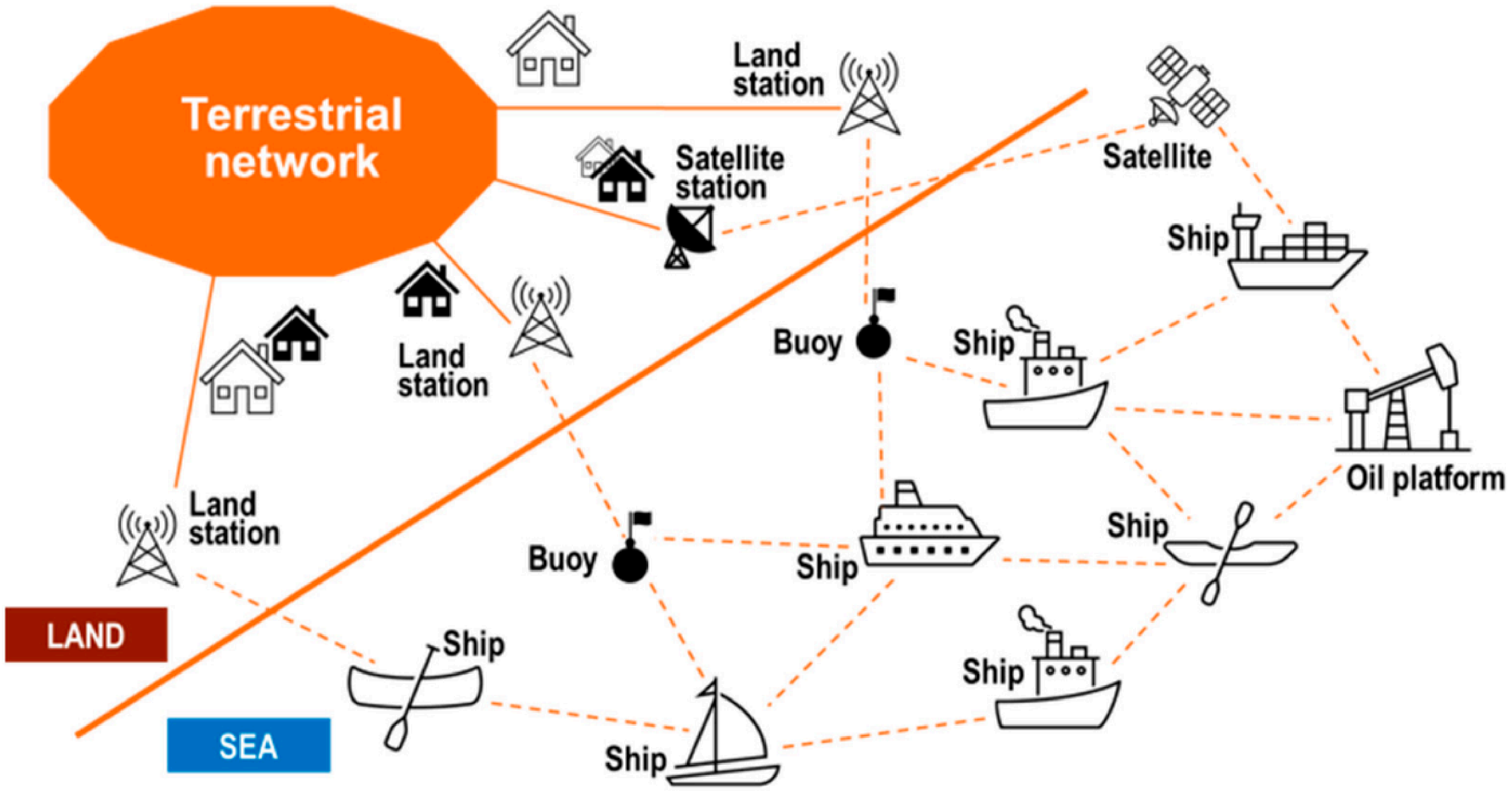
Multi-hop Infrastructure-based:: Wireless Mesh Networks

Application: Military battlefield communication



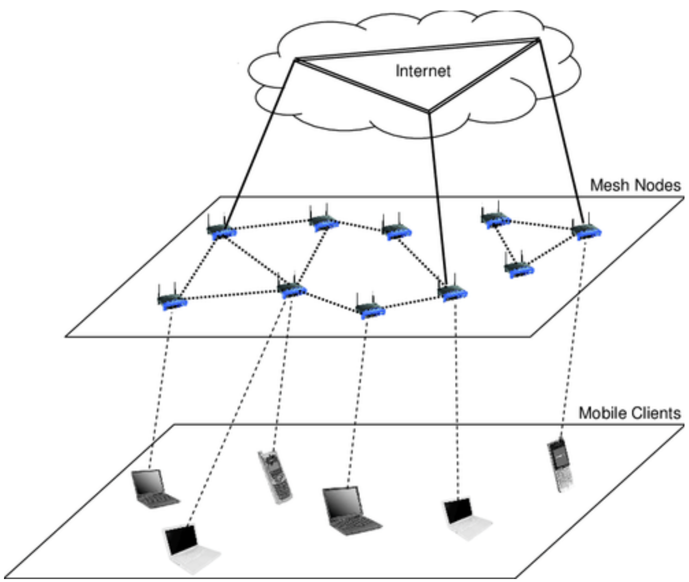
Multi-hop Infrastructure-based:: Wireless Mesh Networks

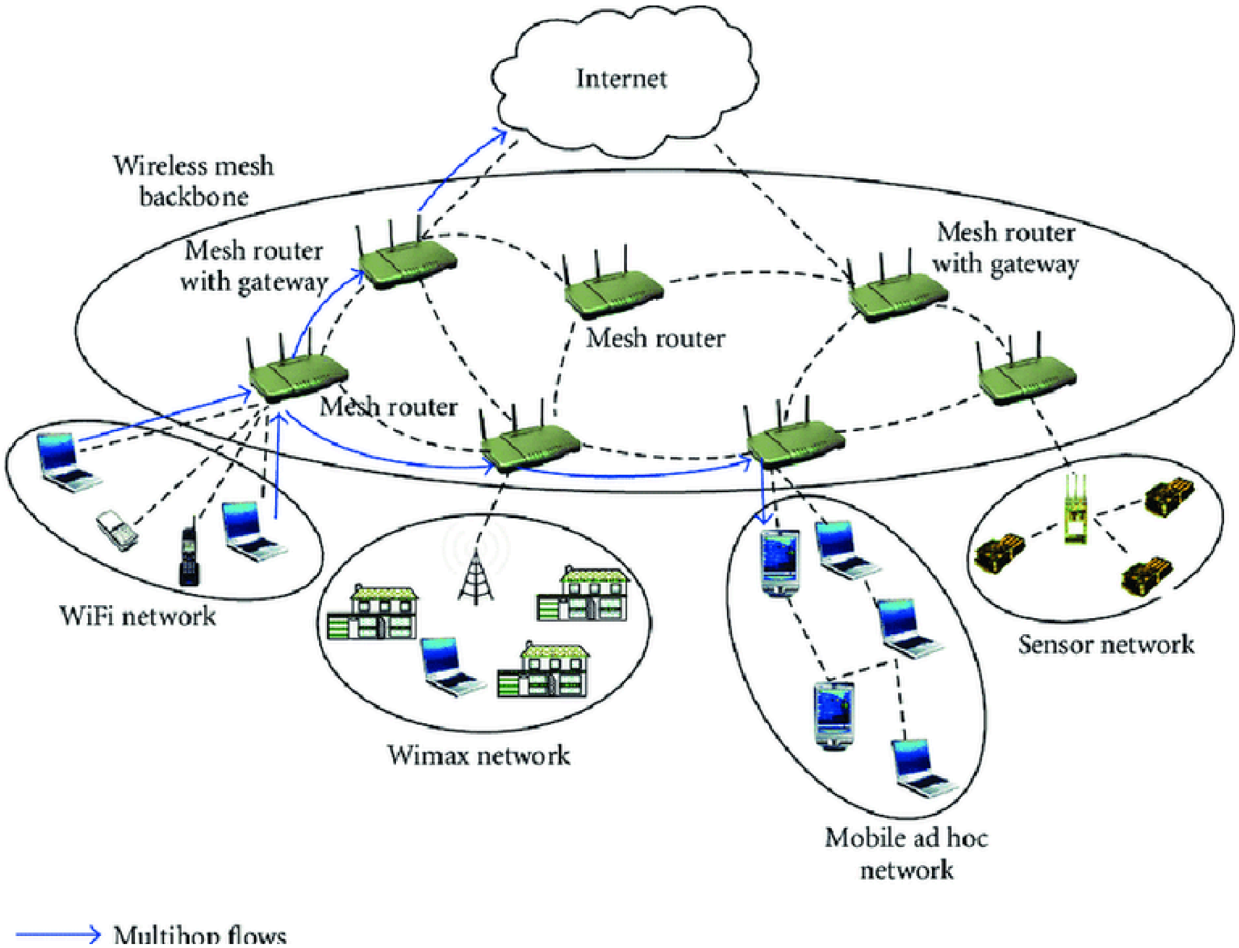
Application: Maritime Mesh Network



Multi-hop Infrastructure-based:: Wireless Mesh Networks

Application: Hybrid Wireless Mesh Network for Backbone for Community Broadband Connectivity





Multi-hop Infrastructure-based::
Wireless Mesh Networks

Application: Drone Mesh Networks (UAV Mesh Networks)

Multi-hop Infrastructure-based:: Wireless Mesh Networks

Application: Very Large-Scale Agricultural Monitoring with Mobile Machineries

Multi-hop Infrastructure-based:: Wireless Mesh Networks

Application: Smart & Connected City with Mobile & Static Devices

Multi-hop Infrastructure-based:: Wireless Mesh Networks

How WMNs Work

1. Network Structure

WMNs consist of two types of devices:

- **Mesh Routers (MRs):**
  - Fixed nodes that form the backbone of the mesh.
  - Often connected to the internet.
- **Mesh Clients (MCs):**
  - End-user devices like laptops, phones, sensors.
  - Connect to the mesh routers or other clients.

2. Multi-hop Routing

- Data travels **hop-by-hop** through multiple mesh routers to reach its destination.
- If a direct path is not available, packets are forwarded through **intermediate nodes**.

3. Self-Healing Capability

- If a node fails or a link breaks, the network **automatically reroutes** traffic through alternative paths.
- Ensures **high reliability and robustness**.

4. Self-Configuration

- New nodes can join the network and start routing **without manual configuration**.
- Each node participates in the routing process and **discovers neighbors dynamically**.

Multi-hop Infrastructure-based:: Wireless Mesh Networks

Multi-hop Infrastructure-based:: Wireless Mesh Networks

1. Interference and Shared Medium
Problem: Nodes use a shared wireless medium, leading to co-channel interference, hidden terminal, and exposed terminal problems.
Impact: Reduces throughput and increases collisions.

2. Scalability
Problem: As the number of nodes increases, routing overhead and contention grow.
Impact: Performance degrades due to increased latency and reduced bandwidth per node.

3. Dynamic Topology & Routing Complexity
Problem: Nodes may join, leave, or move, causing frequent route breaks and topology changes.
Impact: Requires adaptive routing protocols and fast convergence.

4. Limited Bandwidth and Throughput
Problem: Each additional hop in a path reduces available bandwidth due to medium reuse.
Impact: End-to-end throughput suffers, especially in large mesh networks.

Multi-hop Infrastructure-based:: Wireless Mesh Networks

5. Security Threats
Problem: WMNs are vulnerable to eavesdropping, spoofing, DoS attacks, and routing attacks.
Impact: Compromises confidentiality, integrity, and availability.

6. Quality of Service (QoS)
Problem: Hard to guarantee latency, jitter, and packet delivery ratio over unreliable, shared wireless links.
Impact: Affects performance of real-time applications like VoIP and video streaming

7. Synchronization and Coordination
Problem: Distributed nature of WMNs makes scheduling transmissions and synchronizing clocks challenging.
Impact: Causes collisions, inefficient channel use, and poor coordination.

8. Heterogeneity of Devices
Problem: WMNs may include devices with different capabilities, interfaces, and protocols.
Impact: Interoperability issues arise, complicating routing and communication.

Multi-hop Infrastructure-based:: Wireless Mesh Networks

Watch: How do Mesh Networks Works?

Watch: Mesh Networking Idea

Watch: Mesh WiFi Vs. WiFi Extenders

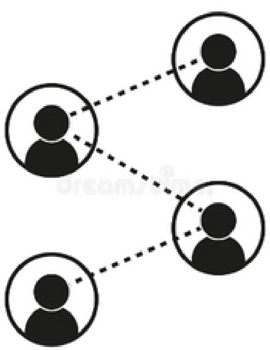
Multi-hop Infrastructure-less
Wireless Access Networks

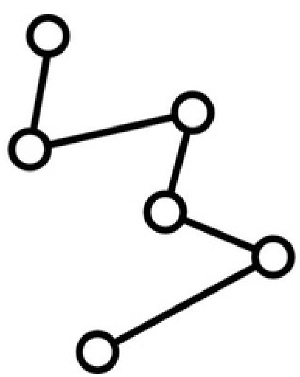
6. Ad hoc wireless Networks

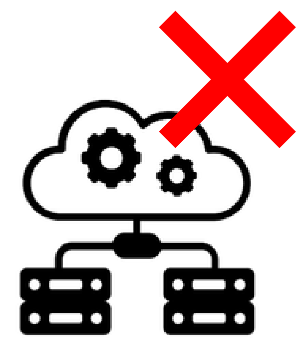
6.1. Wireless Sensor Networks

6.2. Mobile Ad-hoc Networks /Vehicular Ad-hoc Networks

6.3. Delay Tolerant Networks

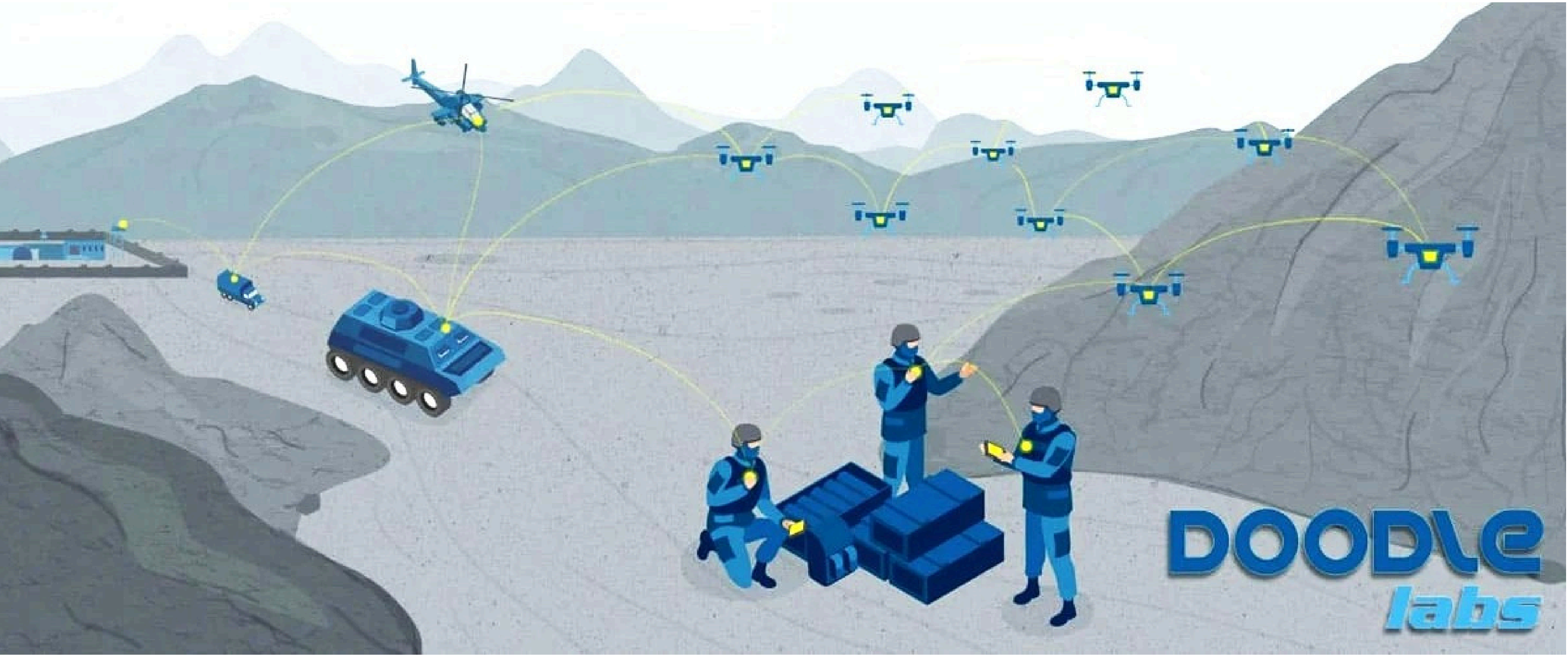






Multi-hop Infrastructure-less::
Wireless Ad-Hoc Networks

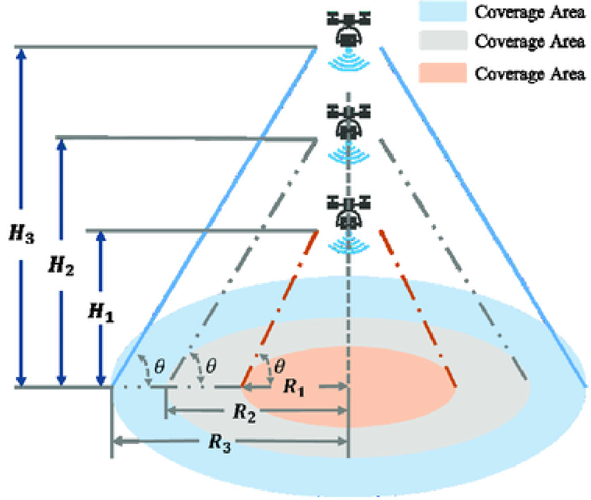
Application: Military battlefield communication in Ad-Hoc Mode

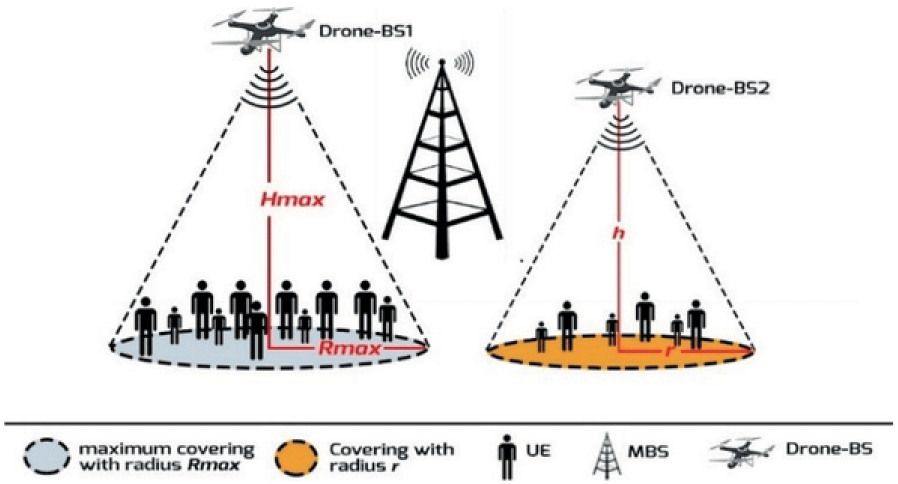


Multi-hop Infrastructure-less::
Wireless Ad-Hoc Networks

Application: Drone Ad-Hoc Networks (UAV Ad-Hoc Networks)

- Temporary Internet Coverage

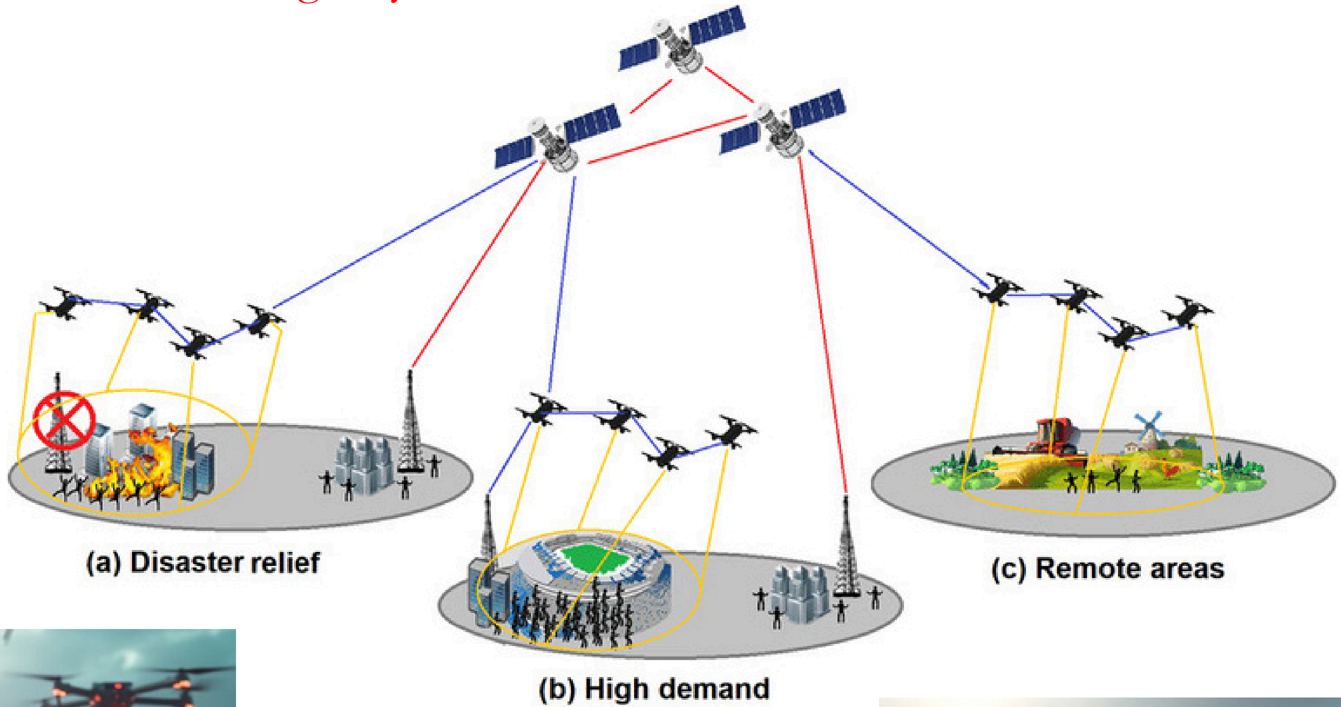




- Search and Rescue in Remote Areas
- Environmental and Wildlife Monitoring
- Disaster Response & Emergency Communications

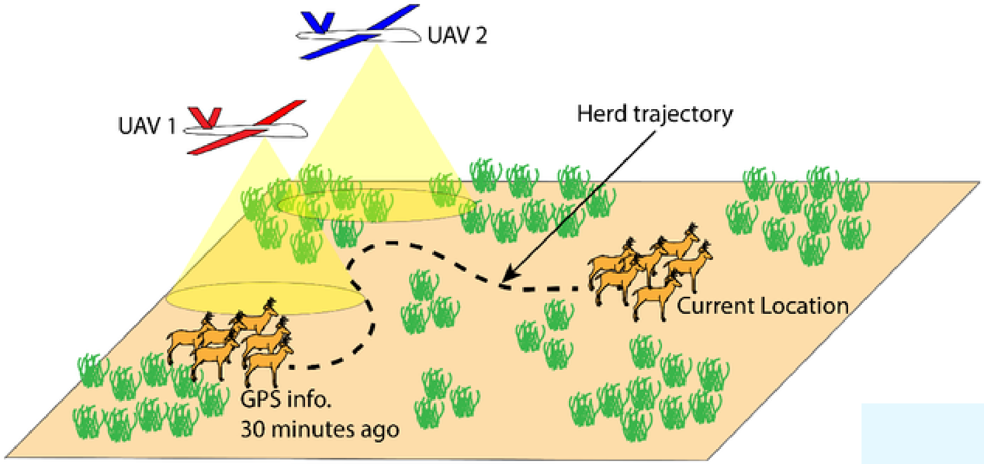
Multi-hop Infrastructure-less::
Wireless Ad-Hoc Networks

Application: Search and Rescue & Emergency Communications in Remote Areas

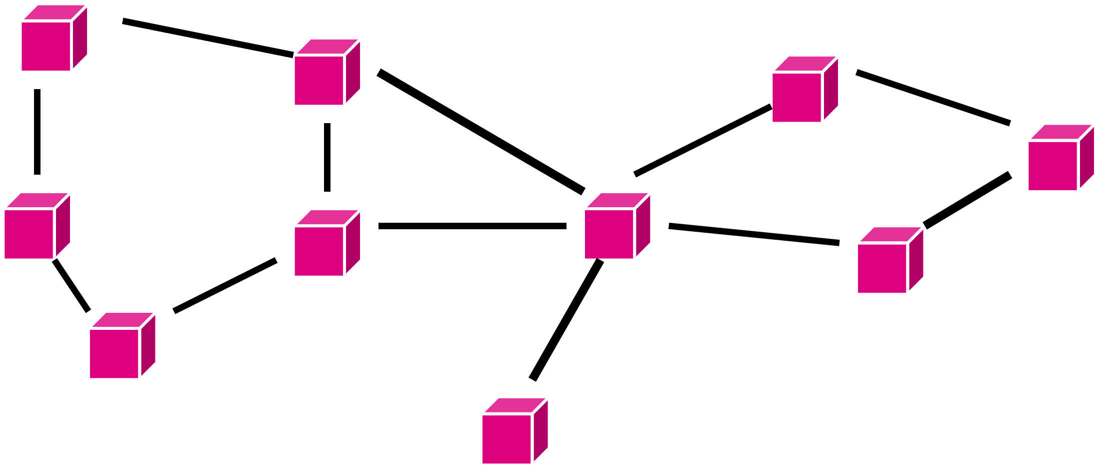


Multi-hop Infrastructure-less::
Wireless Ad-Hoc Networks

Application: Environmental and Wildlife Monitoring



Multi-hop Infrastructure-less:: Ad-Hoc Networks



- Peer-to-peer communications.
- No backbone infrastructure.
- Routing can be multi-hop.
- Topology is dynamic.
- Fully connected with different link SINRs

Multi-hop Infrastructure-less:: Ad-Hoc Networks

- Ad-hoc networks provide a flexible network infrastructure for many emerging applications.
- The capacity of such networks is generally unknown.
- Transmission, access, and routing strategies for ad-hoc networks are generally ad-hoc.
- Cross layer design critical are very challenging.
- Energy constraints impose interesting design tradeoffs for communication and networking.

Multi-hop Infrastructure-less::
Wireless Mesh Network vs Wireless Ad-Hoc Networks

Aspect	**Wireless Mesh Network (WMN)**	**Wireless Ad-hoc Network (WAHN)**
Infrastructure	Semi-infrastructure (uses fixed mesh routers as backbone)	Purely infrastructure-less
Node Roles	Distinct roles: **Mesh routers** (static) and **clients** (dynamic)	All nodes are equal; any node can route data
Deployment	Pre-planned, often permanent	Temporary, spontaneous setup
Routing	Stable and optimized (e.g., HWMP, OLSR)	Dynamic and reactive (e.g., AODV, DSR)
Scalability	Highly scalable due to backbone support	Limited scalability due to overhead and congestion
Mobility	Clients may be mobile; routers are often fixed	All nodes may be mobile
Reliability	More reliable due to fixed routers and redundant paths	Less reliable, frequent topology changes
Use Cases	**Community networks, smart cities, campus-wide networks**	**Military ops, disaster relief, emergency response**

Multi-hop Infrastructure-less::
Wireless Mesh Network vs Wireless Ad-Hoc Networks

Aspect	**Wireless Mesh Network (WMN)**	**Wireless Ad-hoc Network (WAHN)**
Radio Interfaces	Often multi-radio and multi-channel (e.g., 802.11s, WiMAX, LTE)	Usually single-radio, single-channel (e.g., 802.11, Bluetooth)
Backhaul Support	Yes, mesh routers may connect to the Internet or backbone	No backhaul, only peer-to-peer communication
Gateway Capability	Mesh routers often serve as gateways to external networks	No defined gateway; all nodes are peers
Energy Source	Often powered (e.g., fixed routers)	Often battery-operated (mobile nodes)

Multi-hop Infrastructure-less::
Wireless Mesh Network vs Wireless Ad-Hoc Networks

Aspect	**Wireless Mesh Network (WMN)**	**Wireless Ad-hoc Network (WAHN)**
Primary Goal	Provide last-mile connectivity and reliable internet access over large areas	Enable peer-to-peer communication in infrastructure-less and mobile environments
Design Focus	Stability, scalability, and long-term connectivity	Quick deployment, flexibility, and dynamic adaptability
Application Focus	**Community WiFi, smart cities, industrial IoT, public safety**	**Military operations, disaster relief, vehicular networks**
Traffic Pattern	Typically many-to-one or client-gateway	Dynamic patterns; often all-to-all or many-to-many
Performance Goal	**High throughput, fault tolerance, consistent QoS**	**Fast topology adaptation, minimal delay in dynamic environments**

Multi-hop Infrastructure-less::
Wireless Ad-Hoc Networks

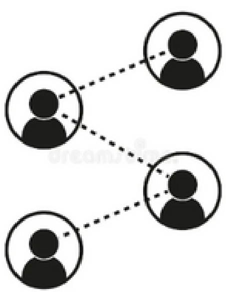
6. Ad hoc wireless Networks

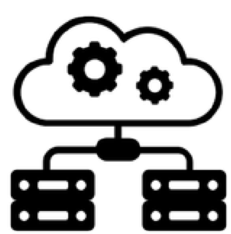
6.1. Wireless Sensor Networks
6.2. Mobile Ad-hoc Networks /Vehicular Ad-hoc Networks
6.3. Delay Tolerant Networks

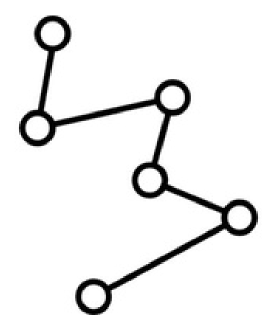
Feature	WSN	VANET	DTN
Infrastructure-less (or minimal)	✅	✅	✅
Multi-hop communication	✅	✅	✅
Dynamic topology	✅ (due to node failure/sleep)	✅ (due to vehicle mobility)	✅ (due to intermittent connectivity)
Self-organizing	✅	✅	✅
Decentralized control	✅	✅	✅

Wireless Sensor Networks

(Structured / Semi-Structured / Un-Structed)



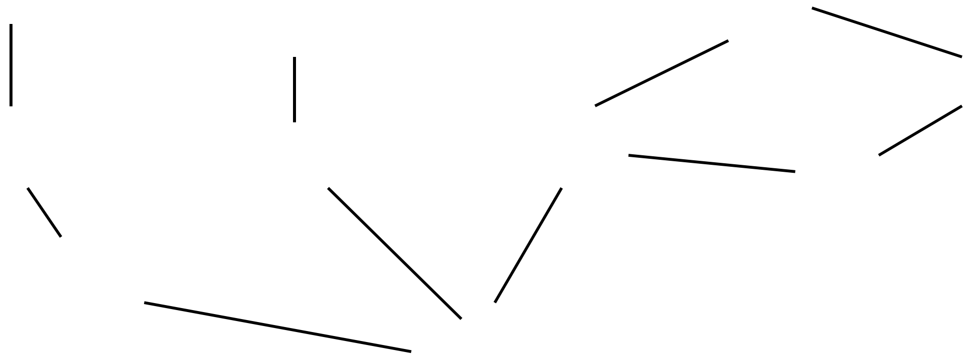




Multi-hop Infrastructure-less:: Wireless Sensor Networks

Energy is the primary constraint

Low power Wireless Sensor Network (Low PAN)[IEEE 802.15]

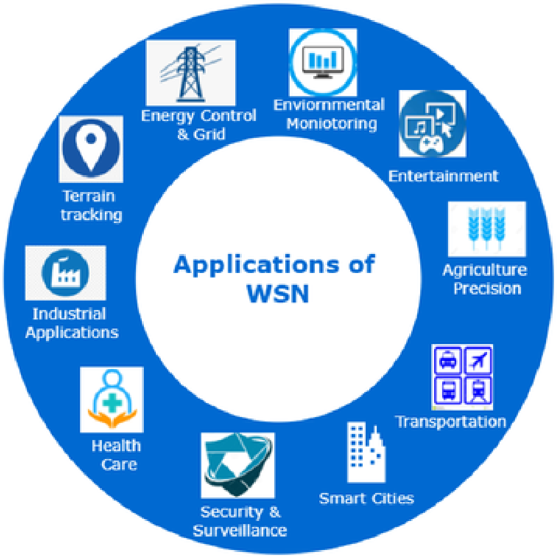


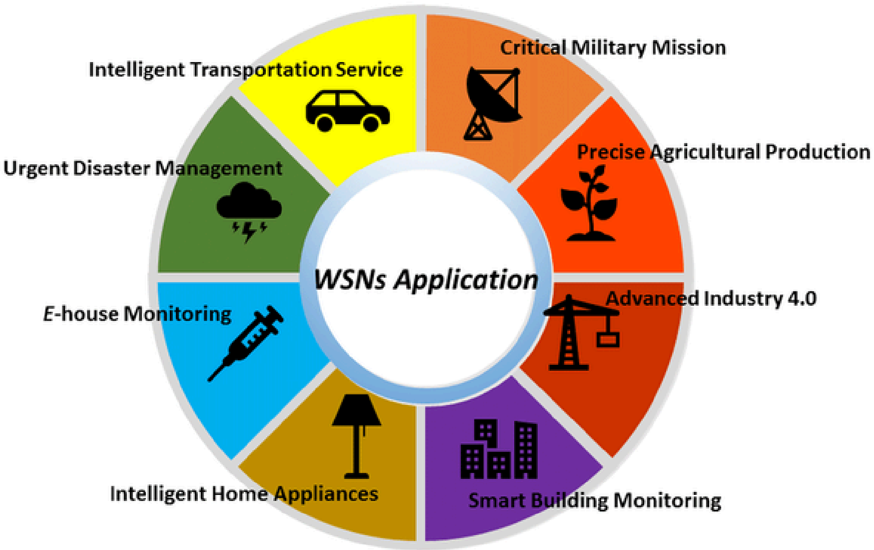
- Nodes powered by non rechargeable batteries
- Data flows to centralized location.
- Low per-node rates but up to 100,000 nodes.
- Data highly correlated in time and space.
- Nodes can cooperate in transmission, reception, compression, and signal processing.

Sensor Networks

- Short-range networks must consider transmit, circuit, and processing energy.
  - Sophisticated techniques not necessarily energy-efficient.
  - Sleep modes save energy but complicate networking.
- Changes everything about the network design:
  - Bit allocation must be optimized across all protocols.
  - Delay vs. throughput vs. node/network lifetime tradeoffs.
  - Optimization of node cooperation.

Sensor Networks :: Applications





Sensor Networks :: Applications

- Constant monitoring & detection of specific events
- Military, battlefield surveillance
- Forest fire & flood detection
- Habitat exploration of animals
- Patient monitoring
- Home appliances

Sensor Networks :: Applications

Precision Agriculture: (Structured / Semi-Structured **)**

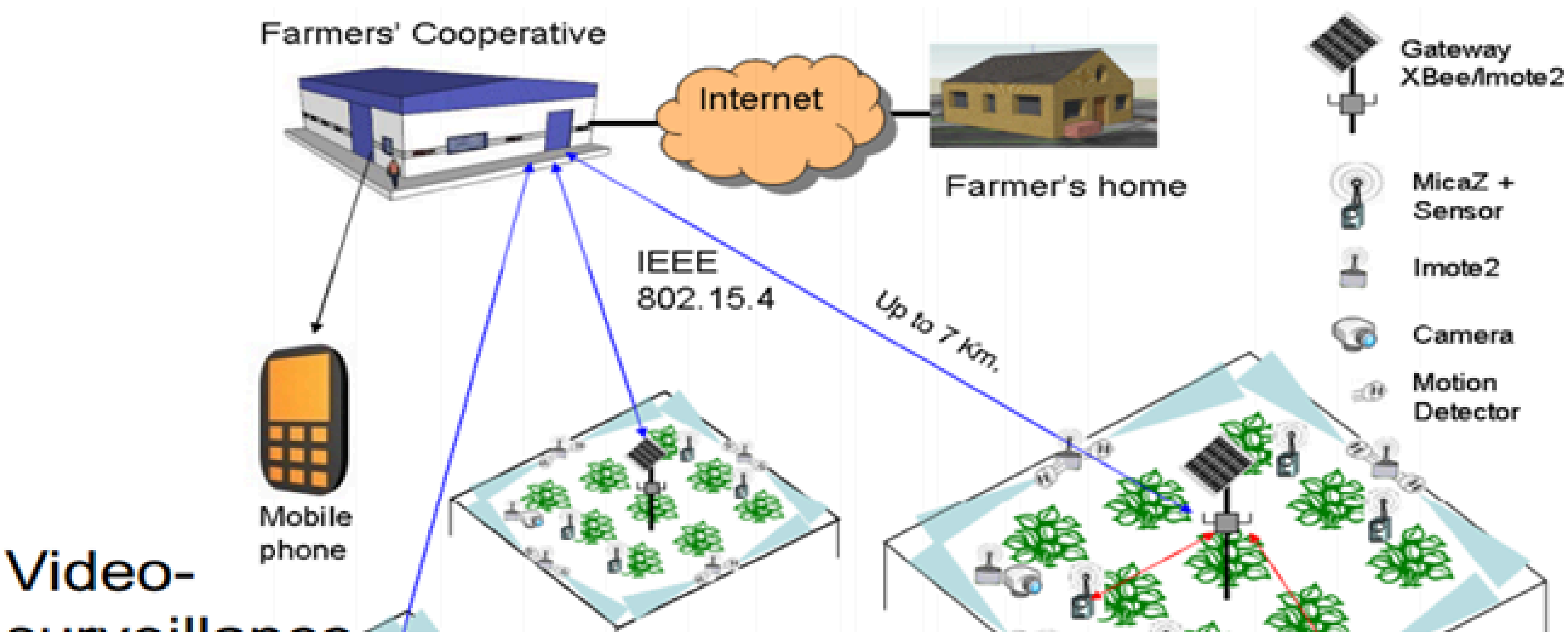


Figure: An example of Low PAN Application

Sensor Networks :: Applications

Remote deployment of sensors for tactical monitoring of enemy troop movements.

Sensor Networks :: Applications

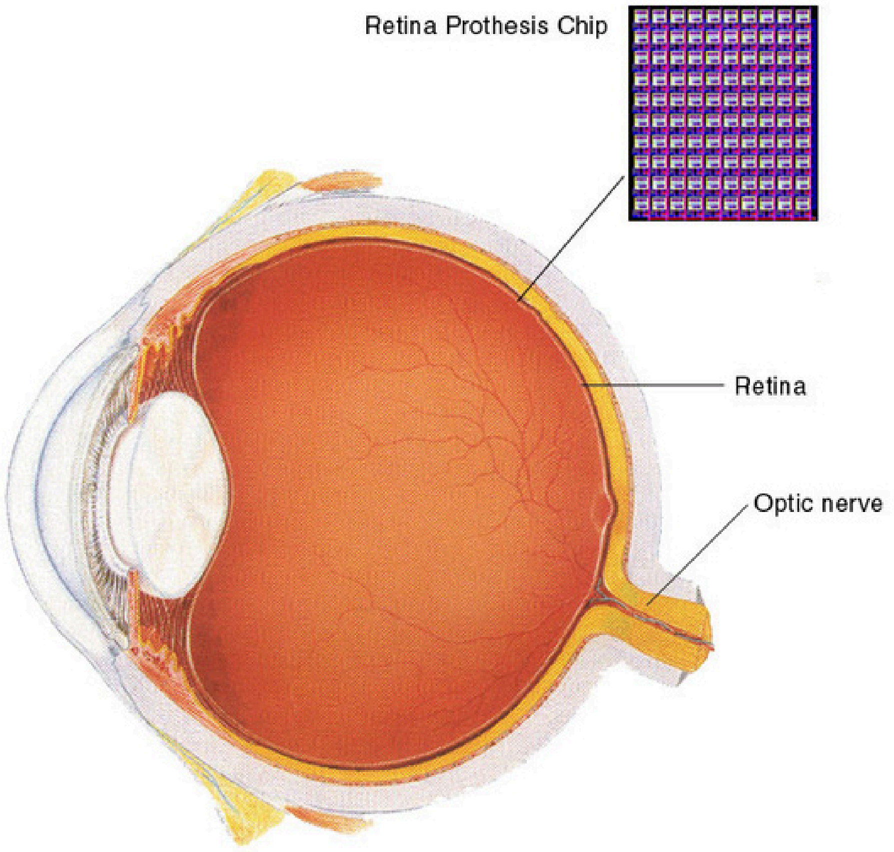
Smart Home/Smart Office: Structured

- Sensors controlling appliances and electrical devices in the house
- Better lighting and heating in office buildings

Sensor Networks :: Applications

Medical/Biomedical: Body Area Network: (Structured)

- Health Monitors
  - Glucose
  - Heart rate
  - Cancer detection
- Chronic Diseases
  - Artificial retina
  - Cochlear implants
- Hospital Sensors
  - Monitor vital signs
  - Record anomalies



Sensor Networks :: Design Issue

- MAC for Sensor networks is specially designed for wireless sensor networks and is derived from IEEE 802.11.

Design Issue: Sensor Networks

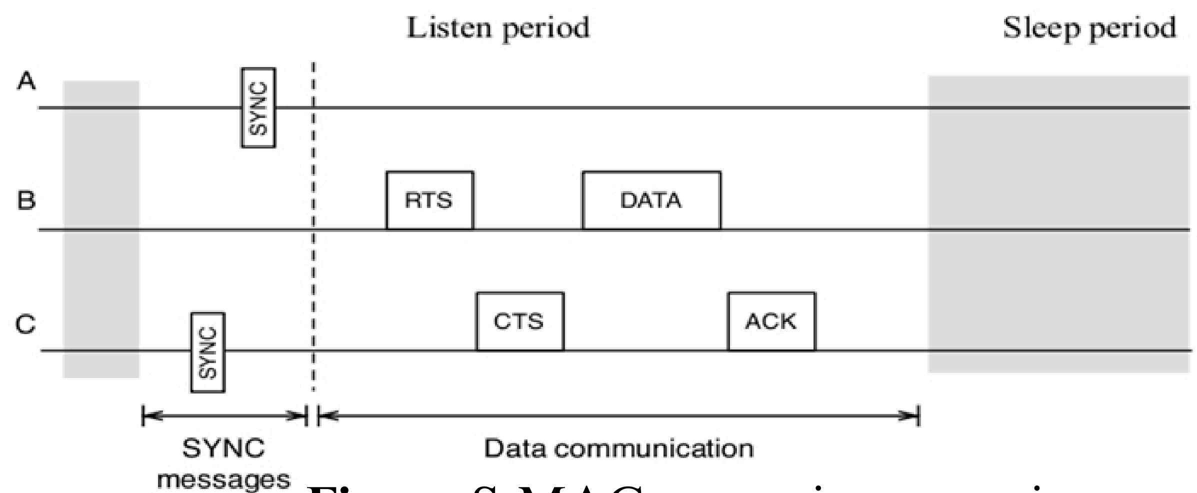
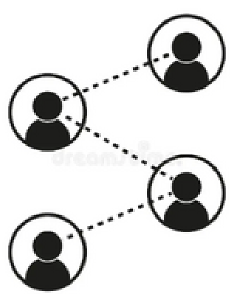
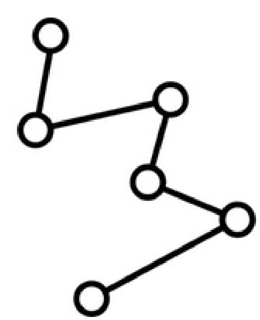


Figure: S-MAC messaging scenario

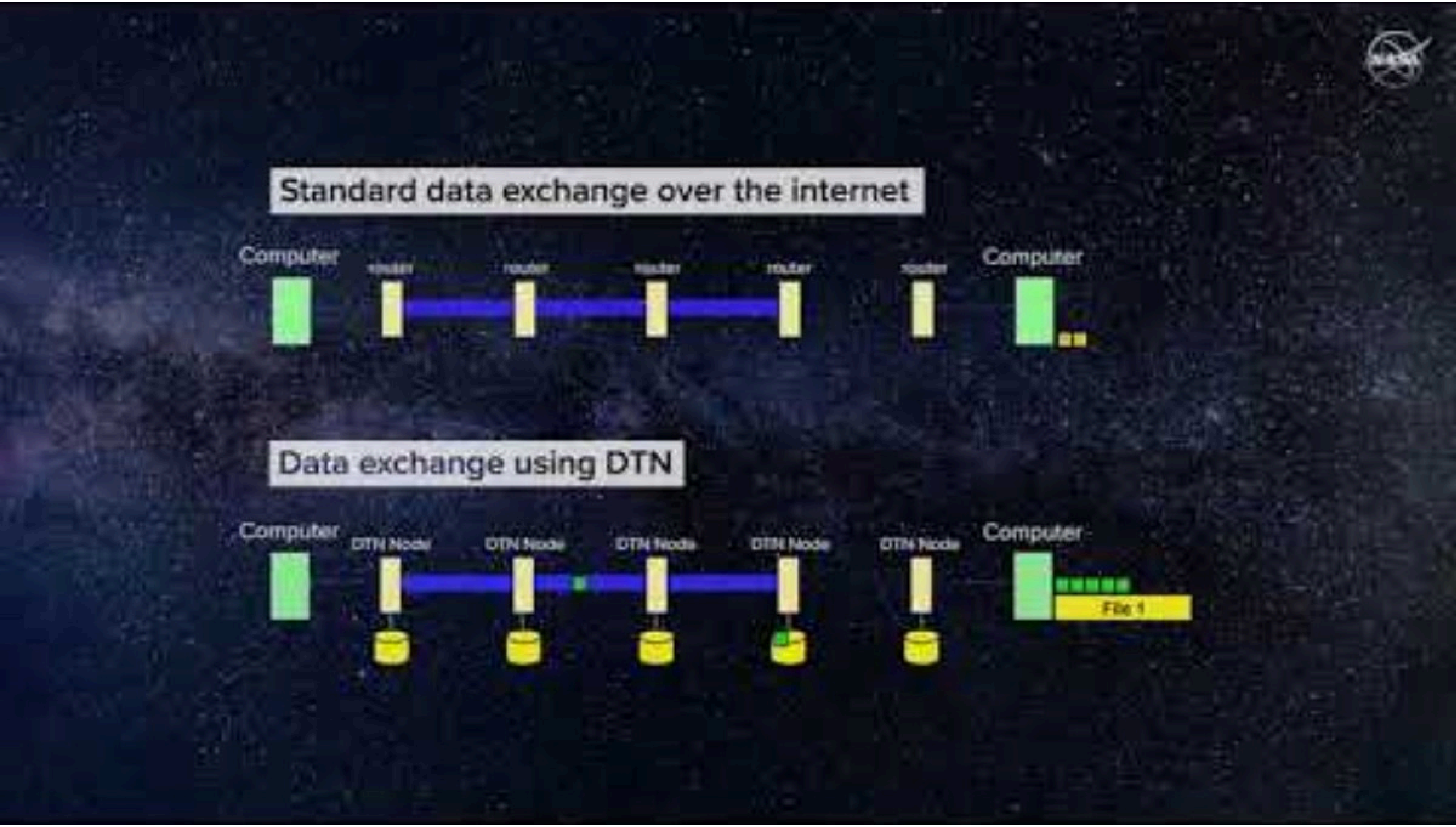
Delay/Disruption Tolerant Network (DTN)

(Un-Structed)

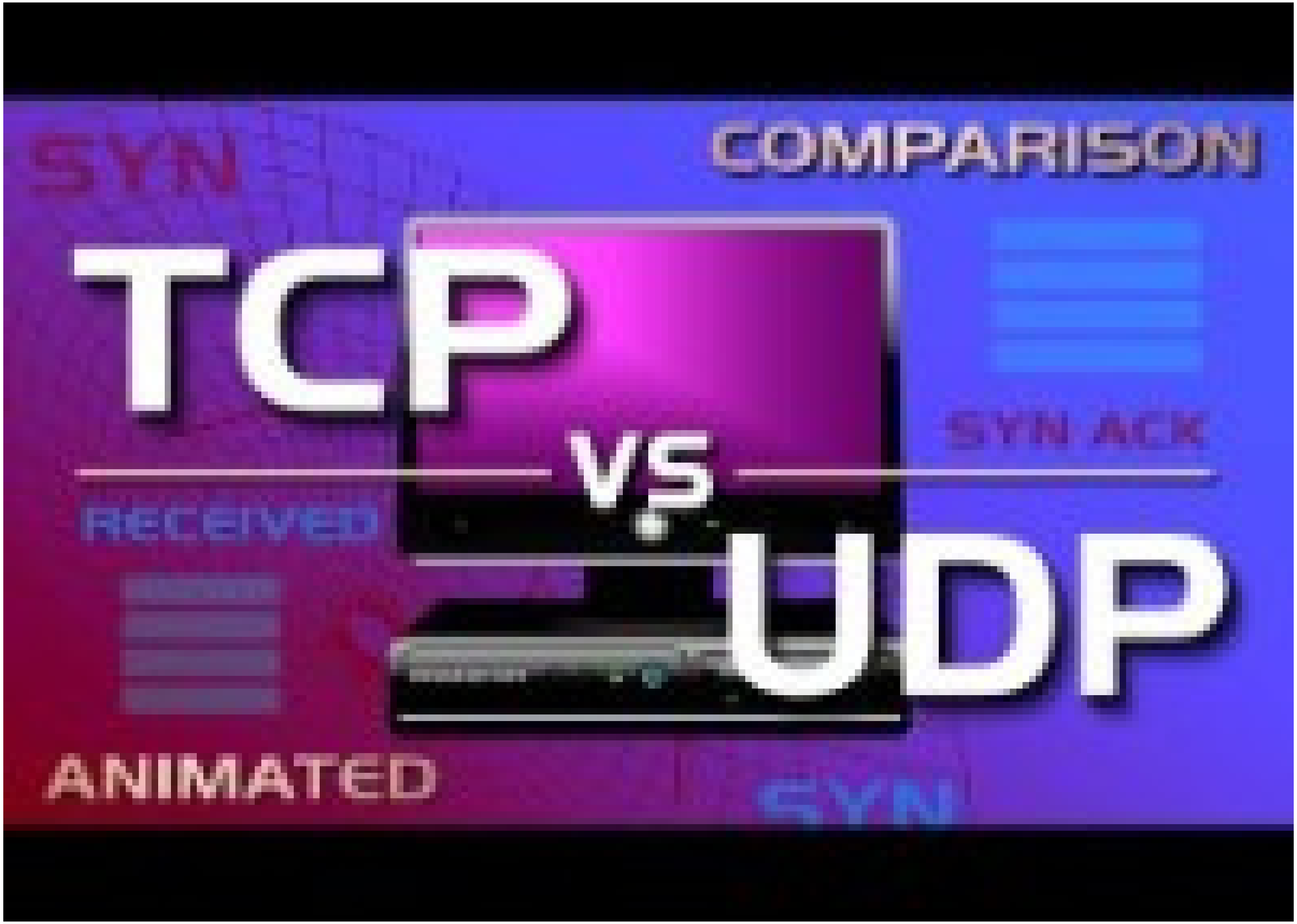




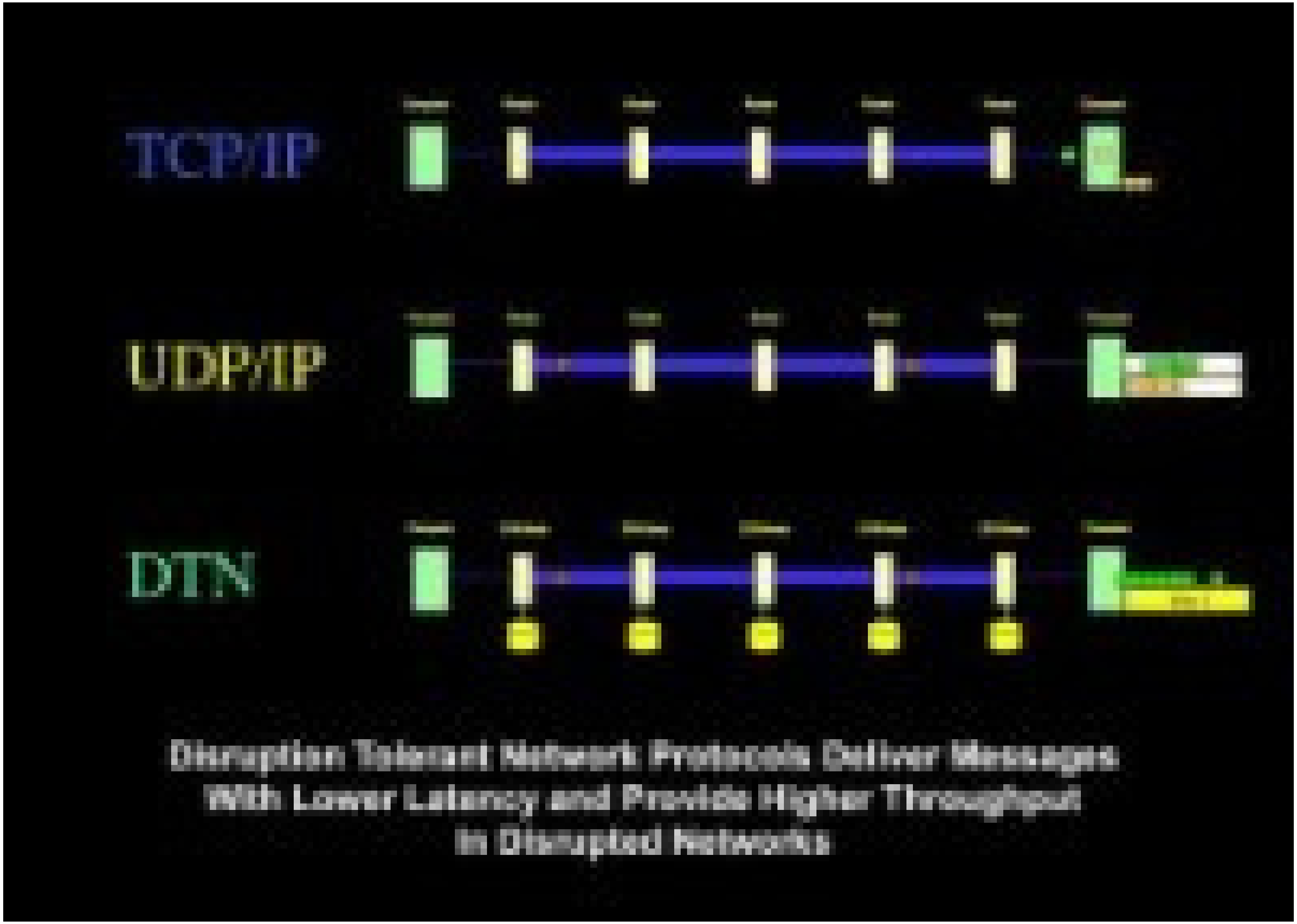
Delay/Disruption Tolerant Network (DTN)



Understanding the TCP and UDP



TCP, UDP and DTN



Delay Tolerant Network (DTN)

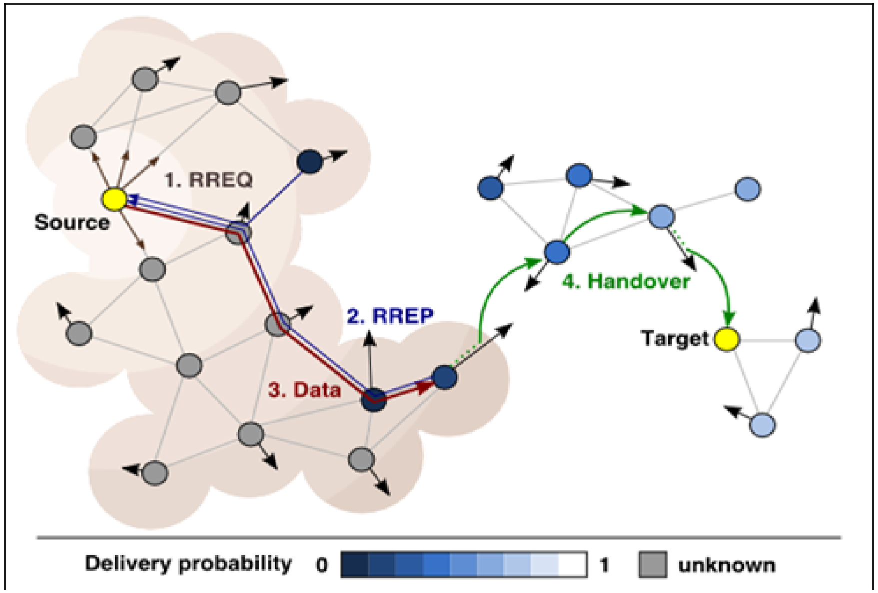


Figure:

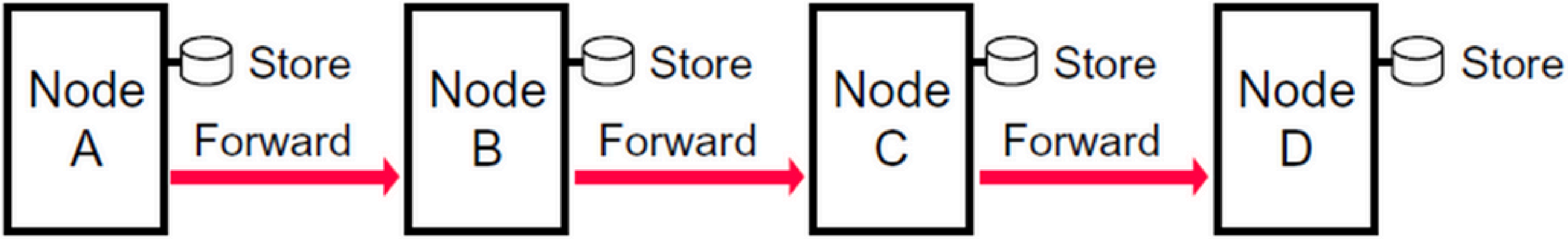


Figure: Store-And-Forward Message Switching

Delay Tolerant Network

- DTN architecture expects nodes to store bundles for an extended period.
- Whenever possible, each received data packet is forwarded immediately.
- If forwarding is not currently possible but is expected to be possible in the future, the packet is stored for future transmission.

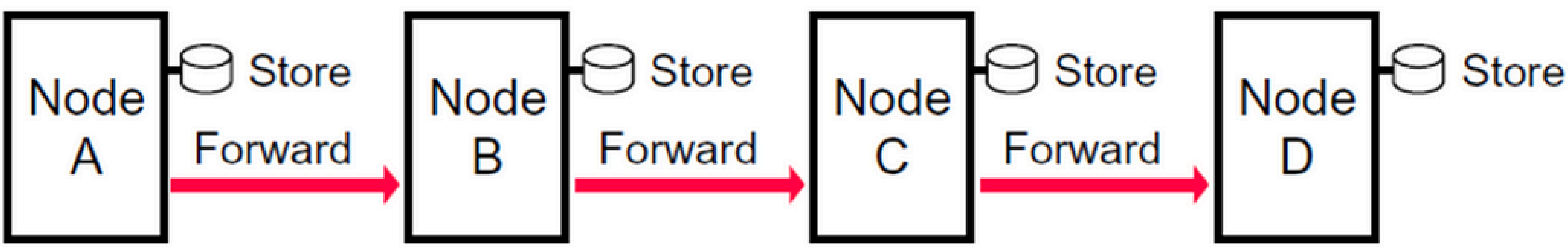
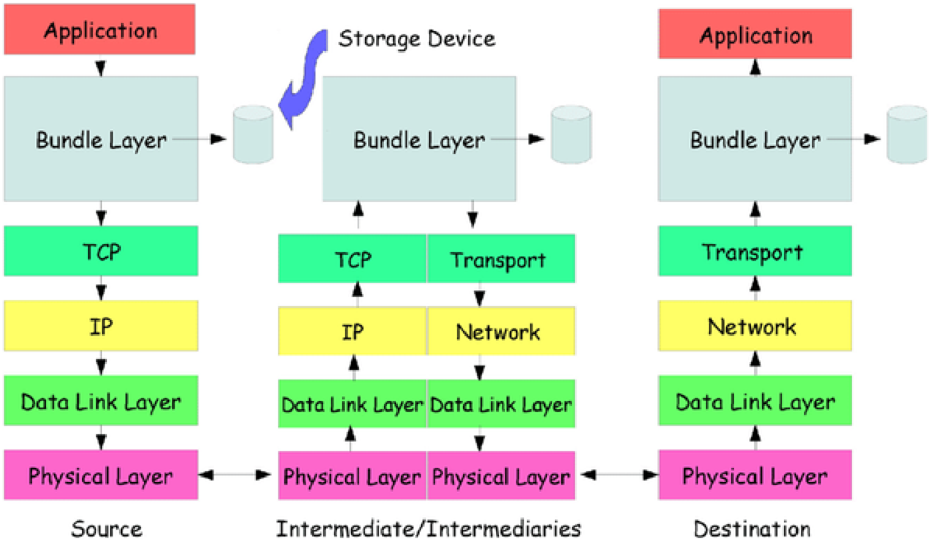


Figure: Store-And-Forward Message Switching

Delay Tolerant Network



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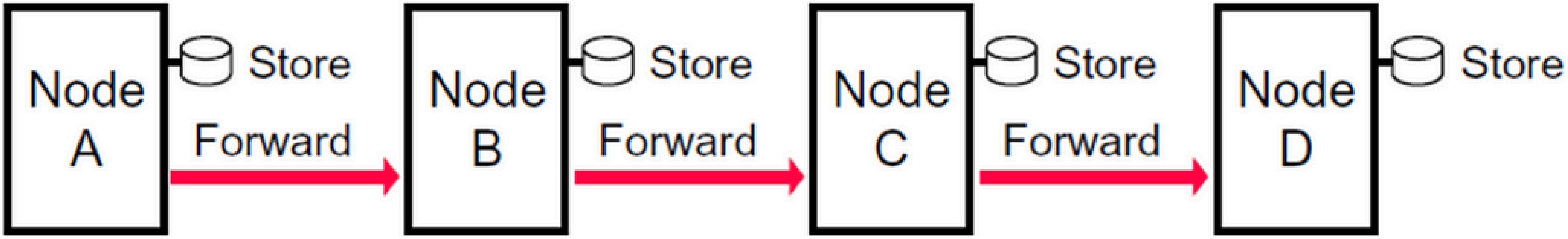
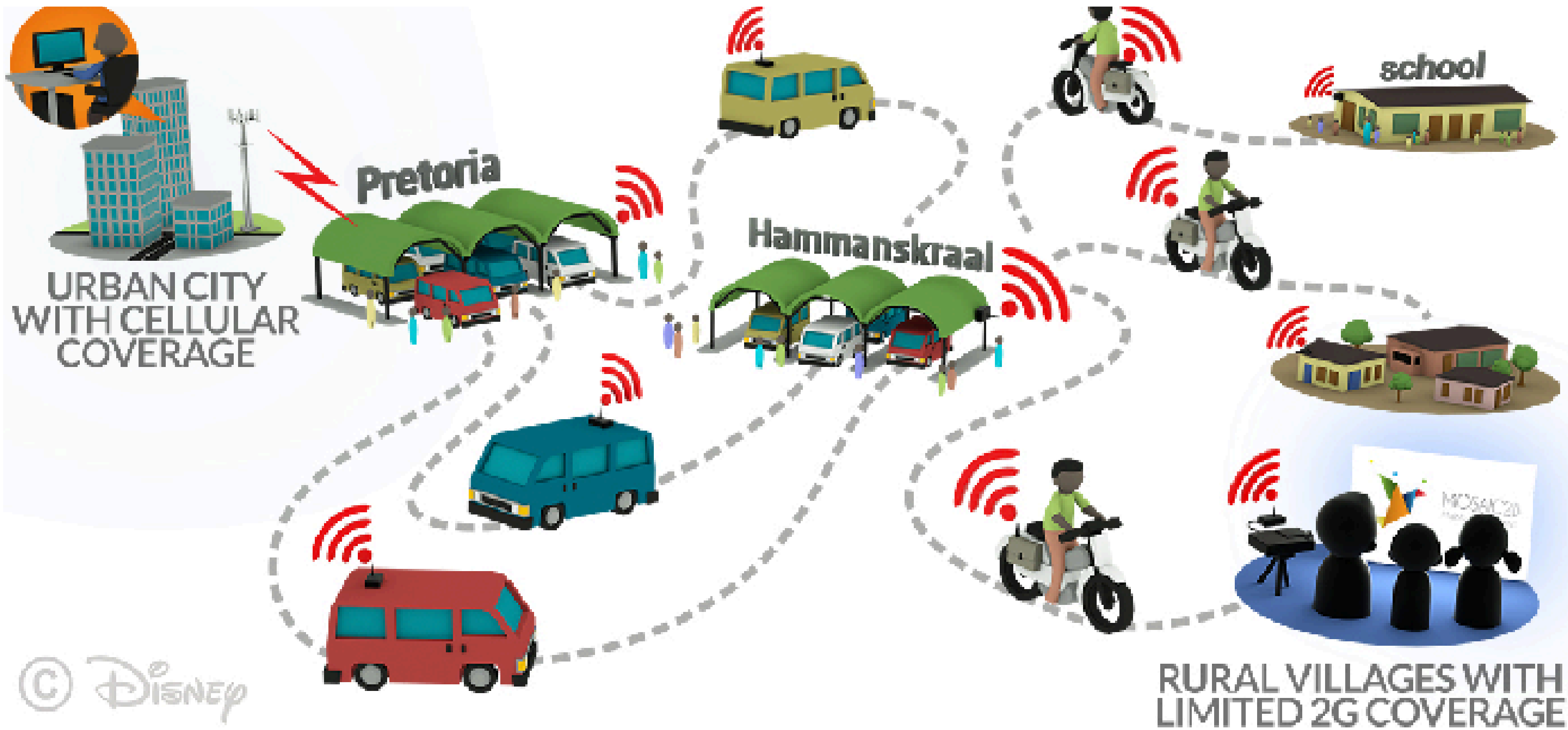


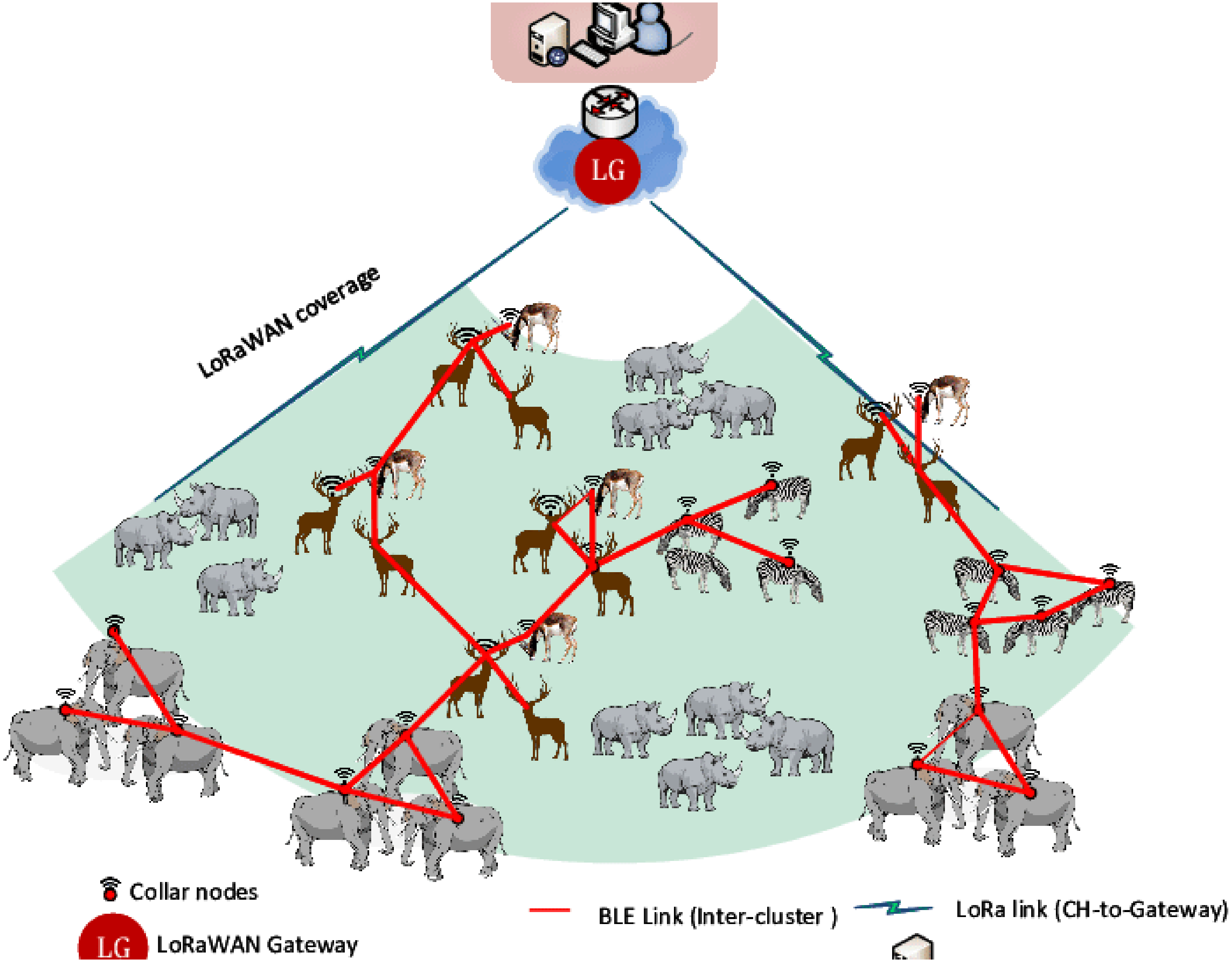
Figure: Store-And-Forward Message Switching

Delay Tolerant Network (DTN):: Applications

Mobile-enabled delay tolerant networking in rural developing regions

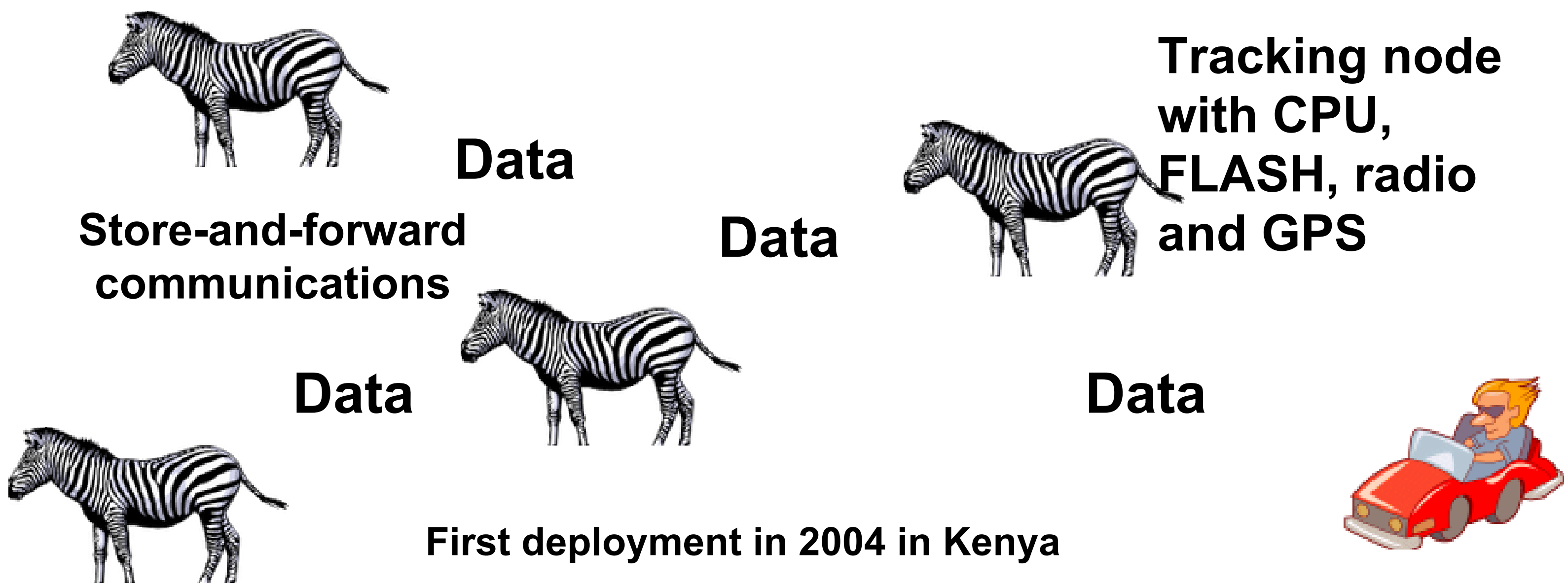


Delay Tolerant Network (DTN):: Applications



Delay Tolerant Network (DTN):: Applications

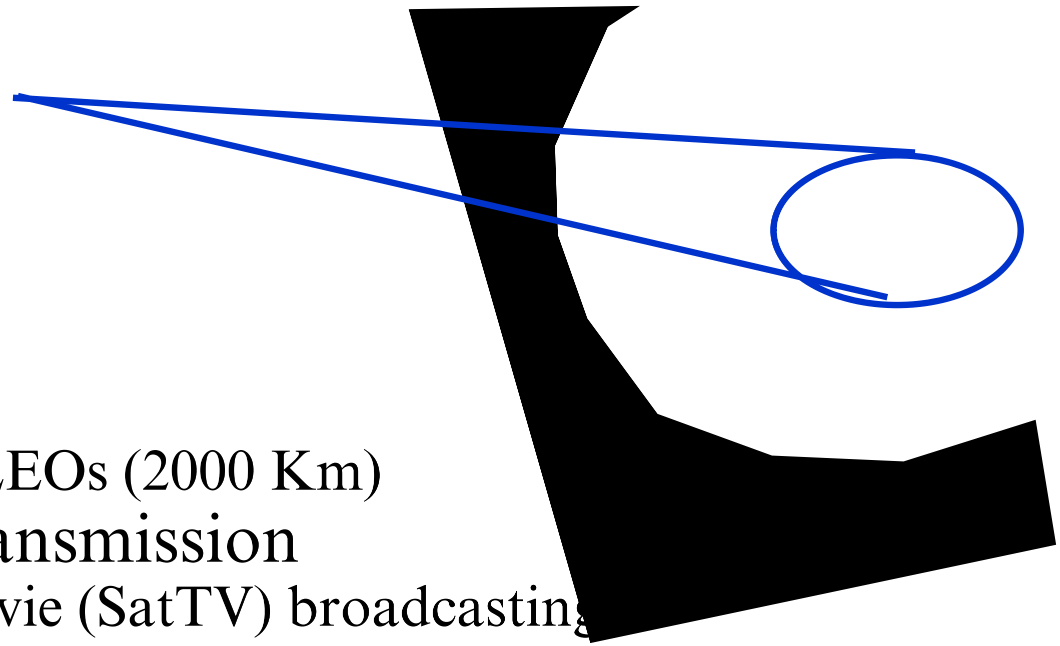
ZebraNet (a real life application)



First deployment in 2004 in Kenya

Delay Tolerant Network (DTN):: Applications

Satellite Systems



- Cover very large areas
- Different orbit heights
  - GEOs (39000 Km) versus LEOs (2000 Km)
- Optimized for one-way transmission
  - Radio (XM, DAB) and movie (SatTV) broadcasting
- Most two-way systems struggling or bankrupt
  - Expensive alternative to terrestrial system
  - A few ambitious systems on the horizon

DTN :: Characteristics

- Internet environment
  - End-to-end RTT is not large.
  - Some path exists between endpoints.
  - E2E reliability using ARQ works well.
  - Packet-switching is the right abstraction.
- DTN characteristics
  - Very large delays.
  - Intermittent and scheduled links.
  - Different network architectures.
  - Conversational protocols fail.
  - No ARQ.

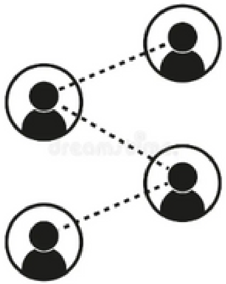
Ref: [2], [3]

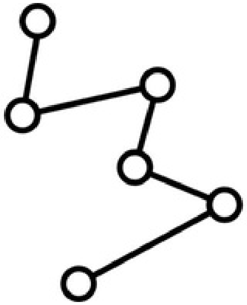
DTN :: Challenges

- Applications assume e2e ‘connections’
- Routing does not understand schedules
- Long-term disconnection causes failures
- End-to-end reliability is poor with high loss rates
- E2E performance is poor with high delays

Vehicular Ad-hoc Networks

(Un-Structed)





Vehicular Ad-hoc Networks

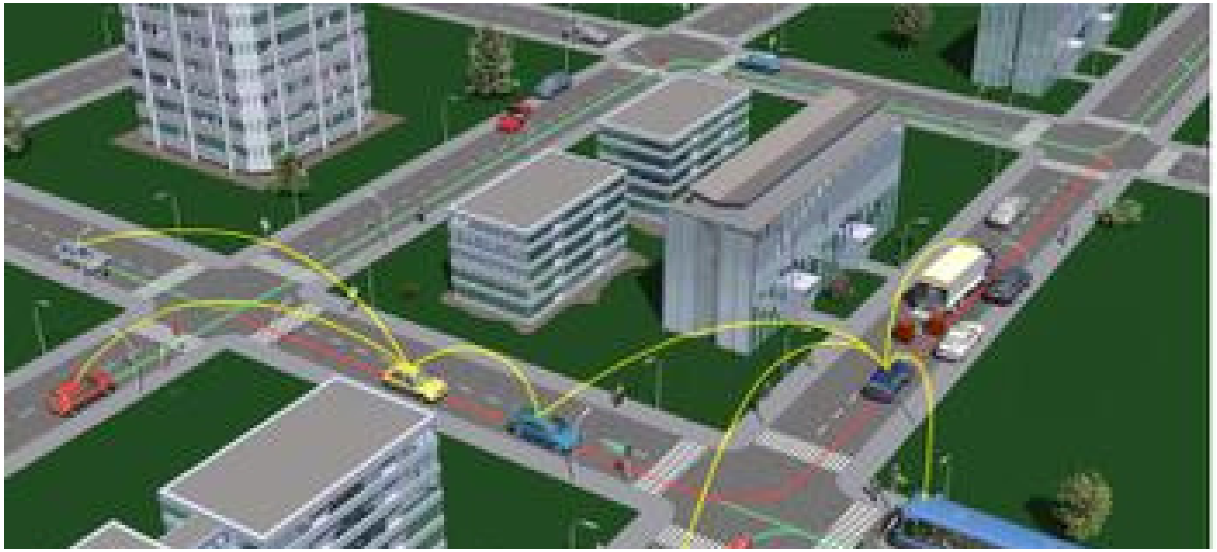
VANET

- a form of Mobile ad-hoc network
- provide communication
  - among nearby vehicles
  - nearby fixed equipment

Vehicular Ad-hoc Networks

How vehicular communications work:

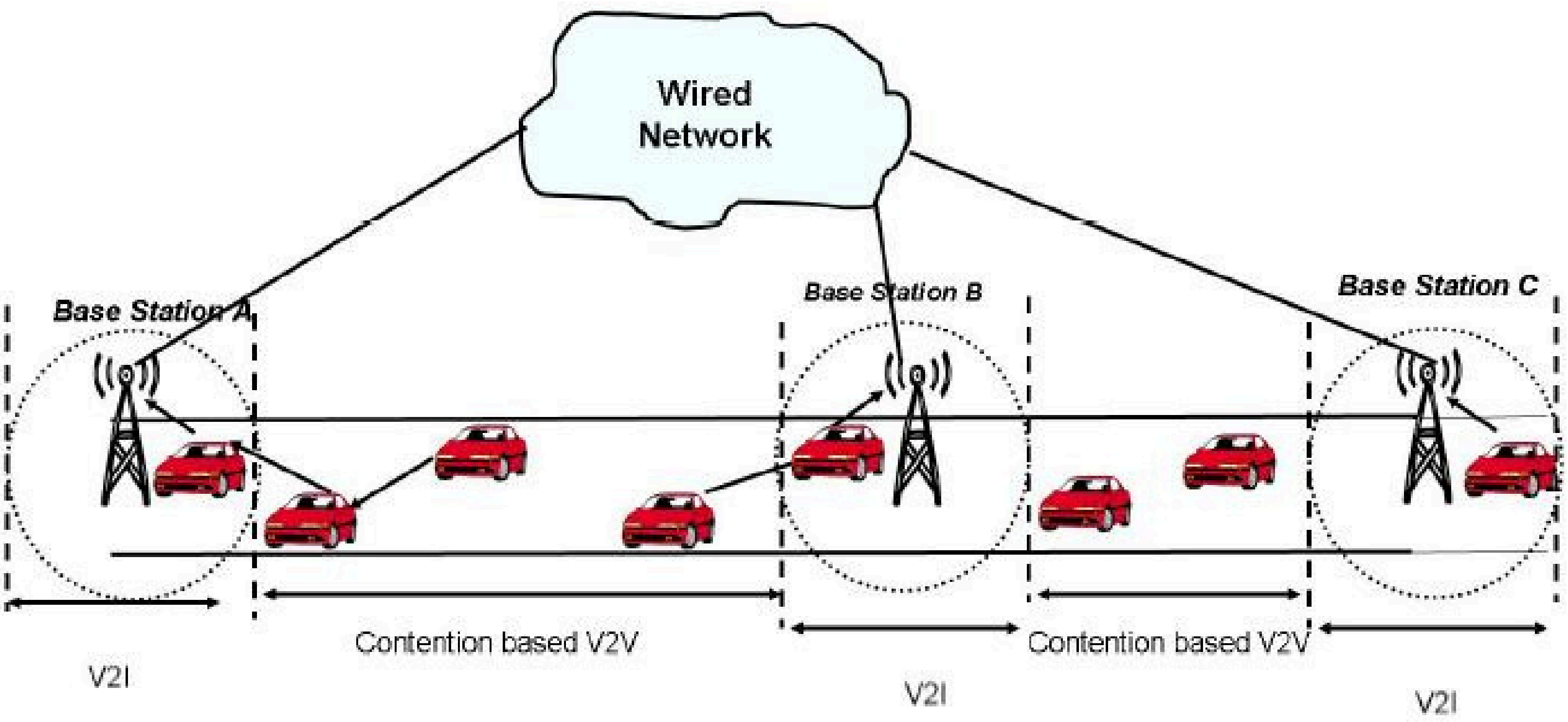
Road-Side infrastructure Units (RSUs), are equipped with **on-board processing and wireless communication modules**



Vehicular Ad-hoc Networks

How vehicular communications work:

- Vehicle-to-Vehicle *(V2V)*
- Vehicle-to-Infrastructure *(V2I)* communication will be possible



Car-to-X (C2X) Communication Patterns

- Vehicle-to-X (V2X),
- Inter-Vehicle Communication (IVC),
- Vehicular ad-hoc network (VANET),
- …

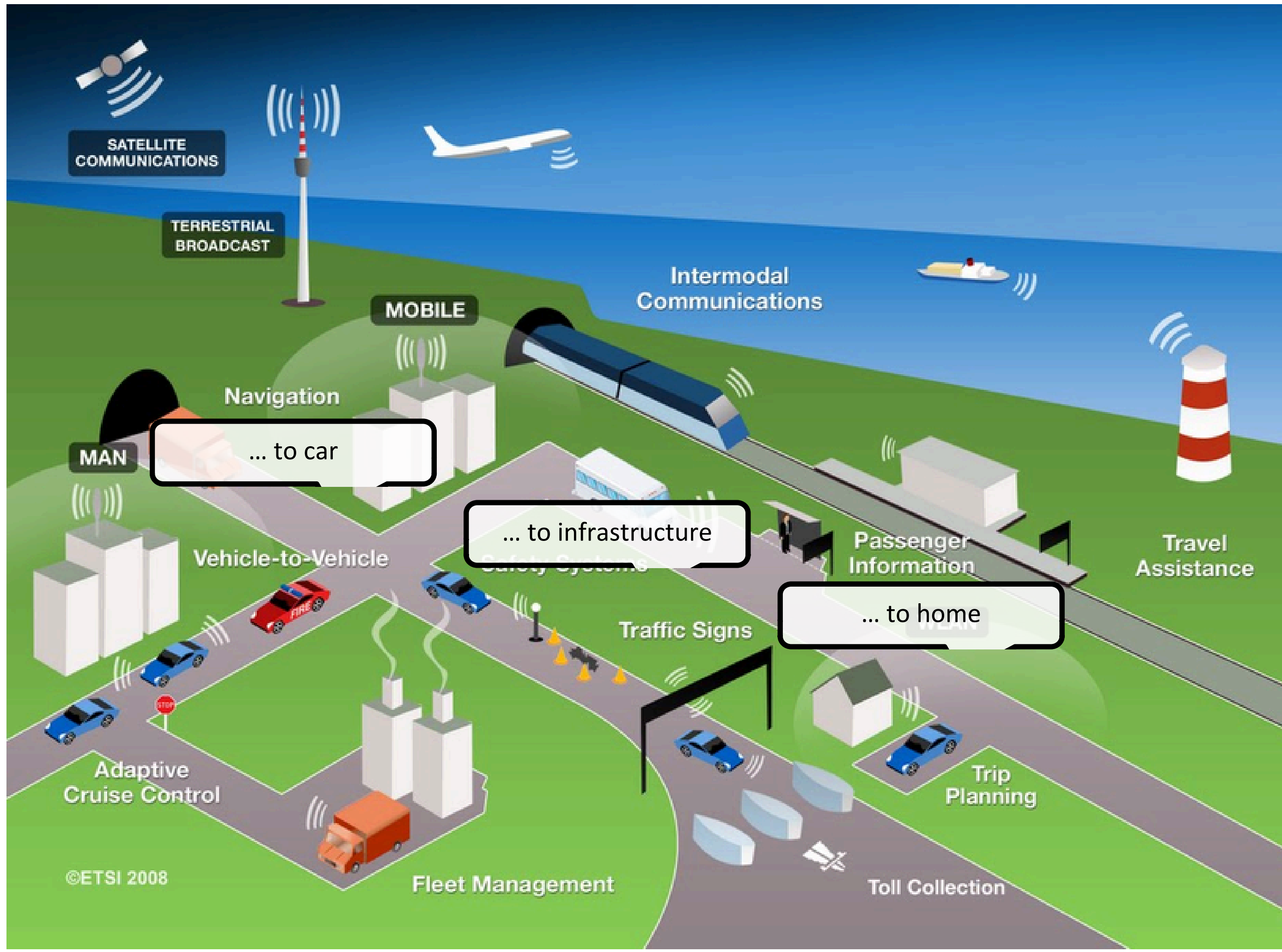
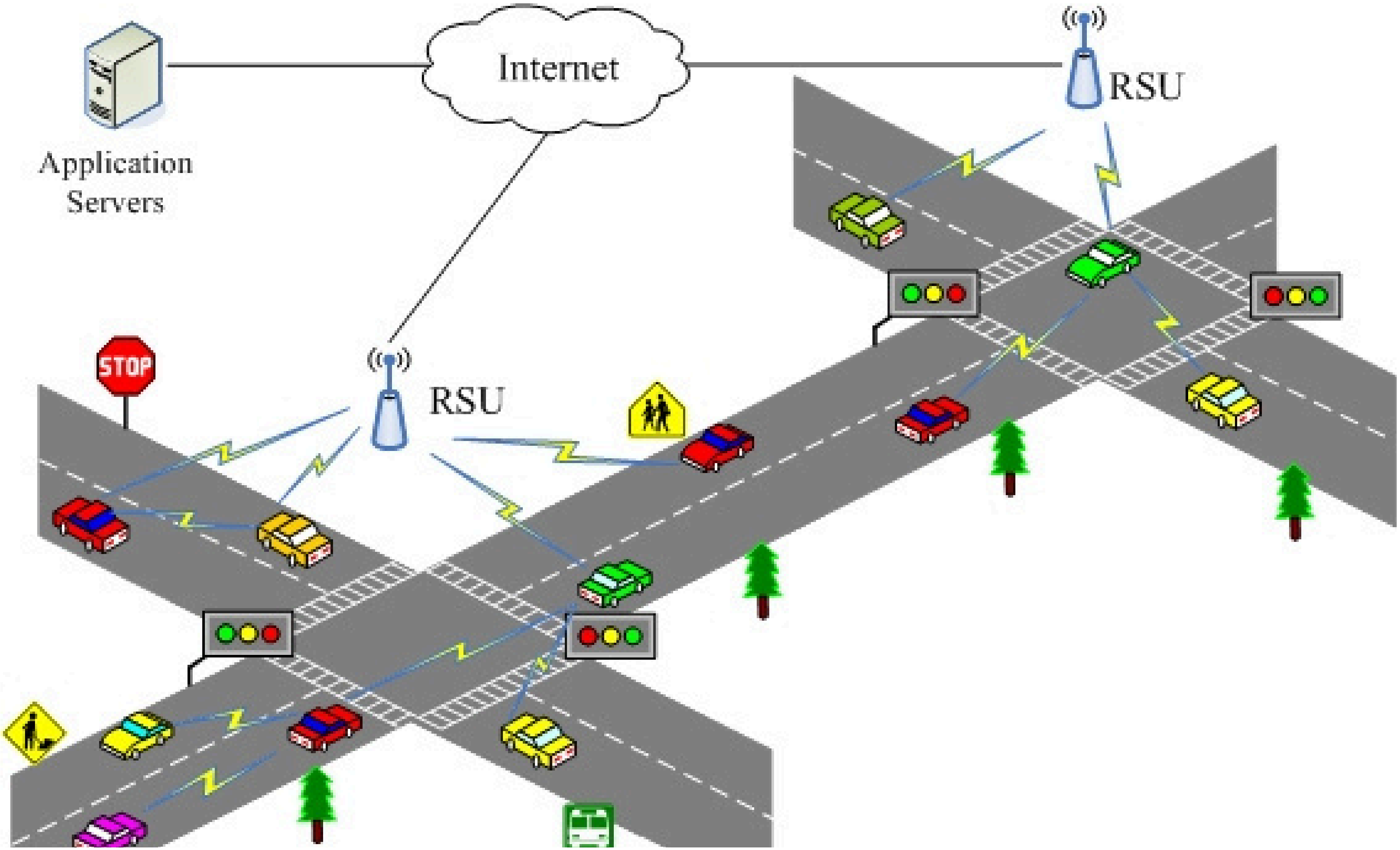


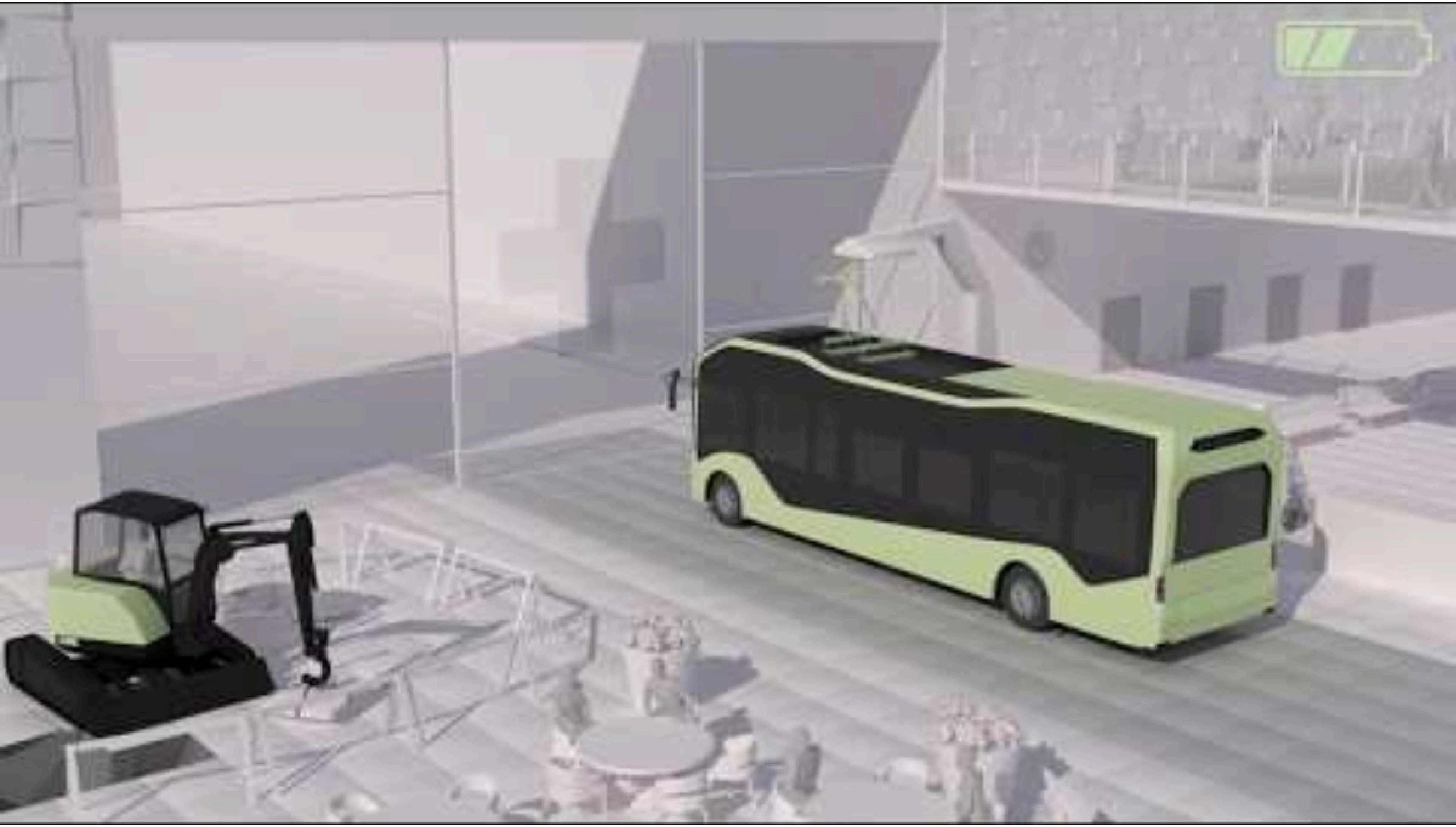
Illustration: ETSI

Car-to-X (C2X) Communication



Intelligent Transportation Systems (ITS)

Intelligent Transportation Systems (ITS)



Intelligent Transportation Systems (ITS)

What can ITS provide ?

- Safety
- Efficiency
- Traffic and road conditions
- Road signal alarm
- Local information

Intelligent Transportation Systems (ITS)

Few Emerging Applications that ITS provide ?

- Autonomous Intersection Management (AIM)
- Vehicle Platooning for Fuel and Space Efficiency
- Eco-Driving and Green Routing
- Cooperative Perception and Safety Sharing
- Autonomous On-Demand Public Transit (Dynamic ride-pooling)
- Automated Smart Parking
- Real-Time Traffic Prediction and Control
- Cooperative Highway Merging and Lane Changing
- Dynamic Speed and Lane Recommendations

Car-to-X (C2X) Communication

- Autonomous Intersection Management (AIM):

However, future AIM will adhere Strictly to Fundamental Traffic Rules, Prioritizing Safety as the Top Priority without Human Intervention

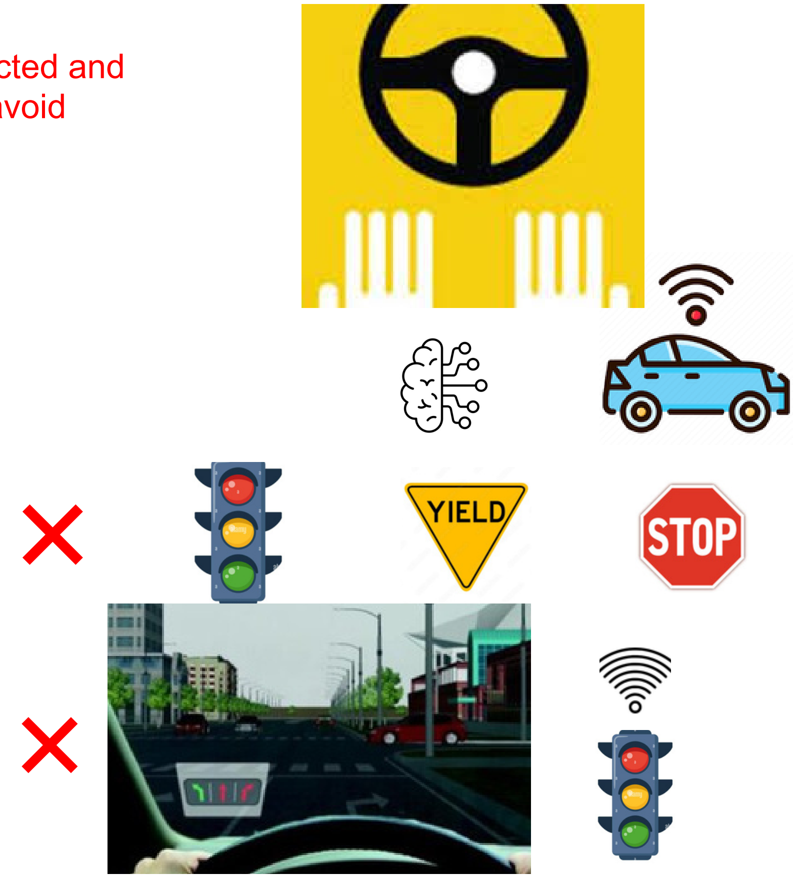
Car-to-X (C2X) Communication

- Autonomous Intersection Management (AIM):

- Replaces traffic lights with autonomous intersection control, where Connected and Autonomous Vehicles (CAVs) communicate with a central coordinator to avoid collisions.

Benefits:

- Minimal vehicle waiting time
- Reduced fuel consumption and emissions
- Elimination of signal delays
- Improved intersection throughput
- Enhanced safety



Research Extensions:

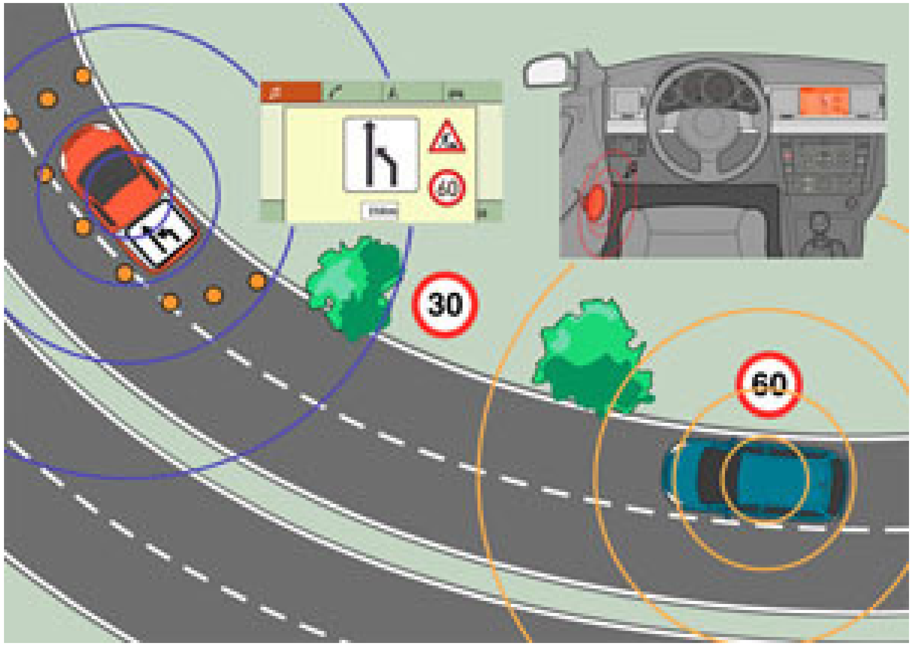
- Robustness to Communication Delays and Packet Loss
- Human-Driven Vehicle (HDV) and CAV Coexistence Models
- Integration with Eco-Driving Strategies
- Scalability in High-Density Urban Grids

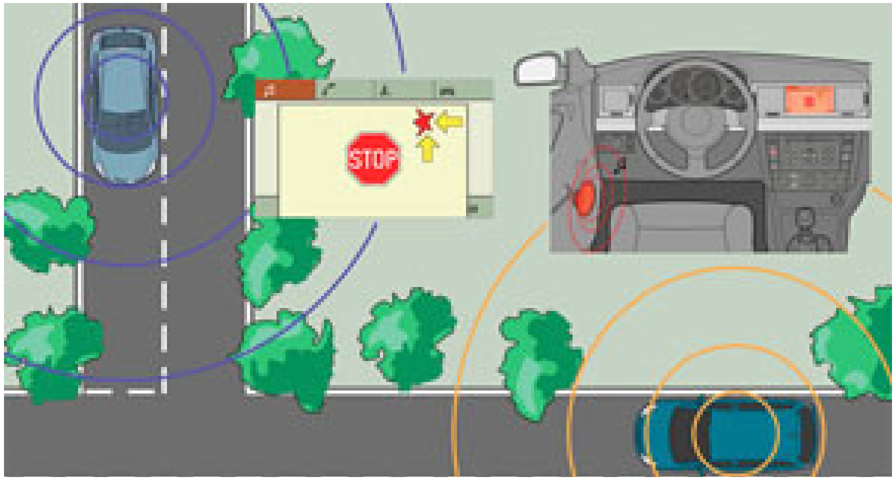
Car-to-X (C2X) Communication

What can Car-to-X (C2X) communication provide ?

Cooperative Perception and Safety Sharing

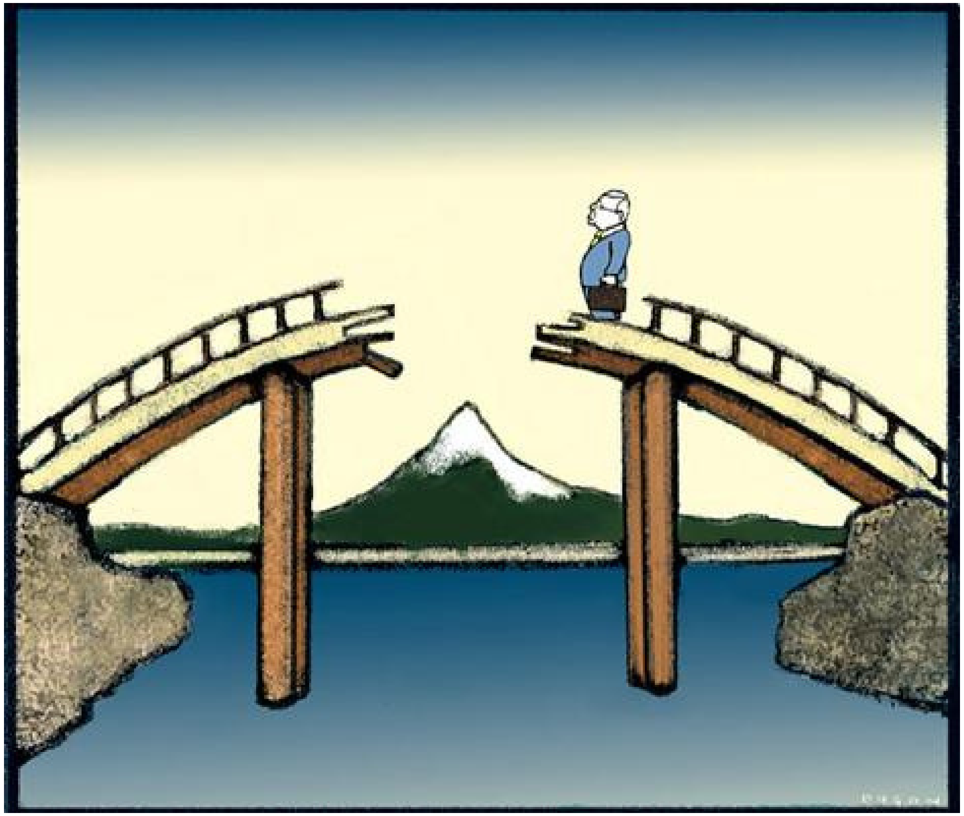
Warning!!





Car-to-X (C2X) Communication

What can Car-to-X (C2X) communication provide ?



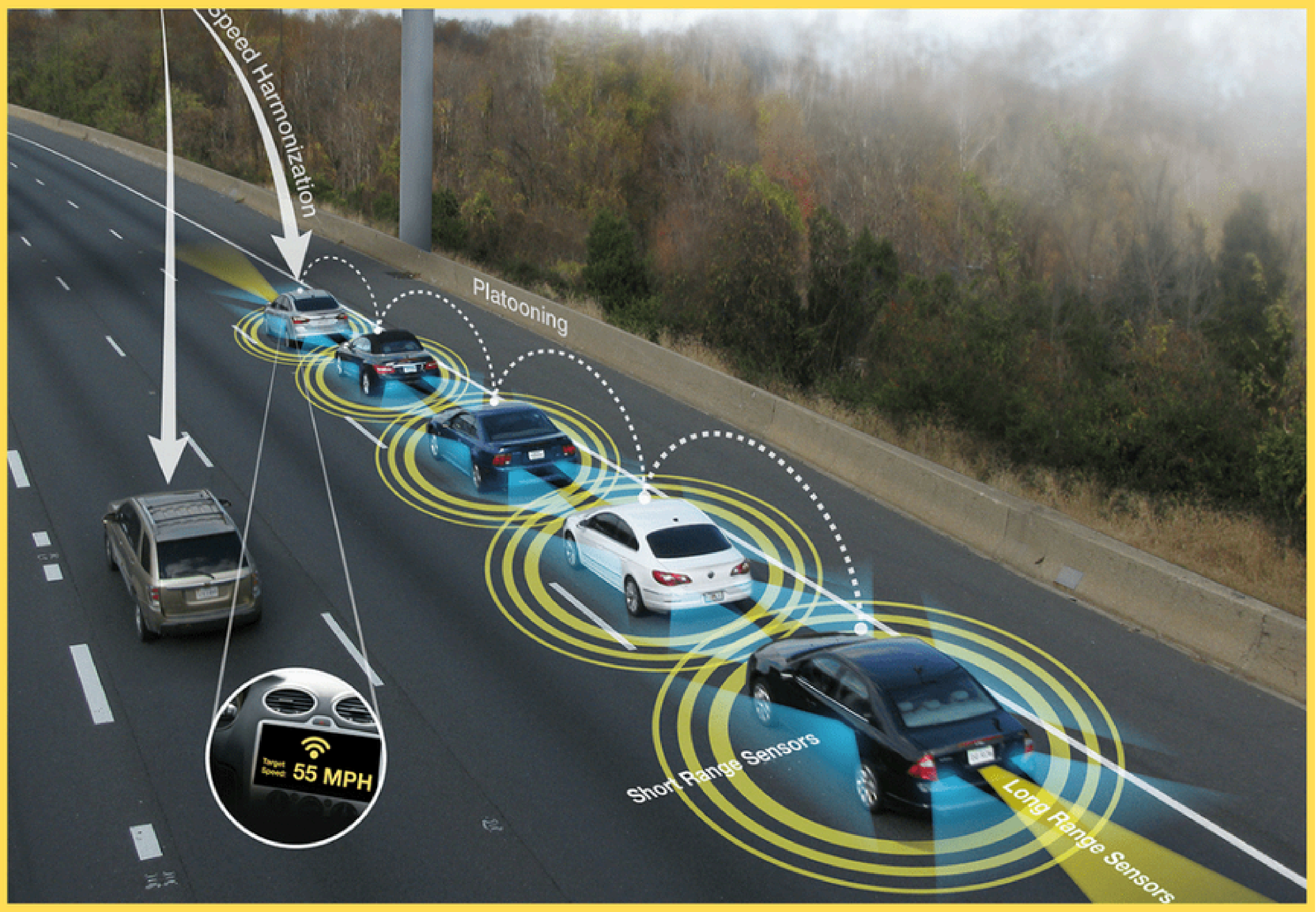
Warnings!!!

Incident Detection and Reporting

Car-to-X (C2X) Communication

What can Car-to-X (C2X) communication provide ?

Vehicle Platooning :: Eco-Driving



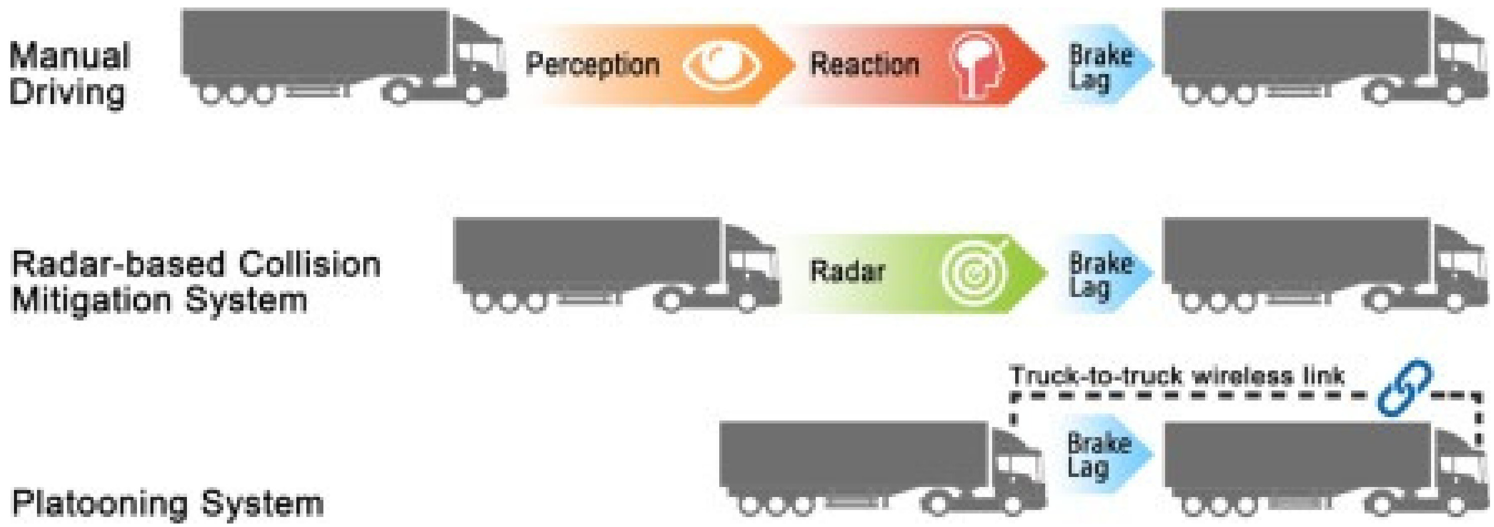
Car-to-X (C2X) Communication

What can Car-to-X (C2X) communication provide ?

Vehicle Platooning:: Eco-Driving

Faster and more synchronized braking response

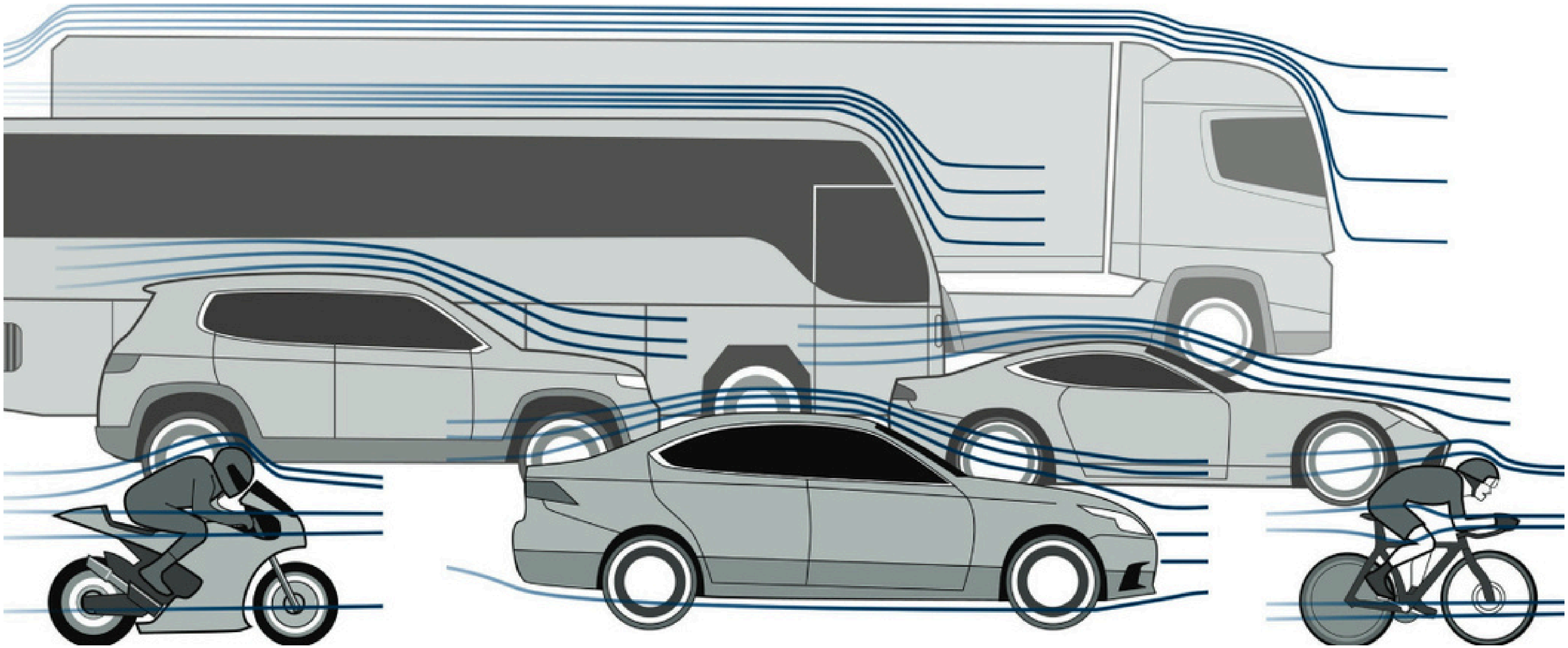
- enhancing safety
- enabling closer following distances
- improving traffic flow efficiency



Car-to-X (C2X) Communication

What can Car-to-X (C2X) communication provide ?

Vehicle Platooning for Improved Fuel Efficiency through Reduced Aerodynamic Drag



Car-to-X (C2X) Communication

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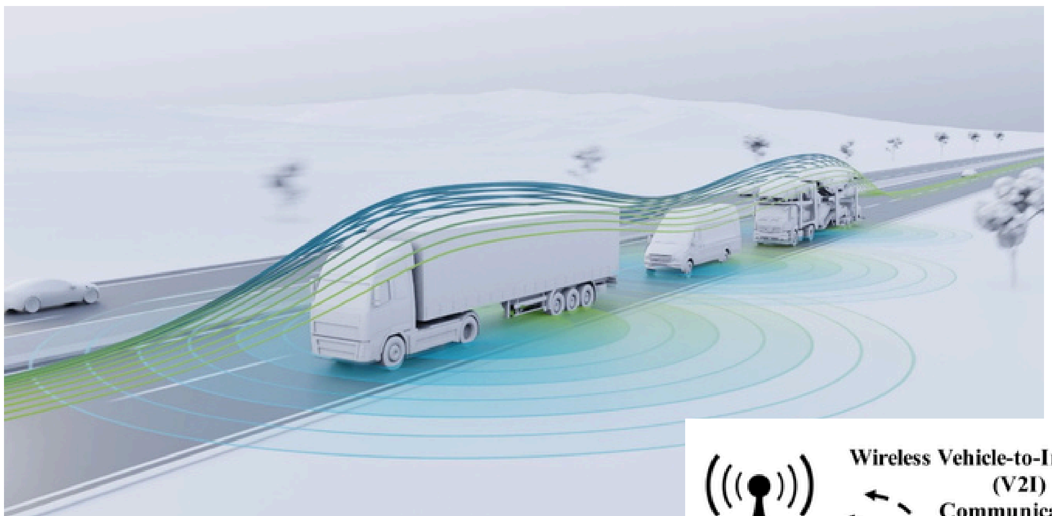
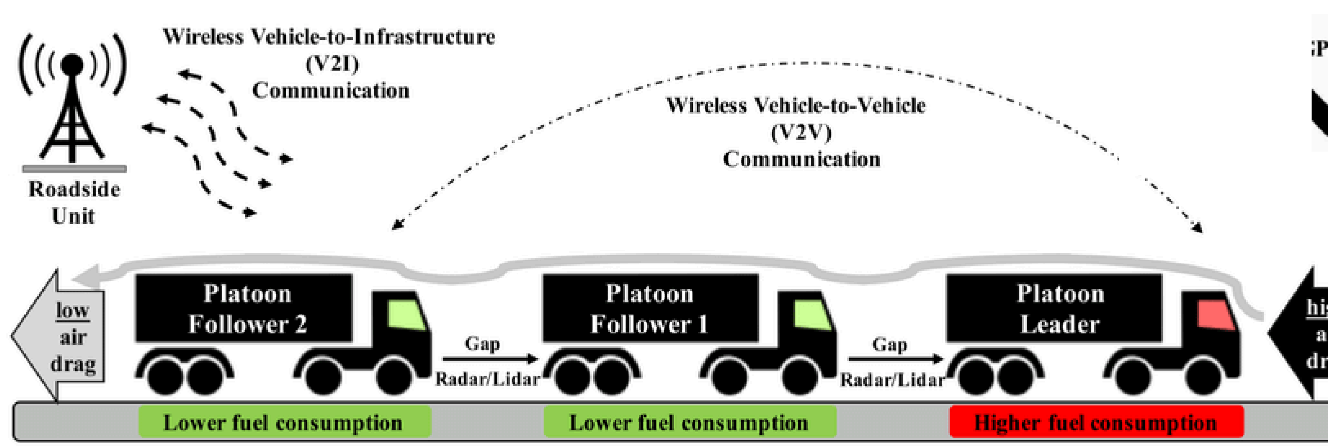




Car-to-X (C2X) Communication

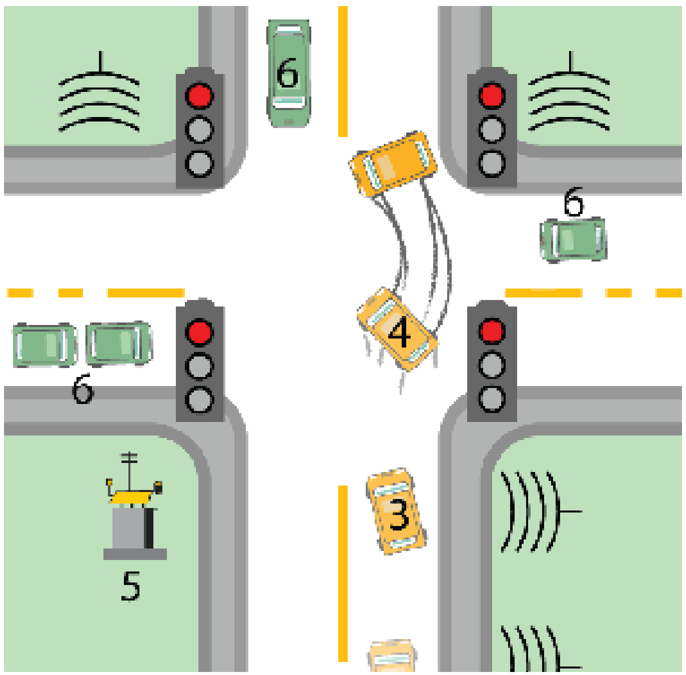
What can Car-to-X (C2X) communication provide ?

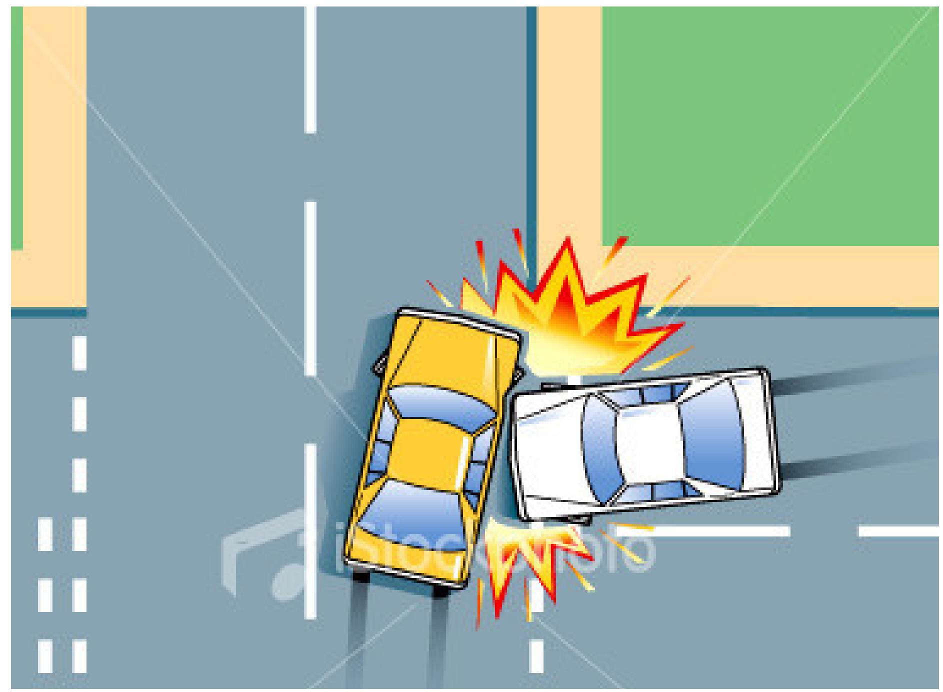
Intelligent Traffic Flow Management

Car-to-X (C2X) Communication

What can Car-to-X (C2X) communication provide ?

Road Traffic Accidents Avoidance

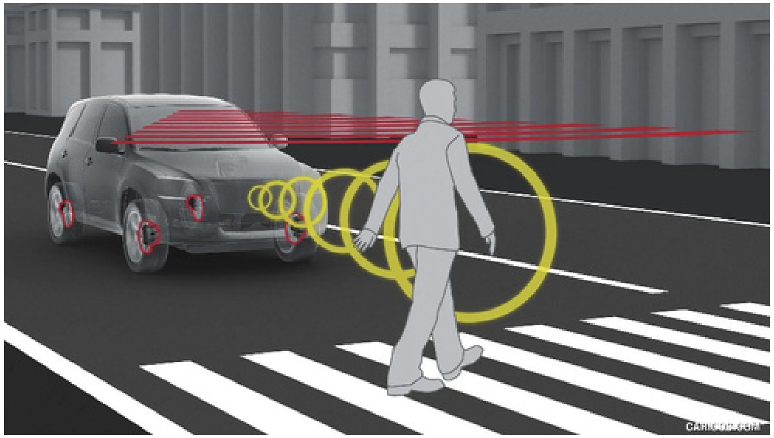




Car-to-X (C2X) Communication

What can Car-to-X (C2X) communication provide ?

Pedestrian and Cyclist Protection



Car-to-X (C2X) Communication:: Taxonomy of Use Cases

- Vehicle-to-X
  - Non-Safety
    - Comfort
      - Contextual Information
      - Entertainment
    - Traffic Information Systems
      - Optimal Speed Advisory
      - Congestion, Accident Information
  - Safety
    - Situation Awareness
      - Adaptive Cruise Control
      - Blind Spot Warning
    - Warning Messages
      - Traffic Light Violation
      - Electronic Brake Light

Car-to-X (C2X) Communication:: Taxonomy of Use Cases

Vehicle-to-X	
Non-Safety	**Safety**
Many messages High data rate	Few messages Small packet size
Low latency demands Low reliability demands	High latency demands High reliability demands

Challenges of C2X Communication

Communication	∘ Highly varying channel conditions ∘ High congestion, contention, interference ∘ Tightly limited channel capacity
Networking	∘ Unidirectional Links ∘ Multi-Radio / Multi-Network ∘ Heterogeneous equipment
Mobility	∘ Highly dynamic topology ∘ But: predictable mobility ∘ Heterogeneous environment
Security	∘ No (or no reliable) uplink to central infrastructure ∘ Ensuring privacy ∘ Heterogeneous user base

Distributed Control over Wireless Links

Automated Vehicles
- **Cars**
- **UAVs**(Unmanned Aerial Vehicles)
- **Insect flyers**

- **Packet loss and/or delays impacts controller performance.**
- **Controller design should be robust to network faults.**
- **Joint application and communication network design.**

Thank You